Beyond Benchmark Performance in Ophthalmic AI: A Review and Evidence Gap Map

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Supplementary Material

Tobias Schlosser[a], Michael Friedrich[b], Richard Vogel[c],
Benny Platte[d], Matthias Vodel[c], Danny Kowerko[b,d], Marc Ritter[d],
Miriam Goldammer[a], Vinodh Kakkassery[e], Ramin Khoramnia[f], Markus Wolfien[a,g]

[a] *Dresden University of Technology, Faculty of Medicine Carl Gustav Carus, Center for Medical Informatics, Fetscherstraße 74, Dresden, 01307, Saxony, Germany*
[b] *Chemnitz University of Technology, Faculty of Computer Science, Junior Professorship of Media Computing, Straße der Nationen 62, Chemnitz, 09107, Saxony, Germany*
[c] *University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Software Development for Media and Application Systems, Technikumplatz 17, Mittweida, 09648, Saxony, Germany*
[d] *University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Media Informatics, Technikumplatz 17, Mittweida, 09648, Saxony, Germany*
[e] *Klinikum Chemnitz, Department for Ophthalmology, Flemmingstraße 2, Chemnitz, 09116, Saxony, Germany*
[f] *University Hospital Carl Gustav Carus Dresden, Department of Ophthalmology, Fetscherstraße 74, Dresden, 01307, Saxony, Germany*
[g] *Dresden University of Technology, Center for Scalable Data Analytics and Artificial Intelligence (ScaDS.AI) Dresden/Leipzig, Strehlener Straße 12, 14, Dresden, 01069, Saxony, Germany*

1. Scope and structure of the supplementary material

This supplementary material provides the publication-level evidence and methodological detail underlying the review of artificial intelligence (AI) involving optical coherence tomography (OCT), OCT angiography (OCTA), and related ophthalmic imaging. It complements the main manuscript with analyses of publication trends, public datasets and benchmark resources, publication-level evidence gap assignments, and an extended interpretive synthesis of the included literature. Evidence maps, coding rules, source-level documentation, and the audit trail connect individual publications and classification decisions to the reported findings.

2. Annual publication trends by category

Source-specific document types were harmonised into five categories, comprising journal articles, conference papers or proceedings, reviews, books or chapters, and other or unclear records. Annual counts were then aggregated by publication year and harmonised category, as shown in Fig. 1 (panel A). Because the corpus was assembled during 2026, the total for that year is incomplete and should be interpreted descriptively.

Panels B–D preserve the annual totals but assign each record a single primary clinical target, method family, and task family, respectively. In panel B, the clinical-target labels match the corresponding axis in panel A. When a record had several targets, disease-specific targets took precedence over broader ones. For the shown figure labels, age-related macular degeneration (AMD), choroidal neovascularisation (CNV), diabetic retinopathy (DR), diabetic macular oedema (DME), and retinal nerve fibre layer (RNFL) are abbreviated.

Panel C groups records by primary method family using the convention in panel B. More specific families, including U-Net or encoder–decoder segmentation, convolutional neural networks (CNNs), vision transformers (ViTs), generative adversarial networks (GANs), foundation models, and large language models (LLMs), took precedence over broader machine-learning (ML) or deep-learning (DL) labels. The other deep-learning category was used only when no more specific method-family target was available. Finally, panel D groups records by primary task family.

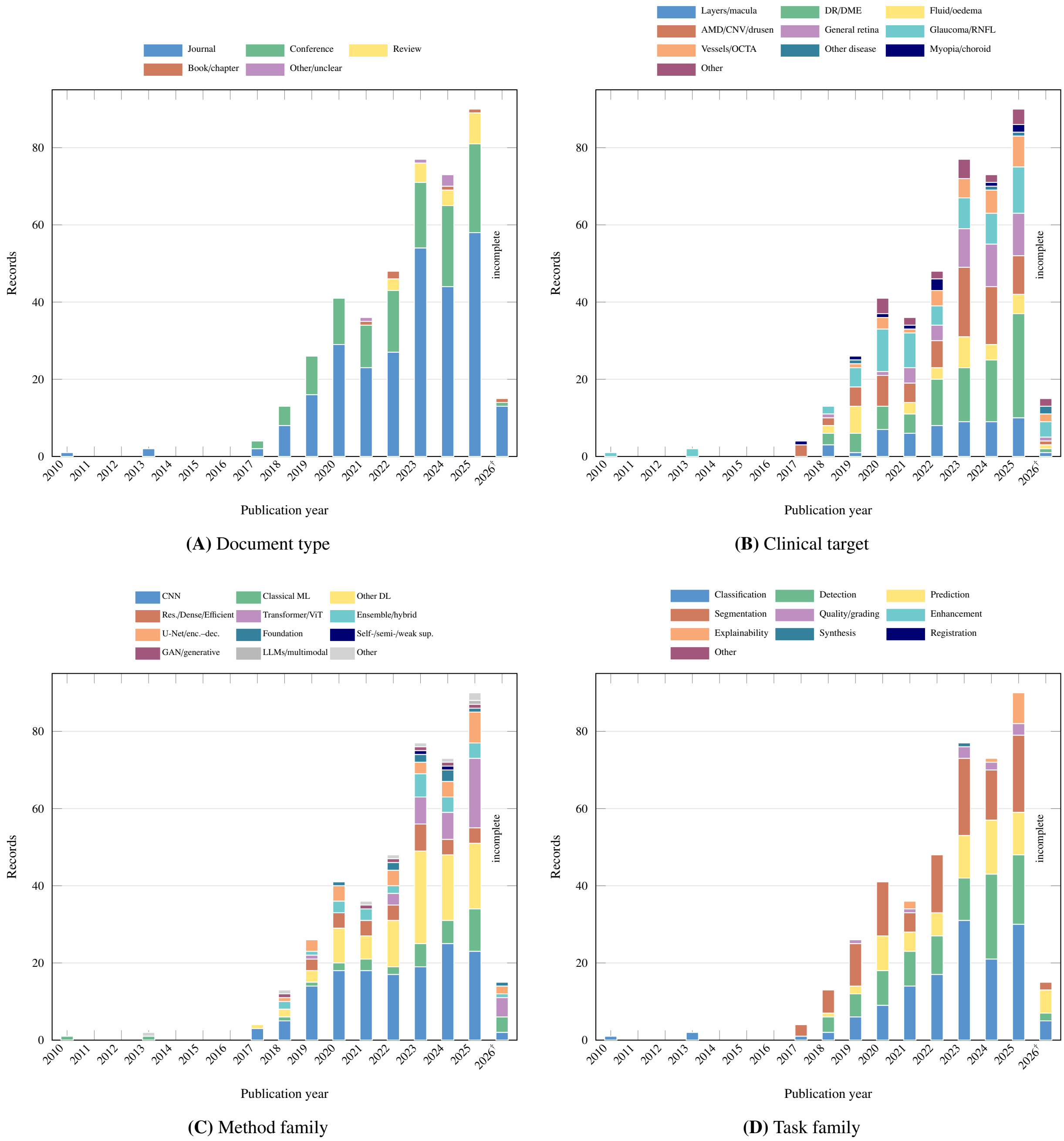
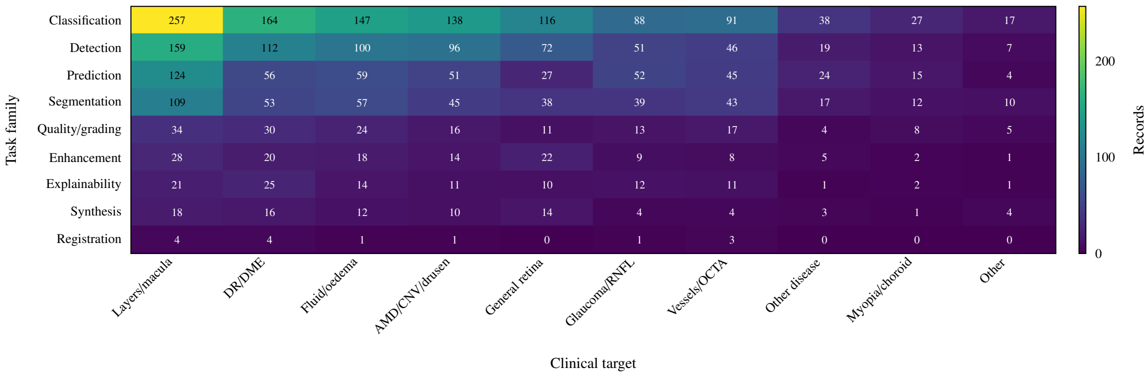
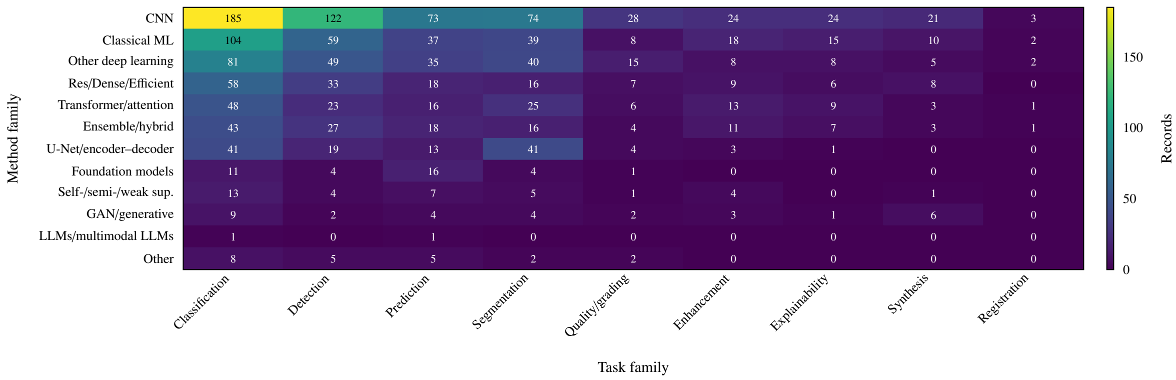


Figure 1: Annual publication trends among the records on artificial intelligence in ophthalmology. **(A)** Harmonised document type. **(B)** Primary clinical target. **(C)** Primary method family. **(D)** Primary task family. Bars show records by publication year, with stacked colours denoting the assigned category in each panel. Each panel assigns one primary label per record, preserving the annual totals. Additional task, clinical-target, and method-family labels were retained separately for audit. The 2026 total is incomplete because the corpus was assembled before year end.



(A) Task family by clinical target



(B) Method family by task family

Figure 2: Heatmaps of task, clinical-target, and method-family assignments among records on artificial intelligence in ophthalmology. **(A)** Task family by clinical target. **(B)** Method family by task family. Cell values represent record-level assignments, not experiment counts or mutually exclusive study counts. Because coding was multi-label, one record may contribute to several cells. The LLM row includes only validated publications in which an LLM or multimodal LLM formed part of the study.

3. Heatmaps of task, clinical-target, and method-family assignments

The two heatmaps in Fig. 2 extend the descriptive assessment of the evidence base. Panel A relates task families to clinical targets, while panel B relates method families to task families. Cell values represent record-level assignments rather than mutually exclusive groups of studies, and a single publication may contribute to multiple cells. This structure highlights both concentrated areas of investigation and gaps in which particular clinical needs are rarely linked to specific methodological approaches. The distribution across panel A shows that task coverage differs substantially among clinical targets, indicating that methodological development is concentrated around particular combinations of clinical question and analytical task rather than distributed uniformly across ophthalmology. Panel B similarly shows that method families are associated with distinct task profiles, reflecting methodological specialisation while also identifying opportunities to evaluate established and emerging approaches.

The method-family heatmap also identifies studies that explicitly used LLMs or multimodal language-model systems. Fields used only for review screening did not count as evidence of study-level LLM use, avoiding conflation of review tools with methods examined in the included literature. This row therefore provides a conservative estimate of language-model use in ophthalmic AI research while maintaining the distinction between the evidence base and the review process. Their limited representation suggests that language-model applications remain an emerging area of ophthalmic AI, with opportunities for further evaluation across clinical tasks and workflows.

4. Identifiable public datasets and benchmark resources

Table 1 consolidates identifiable public datasets and benchmark resources represented in the included literature. It includes named datasets, challenge resources, and reusable benchmark collections while excluding generic labels such as “private dataset”, local hospital cohorts, and unidentifiable study-specific collections. Focusing on traceable resources supports comparison across studies and highlights datasets that can be independently identified and reused.

OCT-based retinal disease classification relies substantially on a small group of public datasets. The Kermany retinal OCT dataset, published under several names including UCSD, Kaggle, Mendeley, OCT2017, and LOCT, is the most widely represented. Other classification resources include OCTDL, OCTID, OCTMNIST, RetinalOCT-C8 / OCT-C8, A2ASDOCT, and SERI-CUHK spectral-domain OCT (SD-OCT). Collectively, these datasets support supervised learning across AMD, CNV, DME, drusen, DR, macular hole, and healthy controls. Their reuse promotes comparison and reproducibility across studies, while evaluation across more diverse imaging distributions, disease labels, acquisition protocols, and preprocessing conventions could further strengthen generalisability.

Segmentation benchmarks constitute a second major group of reusable OCT and OCTA resources. RETOUCH, OPTIMA, AI Challenger OCT, Duke / DUKE SD-OCT / Duke DME OCT, and AROI support segmentation of retinal fluid, layers, and lesions. OCTA-500 / OCTA-3M / OCTA-6M, OCTA-SS, ROSE-1, FAROS, and DRAC 2022 / OCTA-DR broaden this scope to vascular, foveal avascular zone (FAZ), capillary, lesion, image-quality, and DR-grading tasks. Selected fundus benchmarks, including ACRIMA, ORIGA, and EyePACS, further support multimodal OCT or OCTA studies. Together, the resources in Table 1 provide established benchmarks for comparison and reproducibility across imaging tasks. Greater diversity, multimodal coverage, comprehensive documentation, and external validation could further strengthen this shared foundation for ophthalmic AI research.

Table 1: Consolidated overview of identifiable public ophthalmic imaging datasets and benchmark resources represented in the included studies. Dataset names were harmonised where multiple entries referred to the same or closely related resource.

Dataset	Area	Modality	Scale	Typical tasks and notes
A2ASDOCT [1, 2]	Age-related macular degeneration	SD-OCT	Age-Related Eye Disease Study 2 (AREDS2) ancillary SD-OCT cohort with 38,400 B-scans from 384 participants (269 with intermediate AMD and 115 controls)	Public AMD-focused OCT cohort used for quantitative classification of eyes with and without intermediate AMD.
ACRIMA [3, 4]	Glaucoma	Colour fundus photographs	705 fundus images	Optic disc-centred fundus benchmark with 396 glaucomatous and 309 normal images.
AI Challenger OCT dataset [5, 6]	Macular oedema / retinal OCT lesion segmentation	Retinal OCT volumes	AI Challenger 2018 macular oedema dataset with 83 OCT volumes and 10,880 OCT slices	Used for macular oedema-related lesion segmentation, including subretinal fluid and pigment epithelial detachment.
AROI dataset [7, 8]	Retinal layers and fluid in neovascular AMD	Retinal OCT	1,136 annotated B-scans from 25 participants, selected from 3,200 B-scans	A public annotated retinal OCT database used for joint retinal layer, lesion, and fluid segmentation in healthy and neovascular AMD eyes.
DRAC 2022 / OCTA-DR [9, 10]	Diabetic retinopathy	Ultra-widefield OCTA (UW-OCTA)	1,103 UW-OCTA images	Consolidates DRAC 2022, Diabetic Retinopathy Analysis Challenge 2022, DRAC OCTA, and OCTA-DR when these names denote the same DR challenge. Tasks include DR lesion segmentation, image-quality assessment, and DR grading.
Duke / DUKE SD-OCT / Duke DME OCT dataset [11, 12]	Diabetic macular oedema and retinal layer/fluid analysis	SD-OCT B-scans	110 annotated B-scans from 10 participants with DME	Consolidates Duke dataset, DUKE, DUKE OCT dataset, Duke SD-OCT, Duke spectral-domain OCT dataset, and Chiu-style Duke DME references. Used for retinal layer, boundary, and fluid segmentation.
EyePACS [13, 14]	Diabetic retinopathy	Colour fundus photographs	35,126 labelled training images in the Kaggle DR release	Fundus benchmark for diabetic retinopathy grading. Included where studies combine fundus and OCT / OCTA resources.

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Dataset	Area	Modality	Scale	Typical tasks and notes
FAROS [15, 16]	Retinal OCTA microstructure segmentation	OCTA	40 OCTA images from 40 participants	Fully Annotated Retinal OCTA Segmentation dataset. Used for retinal microstructure segmentation and quantitative OCTA analysis.
Kermany / UCSD / Kaggle / Mendeley / OCT2017 / LOCT retinal OCT dataset [17, 18]	Retinal disease (CNV, DME, drusen, and healthy retina)	Retinal OCT B-scans	108,312 labelled OCT images in the original report (approximately 84,495 images in common Kaggle mirrors)	Consolidates UCSD, Kermany, Kaggle OCT, Mendeley OCT, OCT2017, OCT 2017, labelled OCT, publicly available validated OCT, and similar aliases. Used mainly for four-class retinal disease classification.
OCTDL [19, 20]	Multi-disease retinal OCT	Retinal OCT B-scans	2,064 high-resolution OCT images from 821 participants across seven disease or condition groups	Consolidates OCTDL and OCTDL dataset entries. Used for image-level retinal disease classification across AMD, DME, epiretinal membrane (ERM), retinal artery occlusion (RAO), retinal vein occlusion (RVO), vitreomacular interface disease (VID), and healthy classes.
OCTID [21, 22]	Multi-disease retinal OCT	Retinal OCT B-scans	572 OCT images across five classes	Consolidates OCTID benchmark / public entries. Used for classification of healthy retina, AMD, central serous retinopathy (CSR), macular hole, and diabetic retinopathy.
OCTMNIST [23, 24]	Retinal disease benchmark	Downsampled retinal OCT	109,309 source OCT images standardised to a resolution of 28 × 28 pixels	MedMNIST benchmark for four-class OCT classification derived from OCT2017 / Kermany-style OCT data. Kept separate from the clinical-resolution Kermany / UCSD dataset because MedMNIST provides standardised low-resolution benchmark images and predefined splits.
OCTA-500 / OCTA-3M / OCTA-6M [25, 26]	Retinal vasculature and multi-disease OCTA	OCT and OCTA volumes, projections, labels	500 participants with 3 mm × 3 mm and 6 mm × 6 mm fields of view	Consolidates OCTA-500 benchmark / public entries and OCTA-3M / OCTA-6M aliases where used as OCTA-500 subsets. Used for vessel, FAZ, artery, vein, capillary, and retinal layer segmentation.
OCTA-SS [27, 28]	Retinal vasculature	OCTA	55 parafoveal regions of interest (ROIs) from 3 mm × 3 mm OCTA images of 11 participants	Used for OCTA vessel segmentation. Kept separate because it is referenced as a distinct OCTA benchmark and provides detailed superficial-layer vessel annotations.

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Dataset	Area	Modality	Scale	Typical tasks and notes
OPTIMA challenge [29, 30]	Retinal fluid and macular disease	Retinal OCT	A retinal-fluid challenge dataset commonly described as comprising 30 OCT volumes	Consolidates Optima and OPTIMA challenge. Used for intraretinal cystoid fluid segmentation in OCT. Kept separate from RETOUCH because the dataset names and challenge lineage differ.
ORIGA [31, 32]	Glaucoma	Colour fundus photographs	650 fundus images in ORIGA-light (commonly reported as 168 glaucoma and 482 healthy images)	Fundus benchmark for optic disc and cup analysis and glaucoma detection. Included where paired or externally compared with OCT measurements.
RetinalOCT-C8 / OCT-C8 [33, 34]	Multi-disease retinal OCT	Retinal OCT B-scans	24,000 OCT images across eight classes, balanced at 3,000 images per class	Consolidates OCT-C8, OCT-C8 dataset, and RetinalOCT-C8. Used for eight-class retinal OCT disease classification across AMD, CNV, CSR, DME, DR, drusen, macular hole, and healthy retina.
RETOUCH challenge [35, 36]	Retinal fluid in AMD and retinal vein occlusion	Multi-vendor retinal OCT volumes	112 OCT volumes and 11,334 B-scans overall	Consolidates RETOUCH, Re-TOUCH, the Medical Image Computing and Computer Assisted Intervention (MICCAI) RETOUCH challenge, the RETOUCH challenge dataset, and the RETOUCH retinal OCT fluid detection challenge. Used for intraretinal fluid (IRF), subretinal fluid (SRF), and pigment epithelial detachment (PED) detection and segmentation.
ROSE-1 [37, 38]	Retinal OCTA vessel segmentation	En face OCTA	117 OCTA images from 39 participants in ROSE-1	Consolidates ROSE-1 and Retinal OCTA vessel Segmentation-1. Used for OCTA vessel segmentation.
SERI-CUHK SD-OCT [39, 40]	Diabetes-related retinal disease	SD-OCT	The SERI dataset contains 32 SD-OCT volumes, whereas the CUHK dataset contains 43	Used for diabetes-related retinal disease classification, including healthy, DME, and diabetic retinopathy with diabetic macular oedema categories in source studies. Kept separate from A2ASDOCT because the names indicate distinct source datasets.

5. Publication-level mappings for the evidence gap analysis

Tables 3 and 4 provide the publication-level assignments underlying the evidence gap maps. By identifying the publications assigned to each dimension and stance, they extend the aggregate counts with a traceable account of how evidence is distributed across categories. The general reference table covers dimensions relevant to ophthalmic AI research, clinical translation, and implementation, including model performance, technological limitations, real-world impact, data availability and integration, data quality and standardisation, infrastructural demands, financial costs, regulatory requirements, cybersecurity risks, and ethical concerns. Within each dimension, the evidence is classified as predominantly positive, predominantly negative, or balanced.

The ethical reference table applies the same decision framework to privacy, transparency, integrity, security, non-maleficence, trust, accountability, equity, and autonomy. It distinguishes issues reported as caused by AI, resolved through an AI-related approach, or both. Routine ethics approval, consent, anonymisation, restricted data availability, and author-accountability statements were not considered evidence that AI had resolved an ethical issue. Similarly, a technical method received no ethical stance unless the evaluated text explicitly linked it to the relevant ethical dimension. Under these criteria, publications could remain unassigned, and neutral or mixed categories were not used to represent insufficient evidence. Table 2 defines the dimension-level inclusion and exclusion criteria governing every assignment.

Table 2: Operational inclusion, exclusion, and judgement rules for the evidence gap maps. In the general map, positive indicates that a publication presents AI as enabling or improving the relevant dimension, whereas negative indicates a barrier, risk, failure, or constraint created or exposed by AI. Balanced requires evidence in both directions within the same dimension. In the ethics map, caused indicates that AI introduces or amplifies an issue, whereas resolved identifies an AI-related mechanism reported to mitigate it. Caused and resolved requires evidence in both directions.

Dimension	Evidence required for assignment	Evidence not sufficient for assignment
General map		
Model performance	Reported quantitative or qualitative performance, comparison, error, agreement, calibration, or robustness outcome.	Architecture or method description without an evaluated outcome.
Technological limitations	An explicit failure mode or generalisation or robustness constraint affecting an AI model or system, an image artefact, a technical restriction, or a method that explicitly overcomes such a constraint.	A limitation inferred from study design, a sample-size or data limitation without a stated technical consequence, or generic future work.
Real-world impact	A system deployed or used in practice, or a measured patient, clinician, workflow, referral, treatment, access, or service consequence.	Retrospective benchmark performance, claimed clinical relevance, prospective utility, or an unmeasured promise of workload reduction.
Data availability, integration, quality, and standardisation	An explicit AI-linked effect of data scarcity, missingness, quality, representativeness, annotation, access, harmonisation, integration, interoperability, or standardisation.	Merely naming or using a dataset, an ordinary train/test split or preprocessing step, or a routine data-availability statement.
Infrastructural demands	An explicit deployment or integration burden or enabling factor involving hardware, cloud or network capacity, connectivity, staffing, workflow, maintenance, or institutional resources.	A list of devices, software, graphics processing units (GPUs), memory, or runtime in the methods section without a stated implementation consequence.
Financial costs	An explicit consequence involving monetary cost, affordability, reimbursement, economic burden, staffing costs, or health-system resources.	Parameter count, runtime, hardware specification, or "computational cost" without a monetary or affordability consequence.
Regulatory requirements	An explicit AI-related approval, certification, liability, legal, compliance, governance, or post-market obligation or barrier.	Routine research ethics approval, consent, Declaration of Helsinki compliance, or adherence to reporting guidelines.
Cybersecurity risks	An explicit hostile attack, vulnerability, security safeguard, adversarial threat, or resilience claim involving an AI or data system.	Generic privacy, ordinary model robustness, access details, or an inferred security risk.
Ethical concerns	An explicit cross-cutting AI-related ethical harm, benefit, dilemma, or responsible AI implication.	Routine ethics approval or consent, or an ethical issue supported only by procedural boilerplate.
Ethics map		
Privacy	An explicit AI-linked privacy risk or a concrete privacy-specific AI mechanism reported to mitigate that risk.	Routine anonymisation, de-identification, consent, restricted data access, or a non-public dataset without an explicit AI-related privacy effect.

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Dimension	Evidence required for assignment	Evidence not sufficient for assignment
Transparency	An explicit effect on explainability, interpretability, disclosure, traceability, or clinician understanding of an AI decision.	Routine methods reporting, performance metrics, model diagrams, heatmaps, or visualisation without a stated interpretive function.
Integrity	An explicit effect on truthfulness, authenticity, manipulation, corruption, or evidentiary integrity of AI or data outputs.	Ordinary accuracy, ground-truth labelling, validation, reproducibility, data cleaning, or data-quality methods.
Security	An explicit ethical threat or protection affecting the confidentiality, integrity, or availability of an AI or data system.	Generic privacy, robustness, login/access details, or a security effect inferred by the reviewer.
Non-maleficence	An explicit AI-related harm, safety risk, consequence of an incorrect output, or reported harm-prevention mechanism.	A generic diagnostic purpose, accuracy improvement, or patient benefit inferred without a stated harm connection.
Trust	Trust, confidence, acceptance, reliance, or credibility of the AI system is explicitly named, investigated, or measured.	Trust inferred solely from accuracy, robustness, expert agreement, validation, or explainability.
Accountability	Explicit responsibility, liability, oversight, contestability, or answerability for an AI-supported decision.	Author contribution or accountability statements, ethics approval, or expert labelling alone.
Equity	An explicit AI-related subgroup fairness, bias, disparity, representativeness, access, or distributional effect.	Ordinary class balancing unless linked to affected patient groups, access, or outcome inequity.
Autonomy	An explicit effect on patient or clinician agency, informed choice about AI use, or meaningful human control.	The technical term "autonomous", generic personalisation, or clinician involvement without an agency effect.

Table 3: Publication-level reference assignments underlying the general evidence gap map. Each cell is derived from the canonical publication-level decision matrix. A "/" indicates that no publication met the explicit-evidence rule for the corresponding combination of dimension and stance.

General dimension	Positive framing	Negative framing	Balanced
Model performance	[41–379]	/	[380–456]
Technological limitations	[41, 49, 53, 54, 64, 69, 73, 74, 84, 88, 102, 105, 111, 116, 133, 137, 140, 153, 180, 195, 206, 221, 228, 232, 239, 246, 267, 275, 279, 296, 311, 312, 315, 317, 318, 332–334, 336, 339, 340, 342, 348, 351, 367, 376, 384, 402, 420, 427, 443]	[55, 65, 75, 91, 94, 98, 100, 115, 119, 124, 146, 147, 157, 174, 177, 182, 185, 194, 199–201, 210, 218, 220, 231, 237, 244, 251, 258, 268, 280, 282, 286, 290, 291, 300, 323, 327, 337, 344, 345, 347, 354, 356, 360, 366, 381, 385, 390–392, 394, 395, 398, 404, 405, 410, 416, 421, 423, 426, 428, 432, 433, 436, 440, 442, 445, 448, 450–453, 456, 457]	[43, 44, 46–48, 51, 52, 58, 60, 62, 76, 78–80, 86, 95–97, 101, 104, 113, 118, 122, 125, 127, 130, 134, 141, 143, 156, 158–160, 165–167, 169, 170, 172, 176, 181, 183, 188, 189, 191, 196, 203, 204, 208, 211, 214, 215, 224, 225, 227, 229, 230, 234, 245, 247, 250, 253, 255, 256, 259, 264, 269, 271, 273, 278, 285, 288, 293, 295, 297, 299, 306, 309, 314, 320, 322, 324, 325, 335, 338, 341, 357, 359, 361, 363, 372, 373, 377, 378, 380, 382, 383, 388, 389, 393, 396, 400, 401, 406–409, 413–415, 418, 419, 422, 429, 430, 435, 437–439, 441, 446, 447, 454, 455, 458]
Real-world impact	[53, 132, 144, 147, 155, 156, 158, 203, 212, 218, 234, 258, 263, 342, 379, 399, 422]	/	/

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General dimension	Positive framing	Negative framing	Balanced
Data availability, integration, quality, and standardisation	[41, 45, 60, 64, 67, 73, 78, 79, 86, 99, 108, 127, 133, 134, 142, 153, 157, 181, 186, 199, 226, 238, 239, 243, 252, 264, 272, 279, 283, 309, 310, 312, 326, 339, 345, 362, 369, 373, 389, 409, 410, 423, 434]	[52, 53, 55, 70, 74, 91, 97, 114, 119–122, 124, 147, 149–151, 156, 172, 176, 177, 182, 188, 189, 200–202, 209, 218, 220, 224, 229–231, 237, 244, 245, 256, 258, 268, 290, 291, 305, 314, 323, 328, 343, 346–350, 356, 358, 360, 366, 372, 377, 385, 392, 395–397, 402, 407, 413, 416, 418, 419, 425, 427–429, 435, 436, 439, 444, 446, 448, 449, 451, 452, 455]	[43, 44, 46, 49–51, 54, 58, 61, 62, 65, 68, 69, 76, 81, 82, 88, 93, 94, 98, 101, 102, 105, 110, 139, 141, 144, 146, 160, 165, 183, 185, 191, 194, 203, 208, 211, 214, 215, 217, 221, 225, 227, 228, 233, 247, 250, 255, 259, 260, 269, 271, 273, 277, 278, 280, 292, 293, 297–299, 302, 311, 313, 315, 318, 320, 322, 325, 327, 329, 331, 334, 336, 337, 340, 341, 344, 354, 357, 359, 363, 378–380, 382, 387, 393, 394, 404, 406, 414, 420, 422, 430, 431, 437, 438, 441, 442, 445, 447, 456]
Infrastructural demands	[82, 90, 94, 97, 99, 104, 116, 144, 148, 177, 180, 181, 185, 199, 224, 229, 232, 233, 242, 244, 258, 282, 310, 334, 342, 356, 365, 382, 393, 395, 399, 400, 406, 407, 409, 420, 427, 459]	[54, 124, 146, 201, 230, 247, 250, 297, 349, 350, 358, 388, 413, 441]	[41, 53, 59, 65, 156, 158, 203, 223, 281, 341, 379, 418, 429]
Financial costs	[65, 95, 97, 98, 131, 136, 141, 165, 177, 180, 181, 185, 189, 194, 200, 201, 203, 238, 258, 269, 277, 281, 282, 291, 310, 316, 321, 380, 395, 406, 407, 409, 410, 417, 418, 427, 436, 450, 459]	[43, 76, 349, 350]	[77, 153, 158, 323]
Regulatory requirements	[291, 397]	[124, 156, 194, 224, 323, 334, 361, 399, 422, 433]	/
Cybersecurity risks	[230]	[347]	[334]
Ethical concerns	/	[68, 124, 156, 194, 224, 291, 323, 334, 361]	/

Table 4: Publication-level reference assignments underlying the ethical evidence gap map. Each cell is derived from the canonical publication-level decision matrix. A "/" indicates that no publication met the explicit-evidence rule for the corresponding combination of dimension and stance.

Ethical dimension	Caused	Resolved	Caused and resolved
Privacy	[76, 124, 144, 156, 194, 291, 347]	/	[323, 381]
Transparency	[61, 64, 74, 96, 146, 194, 208, 224, 231, 244, 250, 258, 302, 334, 343, 347, 349, 363, 379, 413, 414, 418, 430, 432, 448, 451]	[51, 58, 62, 76, 80, 88, 91, 92, 101, 103, 105, 108, 117, 120, 132, 136, 143, 144, 147, 156, 160, 162, 165, 167, 175, 177, 181, 183, 188, 195, 199–201, 209, 216, 232, 245, 259, 260, 274, 290, 294, 296, 317, 322, 327, 332, 345, 350, 351, 358–360, 368, 372, 380, 382, 394, 407, 411, 416, 425, 431, 438, 440–443, 449, 450, 455]	[68, 124, 171, 173, 178, 179, 193, 203, 214, 223, 238, 291, 293, 297, 316, 323, 329, 335, 381, 397, 427]
Integrity	/	/	/
Security	[347]	[230]	[334]
Non-maleficence	[142, 146, 214, 220, 323, 392, 456]	[62, 165, 174, 189, 263, 271, 282, 296, 310, 354, 356, 380, 439, 450]	[50, 54, 223, 363, 385]
Trust	[96, 144, 146, 194, 237, 244, 291, 347]	[68, 84, 105, 156, 162, 173, 214, 259, 316, 317, 380, 397, 436]	[124, 203, 223, 323, 334, 363, 381]
Accountability	[124]	[53, 68, 277, 341, 394]	/
Equity	[46, 68, 70, 91, 124, 171, 178, 179, 182, 193, 194, 209, 247, 277, 291, 323, 349, 360, 382, 387, 408, 418, 434, 440, 448]	[74, 134, 269, 281, 282, 310, 320, 365, 380, 395]	[62, 156, 441]
Autonomy	/	[53, 277, 341, 350, 394, 399, 456]	/

6. Interpretive synthesis of the evidence base

Collectively, the publications in Table 5 show that ophthalmic AI involving OCT and OCTA extends beyond image classification to a broader range of clinical and analytical tasks. These include segmentation, quantitative biomarker extraction, prognosis, treatment-response prediction, multimodal fusion, and workflow-oriented clinical support [123, 164, 194, 197, 224, 348, 350, 399, 408, 441]. The literature thus extends from disease detection to the generation of clinically meaningful, reproducible, and operationally useful information that could support ophthalmic care [124, 259, 399, 413].

The depth and maturity of evidence vary across this broad range of applications. Retinal disease and glaucoma have comparatively extensive evidence bases spanning diverse methods, clinical questions, and stages of validation, whereas other applications remain more exploratory or specialised [61, 68, 108, 123, 389]. Table 5 therefore depicts a heterogeneous field, with substantial differences across clinical domains in the progression from technical development to validation and clinical translation.

6.1. *Concentration of evidence across ophthalmic domains*

This unevenness is most apparent in the distribution across clinical domains. Much of the evidence in Table 5 concerns AMD, DME, DR, retinal fluid abnormalities, and glaucoma [123, 152, 225, 337, 389]. These conditions already rely on OCT and OCTA for diagnosis, staging, treatment monitoring, or longitudinal surveillance. Although the mapping cannot establish why research has concentrated in these areas, automated analysis could plausibly improve efficiency, standardisation, or sensitivity to change within their established imaging workflows.

The clinical role of imaging also helps explain the questions pursued within each domain. In retinal disease, OCT-based fluid detection, atrophy assessment, lesion characterisation, and monitoring of neovascular activity are integral to routine management, making them natural targets for AI-assisted automation and quantification [123, 152, 261]. In glaucoma, the established role of structural imaging in detection and follow-up similarly motivates studies of classification, staging, progression, and structure–function mapping [61, 389, 408].

Research represented in Table 5 extends beyond these dominant indications, although usually with less evidentiary depth. The included publications address cataract, keratoconus, corneal oedema, dry eye disease, inherited retinal disorders, optic neuropathies, amblyopia, and extra-ophthalmic prediction tasks derived from retinal imaging [108, 199, 346, 392, 451, 452]. This range suggests that OCT and OCTA are being explored as general-purpose phenotyping modalities rather than being confined to a small set of high-frequency classification problems. Even so, the evidence remains deepest where OCT is already embedded in established clinical workflows.

6.2. *From diagnostic classification to quantitative imaging biomarkers*

Across these clinical domains, the technical focus extends from diagnostic classification to quantitative characterisation. Classification remains a prominent and foundational task, distinguishing normal from pathological scans and differentiating CNV, DME, drusen, AMD stages, and glaucomatous change [68, 164, 225]. Beyond classification, Table 5 documents extensive work on segmentation, structural delineation, and quantitative biomarker extraction [123, 234, 261, 267, 311, 348].

Classification condenses image information into a diagnostic label. Segmentation and quantification instead produce structured measurements that reflect anatomy, pathology, and change over time. Reported outputs include retinal layer boundaries, fluid volumes, atrophic regions, ellipsoid-zone metrics, non-perfusion areas, vessel densities, smoothness indices, and other OCT-derived biomarkers [123, 152, 225, 234, 261]. By moving from labels to measurements, these approaches align AI more closely with ophthalmic practice, in which quantification may be as important as categorical diagnosis.

This alignment gives the segmentation and quantification studies potential translational relevance. Biomarker extraction and quantitative lesion assessment can support disease monitoring, treatment decisions, and longitudinal research through outputs that correspond more closely to clinical reasoning than isolated class labels [123, 261, 399]. Their prominence therefore reflects diversification in both the methods and the clinical purposes for which AI is being developed, but does not by itself demonstrate robust measurement across devices and populations or improved clinical decisions and outcomes.

6.3. *Prediction, prognosis, and potential clinical decision support*

The progression from labels to clinically relevant measurements is accompanied by growing interest in prognosis and prediction. Rather than focusing exclusively on recognition of the current disease state, a notable subset of studies in Table 5 addresses treatment response, recurrence, retreatment burden, postoperative visual outcome, progression, or functional decline [61, 197, 236, 337, 408]. These endpoints extend the scope of AI from automated diagnosis towards potentially actionable decision support.

This predictive emphasis is particularly evident in glaucoma. Multiple studies use OCT-derived structural information to estimate visual field loss, mean deviation, or progression risk [61, 124, 408]. By linking structural change with functional impairment, these approaches could complement functional testing and provide clinically relevant information beyond image-based diagnostic classification.

Retinal studies similarly use OCT to predict response to anti-vascular endothelial growth factor (anti-VEGF) treatment, disease recurrence, postoperative outcome, and individualised treatment requirements in AMD, DME, and macular hole settings [197, 236, 337, 379]. In these applications, OCT-derived structure is used both to characterise disease and to anticipate its course or response to therapy. This focus on clinically consequential endpoints is more ambitious than image classification and could support more individualised ophthalmic care.

Greater ambition brings a higher evidentiary standard. Prediction may support earlier or more individualised care, but high discrimination in static datasets is insufficient. Clinical use also requires valid relationships among imaging biomarkers, temporal dynamics, and patient outcomes that matter in practice, together with temporal and external validation. The presence of these studies demonstrates growing interest in anticipatory clinical reasoning and provides a foundation for establishing clinical utility and safety.

6.4. *Methodological diversity, interpretability, and multimodal integration*

The clinical questions described above are addressed using a diverse range of methods. Table 5 spans conventional machine-learning approaches based on handcrafted features, support vector machines, random forests, and transformed OCT-derived parameters [116], as well as CNNs, three-dimensional models, ensembles, transformer architectures, hybrid CNN–transformer systems, self-supervised approaches, uncertainty-aware systems, and multimodal fusion pipelines [68, 225, 301, 441]. This methodological diversity reflects the range of clinical questions, data structures, and technical requirements across ophthalmic AI.

Greater methodological complexity or novelty does not necessarily confer greater clinical value. Comparative studies indicate that more complex architectures may provide only modest improvements over simpler baselines [102]. Their value is strengthened when performance gains are accompanied by greater generalisability, robustness, clinically useful quantification, and reliable performance across relevant populations and settings.

Interpretability, uncertainty modelling, and clinician-facing explanatory tools provide complementary approaches to supporting the clinical use of AI outputs. Gradient-weighted class activation mapping (Grad-CAM), local interpretable model-agnostic explanations (LIME), Shapley additive explanations (SHAP), integrated gradients, uncertainty maps, occlusion testing, and related approaches appear across domains [259, 436]. These tools can make model outputs more transparent and open to scrutiny, although their clinical value depends on whether explanations are faithful, stable, and useful to clinicians. Direct evaluation of how users interpret and act on explanations can help establish their contribution to clinical decision-making.

Multimodal integration extends this methodological range by combining OCT with OCTA, fundus imaging, structured clinical variables, or electronic health record data [68, 197, 199, 441]. Integrating complementary phenotypic signals can provide a more comprehensive representation of disease than single-modality models. In glaucoma, structural, vascular, photographic, and functional inputs capture complementary features of disease [61, 389, 441]. In retinal disease, structural OCT combined with angiographic or clinical information can support more nuanced assessment of disease activity and treatment response [68, 197, 379]. These studies highlight the potential of multimodal systems to integrate clinically relevant information across data sources, with harmonisation, robust handling of missing data, transparent fusion, and validation of each input's added value supporting reliable integration.

6.5. *Validation, translational readiness, and clinical accountability*

The clinical relevance of these methodological advances depends on validation across the conditions in which they are intended to operate. Table 5 includes external testing, multicentre evaluation, cross-device robustness studies, prospective validation, and deployment in routine settings [68, 311, 389, 413]. These designs are particularly informative because OCT and OCTA performance may vary across acquisition devices, patient populations, annotation practices, and image quality. Alongside internal proof-of-concept evaluations, they demonstrate increasing attention to generalisability across the literature [68, 311, 413]. The depth of validation nevertheless varies considerably. Many studies use retrospective institutional datasets, internal splits, public benchmarks, or narrowly defined cohorts [116, 164, 225], while some subdomains have progressed to external or prospective validation. Together, these differences reflect varying levels of translational maturity across the field.

A subset of studies shifts the focus from individual algorithms to their integration within clinical systems. These include graphical interfaces, cloud platforms, real-time systems, point-of-care deployment, image-quality control, anomaly screening, and hybrid human–AI workflows [350, 399, 413, 436]. Embedding AI within these workflows brings safeguards, quality checks, user interaction, and operational reliability into consideration. Evaluation therefore extends beyond algorithmic performance to how reliably and accountably AI functions within the clinical system.

Systematic reviews, scoping reviews, and meta-analyses consolidate a substantial but heterogeneous evidence base spanning diverse tasks, datasets, endpoints, and validation approaches [124, 194, 224, 399]. Collectively, the evidence extends from technical feasibility to clinical translation and accountability. Beyond automated disease recognition, research examines robustness across devices and populations, clinically interpretable quantification, outcome prediction, and safe integration into routine workflows [68, 124, 259, 399].

Overall, the evidence base is substantial and spans clinically important ophthalmic domains, although its depth, validation quality, and translational readiness vary. Retinal disease and glaucoma have the most extensive bodies of work, while other areas show promising but less consolidated evidence [61, 108, 123, 389, 452]. The corpus therefore extends from exploratory research to studies approaching clinical translation. Realising the clinical potential of these advances will require robust validation, clinically meaningful endpoints, interpretable quantification, and effective integration into routine care and research.

Table 5: Overview of the 427 publications included in this review of artificial intelligence in ophthalmology. For each publication, the table gives the cited work and year, rationale for inclusion, and principal contribution.

Work	Year	Basis for inclusion	Principal contribution
[167]: 3D-CNN for Glaucoma Detection Using Optical Coherence Tomography	2019	An end-to-end 3D CNN classifies glaucoma directly from volumetric OCT without retinal segmentation. Performance is assessed by the area under the receiver operating characteristic curve (AUC), and Grad-CAM regions are tested through targeted occlusion.	Doubling the input voxel count relative to a prior downsampled 3D OCT model coincided with improved glaucoma classification. Occluding Grad-CAM-marked regions reduced performance in the reported experiment, a result consistent with, but not proof of, model reliance on those regions.
[389]: A 3D Deep Learning System for Detecting Referable Glaucoma Using Full OCT Macular Cube Scans	2020	In ophthalmic practice, this work trains a 3D CNN on SD-OCT macular cube volumes, uses multimodal clinical data to label referable and non-referable glaucoma, and tests the model externally on independent real-world datasets. Evaluation uses the AUC and cross-site generalisability analyses.	The reported results indicate that standardising OCT cube geometry through layer-segmentation homogenisation improves referable-glaucoma discrimination compared with training on raw volumes. The international validation also examined performance across myopia severity and settings in which glaucoma definitions differed.
[152]: A column-based deep learning method for the detection and quantification of atrophy associated with AMD in OCT scans	2021	The study introduces a column-based CNN to analyse volumetric SD-OCT. It uses vertical A-scan columns within B-scans to detect and quantify atrophy associated with age-related macular degeneration, then projects the atrophic regions onto corresponding infrared images for lesion-level characterisation. Performance was evaluated on 106 clinical OCT scans using metrics reported to approach observer variability.	Its principal contribution is an end-to-end workflow that identifies and measures atrophy lesions against expert annotations. The workflow yields lesion-area and foveal-distance measures for diagnostic, longitudinal, or research analyses. The reported evaluation supports measurement validity on the study sample rather than clinical effectiveness.
[291]: A Comparative Study Between Clinical Optical Coherence Tomography (OCT) Analysis and Artificial Intelligence-Based Quantitative Evaluation in the Diagnosis of Diabetic Macular Edema	2025	This prospective, non-randomised study in an ophthalmic setting evaluates AI-based quantitative analysis on clinical OCT images for diabetic macular oedema, using paraconsistent feature engineering with machine-learning classifiers, and reports diagnostic evidence for both binary DME presence detection and multiclass DME phenotyping.	The authors benchmark multiple conventional classifiers against routine clinical assessment on 700 OCT exams and show that paraconsistent feature engineering can reduce input features while maintaining clinically relevant diagnostic accuracy, supporting AI-assisted DME screening workflows.

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Work	Year	Basis for inclusion	Principal contribution
[102]: A comparison of deep learning U-Net architectures for posterior segment OCT retinal layer segmentation	2022	Eight U-Net-family architectures are compared for pixel-wise retinal-layer segmentation across four OCT datasets spanning different populations, pathologies, acquisition settings, instruments, and annotation schemes. Evaluation is based on Dice scores.	The study isolates the practical impact of architectural changes and convolution depth by running matched U-Net variants (two vs three convolutions per block) across heterogeneous datasets. Vanilla U-Net remained close to more complex variants, with only marginal gains from the latter, informing simpler model selection for OCT retinal layer segmentation.
[98]: A Comprehensive CNN Model for Age-Related Macular Degeneration Classification Using OCT: Integrating Inception Modules, SE Blocks, and ConvMixer	2024	A parameter-efficient framework combining modified Inception modules, depthwise squeeze-and-excitation blocks, and ConvMixer is tested for OCT classification of dry AMD, wet AMD, and normal eyes on a private dataset and the public Noor dataset.	The main technical contribution is a parameter-efficient CNN that combines modified Inception modules, squeeze-and-excitation blocks, and ConvMixer. The authors report high benchmark performance for AMD and normal classification on the private and Noor datasets, but this evaluation does not establish diagnostic effectiveness in clinical practice.
[224]: A Comprehensive Review of Deep Learning-Based Retinal Layer Integrity Assessment in Optical Coherence Tomography (OCT)	2025	This review focuses on deep learning for assessment of the external limiting membrane (ELM) and ellipsoid zone (EZ) in OCT. It synthesises 43 peer-reviewed studies that reported quantitative segmentation or volumetric outcomes and examines their relevance to clinical ophthalmic imaging, model performance, and translational limitations.	The review synthesises trends in AI model families for ELM and EZ segmentation and emphasises key barriers to clinical use, including generalisation, smoothness quantification, and real-time deployment readiness. It also proposes practical directions such as standardised annotations and benchmark development.
[180]: A comprehensive set of novel residual blocks for deep learning architectures for diagnosis of retinal diseases from optical coherence tomography images	2021	This chapter develops a deep learning classification approach for retinal diseases using SD-OCT images, employing residual network blocks designed to address gradient explosion and reporting evidence from performance comparisons on two separate OCT datasets, including real-time prediction versus expert ophthalmologists.	The proposed residual blocks are integrated into a deep architecture for SD-OCT pathology classification. On the evaluated datasets, the authors report real-time predictions that outperformed the participating ophthalmologists. This is a study-specific benchmark rather than evidence of clinical superiority.
[267]: A Deep Ensemble Learning-Based CNN Architecture for Multiclass Retinal Fluid Segmentation in OCT Images	2023	The study develops a convolutional neural network deep-ensemble approach for OCT imaging that segments and differentiates three classes of retinal cysts, with quantitative evaluation on the publicly available RETOUCH challenge dataset.	The contribution is a deep-ensemble pipeline for multiclass retinal cyst segmentation. On the RETOUCH benchmark, the authors report an overall 1.8% improvement over the stated comparators, a dataset-specific result that supports methodological performance but does not itself demonstrate workflow efficiency.

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Work	Year	Basis for inclusion	Principal contribution
[337]: A Deep Learning Approach for Predicting the Response to Anti-VEGF Treatment in Diabetic Macular Edema Patients Using Optical Coherence Tomography Images	2025	The study focuses on diabetic macular oedema in an ophthalmology treatment setting by applying a Siamese deep learning framework to optical coherence tomography images, aiming to predict whether patients will be good versus poor responders to anti-VEGF therapy using post-injection central macular thickness reduction as the evidence for a binary classification task.	The study proposes ESSDP, a Siamese EfficientNetB2 design coupled with feature-distance comparison for OCT-based responder stratification, and reports overall classification results that suggest the approach may inform more individualised anti-VEGF decision-making in DME, including under limited-data conditions.
[454]: A deep learning approach to hard exudates detection and disorganization of retinal inner layers identification on OCT images	2024	In this retrospective, single-centre study, a deep-learning pipeline analyses SD-OCT images from eyes with diabetic macular oedema. You Only Look Once (YOLO) object detection identifies hard exudates, while image classification detects disorganisation of the retinal inner layers (DRIL). Test-set performance was evaluated using standard detection and classification metrics and corresponding agreement tests.	The authors present a dual-branch architecture that combines YOLO-based hard-exudate localisation with an ensemble of CNN classifiers for DRIL, offering an integrated OCT biomarker workflow that supports feature identification and decision support in DME imaging.
[408]: A deep learning approach to predict visual field using optical coherence tomography	2020	An Inception V3 model predicts Humphrey 24-2 visual-field thresholds from combined SD-OCT macular ganglion cell–inner plexiform layer (mGCIPL) and peripapillary retinal nerve fibre layer (pRNFL) thickness maps in eyes with and without glaucoma. Root mean square error (RMSE) on a held-out test set is the primary endpoint.	The results indicate that structure–function learning from combined OCT thickness inputs yields accurate global and region-specific visual-field predictions. Visual-field mean deviation, pRNFL thickness, and mGCIPL thickness were key correlates of prediction error, supporting further evaluation when perimetry is difficult to perform.
[156]: A Deep Learning-Based Eye Disease Diagnosis Using OCT Imaging and SE-Enhanced CNNs	2025	This ophthalmic study develops an OCT-imaging deep-learning classifier using a custom squeeze-and-excitation-enhanced CNN to distinguish cataract, glaucoma, diabetic retinopathy, and healthy controls. Evaluation used stratified five-fold cross-validation and a small independent clinical case series at a private radiology centre, with Grad-CAM and saliency-based interpretation.	DEEPSIGHT couples a squeeze-and-excitation-enhanced classifier with clinician-facing saliency outputs and a prototype interface. The authors report high classification performance and a small independent case evaluation. These findings motivate further study of interpretable decision support but do not establish practical usability.

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Work	Year	Basis for inclusion	Principal contribution
[350]: A Deep Learning-Based Graphical User Interface for Predicting Corneal Ectasia Scores from Raw Optical Coherence Tomography Data	2026	In an ophthalmic setting, the study builds a Tkinter-based graphical user interface (GUI) around a modified EfficientNet-B0 convolutional model. It ingests raw Casia2 anterior-segment OCT 3dv data, arranged as cropped and resized 16-image stacks, to predict continuous corneal ectasia scores and derive two- and three-class categories based on the Ectasia Screening Index. Evaluation used test-set regression and classification.	The end-to-end workflow outputs a numerical ectasia score and corresponding diagnostic labels from raw OCT files. It also lets users review the extracted images alongside the predicted scores and classes, helping maintain consistency across software updates.
[348]: A Deep Learning Framework for Segmentation of Retinal Layers from OCT Images	2017	In an ophthalmic OCT setting, the study uses a CNN followed by bidirectional long short-term memory (LSTM) to segment multiple retinal-layer boundaries. It first extracts regions and edges for the layers of interest and then traces eight continuous boundaries. Performance was assessed using pixel-wise mean absolute error (MAE) on healthy and AMD data against inter-marker baselines and dataset-provided benchmarks.	Learned CNN feature extraction and recurrent continuity modelling jointly replace conventional preprocessing and boundary tracing. Training on mixed healthy and AMD cases produced lower pixel-level boundary error than inter-marker disagreement on the public OCT datasets.
[274]: A Deep Learning Model Using VGG19 and Random Forest for AMD Stage Classification on Retinal OCT Images	2025	An ophthalmic study evaluates a deep learning workflow that uses VGG19-derived CNN features with a random forest (RF) classifier to stage age-related macular degeneration from retinal OCT images, with performance evaluated on a public OCT dataset for the classification task across Early, Intermediate, and Advanced stages.	The VGG19–RF pipeline combines global average pooling and VGG19 penultimate-layer features with a random-forest classifier. Its reported accuracy on the public dataset permits comparison with the standard CNN baseline. Computational efficiency is proposed but not independently evaluated.
[447]: A deep learning multi-capture segmentation modality for retinal OCT imaging	2022	The study targets retinal and choroidal layer segmentation on OCT using a deep-learning approach that combines CycleGAN image-to-image translation to standardise multi-capture focus and denoising conditions with a vision-transformer-based TransU-Net. Segmentation performance is compared with that of a baseline U-Net using the Dice coefficient.	CycleGAN standardisation followed by TransU-Net is the article's central pipeline. The reported Dice comparison favours TransU-Net over U-Net for enhanced-depth-imaging OCT. The authors interpret this result as support for adaptation across capture conditions, although instrument-agnostic performance requires broader external evaluation.
[218]: A deep learning system for the detection of optic disc neovascularization in diabetic retinopathy using optical coherence tomography angiography images	2025	Using a deep-learning pipeline on OCTA, the authors detect, quantify, and visualise optic disc neovascularisation (NVD) in diabetic retinopathy. They compare its detection and region-identification performance with other deep-learning methods and assessments by retina specialists.	The system performs optic disc boundary detection, identifies NVD regions, and generates 3D OCTA visualisations of NVD location and blood-flow information. These outputs may support more intuitive automated assessment for clinical interpretation.

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Work	Year	Basis for inclusion	Principal contribution
[87]: A Deep Learning System Model For Detecting Neurodegenerative Diseases From Retinal Optical Coherence Tomography	2025	Enhanced adaptive filtering is combined with a hybrid CNN to classify abnormalities associated with neurodegenerative disease in OCT-derived retinal images. The reported accuracy is 92.3%.	The system integrates enhanced adaptive filtering and data augmentation with hybrid-CNN feature extraction. The reported classification result establishes performance on the evaluated retinal images, while detection of subtle neurodegenerative changes outside that setting remains untested.
[329]: A Deep Learning System Using Optical Coherence Tomography Angiography to Detect Glaucoma and Anterior Ischemic Optic Neuropathy	2023	This retrospective ophthalmology study trains a ResNet50 CNN with integrated gradients on OCTA images of the macular superficial capillary plexus (SCP) and radial peripapillary capillary plexus (RPC). The model classifies glaucoma, non-arteritic anterior ischaemic optic neuropathy, and healthy controls. Test-set discrimination was evaluated using the AUC and agreement metrics.	The reported results indicate that combining SCP and RPC inputs yields better three-class performance than either plexus alone. Model-attribution maps also highlight diagnostically relevant regions, suggesting potential support for differentiating optic neuropathies in difficult cases.
[334]: A fusion of deep neural networks and game theory for retinal disease diagnosis with OCT images	2024	The authors combine Inception-based feature extraction, GAN-based classification, and game-theoretic evaluation to classify retinal disease in OCT images and examine model behaviour under adversarial conditions.	GIGT integrates OCT preprocessing, hybrid classification, and a game-theoretic assessment in a Python pipeline. The authors claim improved diagnostic accuracy from the reported experiments, but the available evaluation does not establish reliability under clinical distribution shift.
[301]: A GCN-assisted deep learning method for peripapillary retinal layer segmentation in OCT images	2021	The study introduces a graph convolutional network (GCN)-assisted U-shaped architecture for peripapillary retinal-layer segmentation. It targets nine layers, including the RNFL, and is trained and tested on a collected dataset with Dice-based comparisons against earlier methods.	The study refines an ophthalmic segmentation pipeline by inserting a graph-reasoning block between the encoder and decoder to exploit the retina's stratified organisation around the optic disc. This approach improved RNFL and multilayer delineation in the evaluated peripapillary images.
[312]: A Hybrid CBAM-Vision Transformer for Enhanced Classification of Retinal Diseases in Optical Coherence Tomography Imagery	2025	In this ophthalmic computer-vision study, a convolutional block attention module (CBAM)-augmented ViT is evaluated for retinal-disease classification using OCT images. The method addresses low resolution and class imbalance, with performance assessed using test-set metrics on OCTMNIST.	The CBAM-ViT-tiny hybrid combines attention-refined embeddings, weighted loss, augmentation, and full fine-tuning. It achieved the best reported test performance among the stated OCTMNIST comparators, a benchmark finding that does not establish effectiveness on other datasets.

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Work	Year	Basis for inclusion	Principal contribution
[369]: A Hybrid Deep Learning Classification of Perimetric Glaucoma Using Peripapillary Nerve Fiber Layer Reflectance and Other OCT Parameters from Three Anatomy Regions	2024	This glaucoma diagnostic study applies a hybrid deep-learning framework to SD-OCT data from peripapillary and macular regions. It uses peripapillary nerve fibre layer (NFL) reflectance and thickness maps with optic nerve head parameters to classify healthy eyes and perimetric glaucoma. Performance on a held-out test set was compared with logistic regression using receiver operating characteristic (ROC) analysis.	A CNN processes superpixelised NFL reflectance and thickness, while a fully connected network processes OCT-derived disc and macular measures and demographics. Incorporating NFL reflectance improved discrimination beyond logistic regression and remained competitive when reflectance was omitted, informing how OCT structural signals can be fused for screening-oriented glaucoma classification.
[116]: A Hybrid Machine Learning Approach Using LBP Descriptor and PCA for Age-Related Macular Degeneration Classification in OCTA Images	2020	The authors apply a local binary pattern (LBP) descriptor and principal component analysis (PCA) pipeline with a support vector machine (SVM) classifier to OCTA images. Stratified k-fold cross-validation and AUC assess binary discrimination between AMD groups.	The study illustrates how rotation-invariant texture features from full-field OCTA, followed by PCA decorrelation, can distinguish healthy from neovascular AMD and separate CNV-related categories from non-neovascular disease using AUC-based evaluation.
[97]: A Hybrid Model Composed of Two Convolutional Neural Networks (CNNs) for Automatic Retinal Layer Segmentation of OCT Images in Retinitis Pigmentosa (RP)	2021	For SD-OCT B-scans in retinitis pigmentosa (RP), this hybrid deep-learning study uses a CNN-based semantic segmentation network (U-Net) plus a sliding-window refinement CNN to delineate retinal layer boundaries and derive photoreceptor metrics, including EZ measurements. Performance was evaluated against Spectralis manual grader corrections using pixel-wise, Bland-Altman, and correlation evidence.	The reported results indicate that combining fast U-Net semantic segmentation with sliding-window error correction improves boundary localisation and yields more accurate EZ and photoreceptor outer-segment measurements than either component alone, which can reduce manual effort for SD-OCT-based RP assessment.
[365]: A Lightweight CNN Net for AMD Detection Using OCT Volumes	2022	The article describes a lightweight convolutional neural network trained on optical coherence tomography volumes to classify whether a patient has age-related macular degeneration, providing evidence from reported model performance aimed at enabling diagnosis in rural and remote ophthalmic settings.	The article specifies an efficient AMD detection CNN with a small parameter and compute footprint and reports quantisation-aware fixed-point results, which is relevant for deploying AI on resource-limited hardware.
[43]: A lightweight deep learning model for retinal optical coherence tomography image classification	2023	Using a lightweight deep learning approach for OCT image classification, the authors detect and grade diabetic macular oedema and drusen macular degeneration, with evaluation reported on publicly available Mendeley OCT and Duke datasets to provide evidence for retinal disease diagnosis in an ophthalmic imaging setting.	Residual blocks and channel-attention modules form a lightweight end-to-end classifier. The authors report high accuracy on the two public datasets. The findings support parameter efficiency on those benchmarks, whereas screening performance in practice remains untested.

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Work	Year	Basis for inclusion	Principal contribution
[92]: A machine learning classification model for primary open-angle glaucoma based on optical coherence tomography optic disc parameters and interocular asymmetry features	2026	This prospective case-control study applies OCT-based machine learning classification to distinguish primary open-angle glaucoma (POAG) from healthy controls using peripapillary pRNFL and Bruch's membrane opening–minimum rim width (BMO-MRW) quadrant measurements together with interocular asymmetry indices, validated with discrimination metrics (AUC, accuracy, F1) and feature attribution via SHAP for interpretability.	Elastic-net selection followed by a support-vector machine identifies five OCT-derived features, with temporal and superotemporal BMO-MRW asymmetry contributing prominently to POAG-control separation. The SHAP analysis makes this imaging signal interpretable within the case-control sample, not prospectively prognostic.
[166]: A Multi-branch and Attention Based CNN Architecture for the Classification of Retinal Diseases from OCT Images	2023	A multi-branch CNN augmented with CBAM is tested for multiclass retinal disease classification on OCT images. The reported results are 96% accuracy and 96% F1-score relative to the stated comparators.	A CBAM-enhanced multi-branch design combines deep and shallow feature extraction. The branches emphasise informative regions and suppress irrelevant signals, and the study reports improved multiclass OCT classification relative to its stated comparators.
[139]: A multi-scale anomaly detection framework for retinal OCT images based on the Bayesian neural network	2022	The authors apply a Bayesian neural network with a multi-scale Bayesian U-Net to retinal OCT images for unspecific anomaly detection, using epistemic uncertainty to separate healthy from non-healthy cases and reporting quantitative evaluation on a public UCSD dataset plus a private dataset.	Training only on healthy OCT data allows the pipeline to model uncertainty in healthy tissue and flag anomalous inputs with a threshold rule. This design is relevant when labelled abnormal examples are limited, although screening utility was not evaluated.
[388]: A Neural Network Approach to Quantify Blood Flow from Retinal OCT Intensity Time-Series Measurements	2020	This Scientific Reports study uses a convolutional neural network trained on forward-modelled retinal OCT intensity time-series to estimate vessel blood-flow rates in an ophthalmic OCT angiography setting, with validation performed against Doppler OCT in blood-flow phantoms and further checked in vivo for flow conservation across retinal vessel branch points and expected power-law scaling with vessel diameter.	The framework converts OCT intensity time series into label-free, vessel-level flow-rate maps and incorporates signal-to-noise ratio (SNR)-aware model selection and spatial masking. Phantom and in vivo checks examine sensitivity to Doppler angle, haematocrit, and signal-to-noise conditions, but do not alone establish deployment readiness.
[316]: A new computer-aided diagnosis tool based on deep learning methods for automatic detection of retinal disorders from OCT images	2024	A deep learning computer-aided diagnosis tool uses ophthalmologist-provided semantic descriptions to classify retinal disorders on a UCSD-derived OCT dataset, with test-set evaluation of both classification and interpretability for screening and second-opinion workflows.	The reported results indicate that embedding clinician-derived semantic information into the network improves classification on the test OCT images, while heat maps highlight abnormal regions. These findings support technical classification and visual inspection within the evaluated dataset, but do not establish screening effectiveness or clinical decision-support utility.

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Work	Year	Basis for inclusion	Principal contribution
[332]: A new convolutional neural network based on combination of circlets and wavelets for macular OCT classification	2023	This ophthalmic deep-learning study uses a two-dimensional (2D) CNN on macular OCT B-scans represented in a non-data-adaptive multi-scale circlet and wavelet time-frequency transform domain, aiming to classify normal, AMD, and DME with evidence from subject-wise 5-fold nested cross-validation and external testing on a second OCT dataset plus Grad-CAM interpretability.	The study proposes the CircWave transform and corresponding CircWaveNet classifier by concatenating circlet and two-dimensional discrete wavelet transform (2D-DWT) sub-band channels, then demonstrates improved discrimination over original images and single-transform inputs while producing class-relevant Grad-CAM attention for DME-associated circular features and linear features in normal OCT scans across datasets.
[221]: A new intelligent system based deep learning to detect DME and AMD in OCT images	2024	Using a computer-aided diagnosis (CAD) approach based on a convolutional neural network on OCT retinal images, the authors classify eyes into AMD, DME, and normal categories, evaluating multiple CNN architectures (including transfer learning models) on two datasets as evidence for automated ophthalmic image analysis.	Among the evaluated architectures, the custom-CNN features paired with Inception_V3 achieved the highest reported DUKE-dataset accuracy (99.53%). This result establishes comparative benchmark performance, not diagnostic efficiency or clinical effectiveness.
[54]: A novel deep learning approach for diabetic retinopathy classification using optical coherence tomography angiography	2025	In an ophthalmic DR screening setting, the study applies a CNN-based classification model to OCTA-derived retinal images, supported by conditional generative adversarial network (cGAN)-driven augmentation and a dedicated preprocessing pipeline to address limited data and imaging quality, with evidence from reported performance results.	The study offers an automated, noninvasive DR versus non-DR pipeline that couples cGAN-enhanced dataset expansion with contrast enhancement, denoising, and vessel segmentation before CNN inference, aiming to improve early detection reliability from OCTA.
[452]: A Novel Deep Learning Method for Nuclear Cataract Classification Based on Anterior Segment Optical Coherence Tomography Images	2020	Using GraNet on anterior-segment OCT (AS-OCT) images, the authors automate nuclear cataract grading and compare focal-loss plus cross-entropy cross-training with the stated baseline approaches.	GraNet adds a grading block that learns high-level features from AS-OCT. Focal loss and cross-entropy cross-training are then combined to improve reported classification accuracy relative to the stated methods.

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Work	Year	Basis for inclusion	Principal contribution
[441]: A Novel Multimodal Implementation of a Foundation Artificial Intelligence Model Using Optic Nerve Head Fundus Photographs and OCT Imaging for Glaucoma Detection	2026	In an ophthalmic diagnostic evaluation, the study applies a self-supervised RETFound foundation model to glaucoma detection using paired optic nerve head colour fundus photographs and OCT images, comparing multimodal versus unimodal (fundus-only or OCT-only) performance with evidence based on AUC, precision, and recall across key clinical subgroups (race, age, and disease severity).	The reported results indicate that multimodal fundus-OCT RETFound improves discrimination over the fundus-only model while remaining statistically comparable to the OCT-only model, with subgroup results showing no significant race or age effects and stronger detection for moderate-to-severe versus mild glaucoma.
[381]: A Review of Machine Learning Algorithms for Retinal Cyst Segmentation on Optical Coherence Tomography	2023	This 2023 review examines machine learning for retinal cyst and fluid segmentation from OCT, covering classical and deep-learning methods across a three-stage pipeline comprising denoising, retinal-layer segmentation, and cyst/fluid segmentation, as well as the available public OCT datasets. Its evidence consists of synthesised algorithmic and dataset overviews relevant to quantitative assessment of structural changes in ophthalmic disease.	The study distills the technical design choices for an OCT cyst/fluid assessment system by surveying denoising and layer-specific strategies alongside cyst/fluid segmentation and by compiling dataset resources and future challenges, helping researchers benchmark model development targets and pipeline components for this imaging task.
[413]: A robust deep learning classifier for screening multiple retinal diseases on optical coherence tomography	2025	The study develops FlexiVarViT, a high-resolution transformer-based classifier for OCT B-scans that is tailored to variable slice count and native in-plane resolution to screen multiple retinal disease categories, trained and cross-validated on a multi-disease Spectralis dataset and evaluated on two external OCT datasets from different devices and populations, providing evidence of robustness and generalisability via quantitative out-of-domain testing.	Dynamic patching allows FlexiVarViT to retain native resolution, while Kermany-supervised pretraining and volume-level attention produced the best reported cross-dataset performance among the tested variants. The external datasets broaden the benchmark, but do not by themselves establish screening performance under real-world variability.
[194]: A scoping review of the use of artificial intelligence models in automated OCT analysis and prediction of treatment outcomes in diabetic macular oedema	2025	This Cochrane-methodology scoping review maps AI model studies using SD-OCT imaging for diabetic macular oedema, covering diagnostic classification, automated OCT biomarker segmentation/detection (e.g., IRF, SRF, hyperreflective foci (HRF)), and prediction of visual or anatomical treatment outcomes, based on a synthesis of 40 eligible human primary research reports selected after quality screening.	The review synthesises the current landscape of OCT-AI workflows in DME by distinguishing evidence for diagnosis and biomarker analysis from that for prognostic and treatment-outcome prediction, highlighting the limited external validation and the need for standardised evaluation and more robust prognostic modelling to support clinical translation.

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Work	Year	Basis for inclusion	Principal contribution
[225]: A Vision Transformer Based Deep Learning Architecture for Automatic Diagnosis of Diabetic Retinopathy in Optical Coherence Tomography Angiography	2023	Using a vision transformer deep learning architecture on optical coherence tomography angiography images, the study targets diabetic retinopathy by performing lesion segmentation, image quality assessment, and DR grading, matching the DRAC 2022 multitask evaluation structure and providing competition-level evidence from an OCTA dataset.	For all three DRAC 2022 subtasks, the authors report segmentation, agreement, and grading metrics that characterise performance within the competition. Completion of the contest tasks demonstrates technical integration, not validation of an end-to-end screening workflow.
[440]: Accuracy of a deep convolutional neural network in the detection of myopic macular diseases using swept-source optical coherence tomography	2020	Deep convolutional neural networks are tested on swept-source OCT for binary and ternary classification of myopic macular diseases, including myopic choroidal neovascularisation and retinoschisis, relative to high myopia and non-high myopia. Performance is estimated by k-fold cross-validation.	The ensemble combines nine CNN architectures with Grad-CAM visualisations. The authors found no statistically significant performance difference from three ophthalmologists on the same datasets. Absence of a detected difference is not an equivalence test and does not establish clinical interchangeability.
[149]: Accuracy of generative deep learning model for macular anatomy prediction from optical coherence tomography images in macular hole surgery	2024	In this 2024 study, a conditional variational autoencoder predicts postoperative 6-month swept-source OCT macular anatomy from preoperative OCT in patients with successfully closed full-thickness macular holes. Image synthesis is evaluated against postoperative OCT using agreement and accuracy measures for neurosensory retinal areas and foveal structural metrics.	The study quantifies how closely generated AI-OCT preserves foveolar height and mean foveal thickness while also estimating ELM and EZ restoration, providing evidence that the model can generate clinically interpretable postoperative retinal layer reconstructions rather than only scalar surgical outcomes.
[313]: Accurate Detection of Non-Proliferative Diabetic Retinopathy in Optical Coherence Tomography Images Using Convolutional Neural Networks	2020	A transfer-learning computer-aided diagnosis approach classifies non-proliferative diabetic retinopathy (NPDR) versus normal OCT B-scans after fovea-guided extraction of nasal and temporal retinal patches. Five-fold cross-validation estimates diagnostic performance.	The study identifies an optimised deployment strategy in which two independently trained, ImageNet-pretrained AlexNet models, fed by nasal and temporal patches aligned using the outer nuclear layer (ONL), extract pool5 features that are fused in a linear SVM, achieving the reported 94% accuracy and highlighting how preprocessing and feature-layer choices impact OCT-based NPDR detection.
[259]: Adaptive and Explainable Deep Learning for Automated Retinal Disease Diagnosis Using OCT Images	2025	The study proposes an end-to-end, adaptive and explainable deep-learning approach that classifies retinal disease states from OCT imaging, using an MSRA-CNN with hierarchical Grad-CAM++ plus uncertainty weighting to generate clinician-interpretable heatmaps and reporting quantitative evaluation with inter-observer agreement for evidence of clinical relevance.	Adaptive illumination normalisation, elastic deformation, adversarial perturbation, hierarchical Grad-CAM++, and uncertainty weighting are combined in one OCT classification pipeline. The study reports performance for CNV, DME, drusen, and normal images together with visual explanations. Reliability in clinical decision-making remains to be externally established.

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Work	Year	Basis for inclusion	Principal contribution
[159]: ADCFormer: A hybrid model of Adaptive Depth-wise Convolution and Transformer for retinal macular edema segmentation in OCT images	2025	The study presents a transformer/CNN hybrid, ADCFormer, for segmenting diabetic macular oedema lesion regions in OCT images, using global efficient self-attention (ESA) with adaptive depth-wise convolution (ADWConv) and reporting comparative evidence from multiple external and hospital-collected OCT datasets.	ADCFormer combines ESA, ADWConv, and global–local feature fusion (GLFF) components with a residual multiscale feed-forward design to fuse global and local features. Cross-dataset results quantify DME lesion segmentation under noisy OCT conditions, but do not establish practical clinical value.
[215]: Advanced Deep Learning Techniques for Evaluating OCT Image Quality and Detecting Retinal Pathologies	2025	This study combines a pretrained image-quality assessment model (ARNIQA) with ResNet50 and Xception CNN classifiers. Using a Tunisian OCT dataset, it examines how dataset composition and image quality affect three-class retinal pathology classification across high-quality, low-quality, and unmodified image sets.	The contribution is a quality-aware workflow linking ARNIQA-based image assessment with the two CNN classifiers. Stratifying inputs by measured quality permits direct analysis of quality-related performance differences, although the reported results do not establish reliability outside the Tunisian dataset.
[395]: Advancement of cataract classification with artificial intelligence using anterior segment optical coherence tomography images with self-supervised vision transformer	2026	This clinical investigation evaluates a self-supervised vision transformer (SS-ViT) trained on uncropped AS-OCT images. Using Lens Opacities Classification System III (LOCS III) labels, the model grades nuclear, cortical, and posterior subcapsular cataract severity. Performance across the supervised tasks is assessed primarily by the area under the precision–recall curve (AUPRC).	The study establishes the usefulness of AS-OCT domain self-supervised pretraining for ViT-based cataract grading by comparing SS-ViT against ImageNet-pretrained and non-pretrained baselines across five LOCS III-based tasks, including subregion-specific assessments and input configurations that leverage all angular cross-sections.
[349]: Advancing Diabetic Retinopathy Diagnosis: Leveraging Optical Coherence Tomography Imaging with Convolutional Neural Networks	2023	This review evaluates ophthalmic applications of convolutional neural networks using optical coherence tomography retinal imaging to automate diabetic retinopathy lesion detection, microaneurysm segmentation, and severity grading, summarising evidence across the OCT-to-CNN workflow and its reported diagnostic potential.	The review synthesises the technical considerations for OCT-based CNN pipelines and highlights practical implementation issues, such as data availability and model interpretability, along with future directions like multimodal OCT integration and real-time point-of-care screening.
[323]: Advancing Diabetic Retinopathy Screening: A Systematic Review of Artificial Intelligence and Optical Coherence Tomography Angiography Innovations	2025	In the ophthalmic setting of diabetic retinopathy screening, this systematic review synthesises evidence on AI methods, predominantly deep learning alongside related ML, applied to optical coherence tomography angiography data, targeting DR detection and severity classification using pooled diagnostic performance outcomes reported across studies with OCTA acquisition and parameter variations.	By consolidating 32 controlled human studies and extracting how different OCTA image types, tasks, and model families perform using metrics such as accuracy, sensitivity, specificity, and AUC, the review clarifies which AI-OCTA approaches have the strongest reported diagnostic evidence and where cross-study variability may affect clinical translation.

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Work	Year	Basis for inclusion	Principal contribution
[213]: Advancing Ophthalmic Diagnostics: Employing Deep Learning for Precision Detection of Retinal Damage in OCT Images	2024	Using VGG16 and InceptionV3 on retinal OCT images, the authors classify diabetic macular oedema, age-related macular degeneration, choroidal neovascularisation, and healthy eyes, reporting performance on an 85,000-image dataset to support automated diagnostic decision-making.	The authors compare two CNN families and identify model-specific strengths, with VGG16 favoured for CNV and InceptionV3 for normal retinas. The reported overall accuracy of 92.76% summarises performance on the study data, not precision in routine diagnosis.
[281]: Agreement of a Novel Artificial Intelligence Software With Optical Coherence Tomography and Manual Grading of the Optic Disc in Glaucoma	2023	In this prospective cross-sectional study, offline AI running on a smartphone fundus camera estimates the vertical cup-to-disc ratio (VCDR) from optic disc photographs. Agreement is assessed against SD-OCT-derived VCDR and expert manual grading of stereoscopic fundus images from glaucomatous and healthy eyes.	Across 473 eyes, the new AI VCDR estimates showed quantitative concordance with SD-OCT and specialist manual segmentation. On-device inference also demonstrates that the measurements can be generated without cloud processing, although workflow benefit was not evaluated.
[422]: An artificial intelligence cloud platform for OCT-based retinal anomalies screening system in real clinical environments	2025	AI-PORAS, an ophthalmic cloud platform combining domain-adaptive object detection and classification on OCT B-scans, localises and screens for 15 retinal anomalies. Validation covers 165,384 eyes, and deployment data come from 207 medical institutions in China.	The article details the end-to-end clinical workflow and cross-scanner adaptation strategy for OCT-AI and reports external real-world screening outcomes for 116,717 diagnosed patients, providing practical support for scalable remote retinal anomaly assessment in routine care settings.
[319]: AN EARLY RETINAL DISEASE DIAGNOSIS SYSTEM USING OCT IMAGES VIA CNN-BASED STACKING ENSEMBLE LEARNING	2023	Using OCT retinal imaging and CNN-based feature extraction with stacking ensemble learning, the authors detect diabetic macular oedema, drusen, and choroidal neovascularisation, providing evidence in an ophthalmic diagnostic setting via two publicly available OCT datasets (Duke and UCSD).	Fine-tuned AlexNet features are combined through homogeneous and heterogeneous stacking ensembles. The authors report near-unity metrics on both public datasets and improvements over prior approaches. The unusually high benchmark estimates warrant independent, externally sampled evaluation.
[460]: An Efficient Retinal Layer Segmentation Based on Deep Learning Regression Technique for Early Diagnosis of Retinal Diseases in OCT and FUNDUS Images	2023	A deep learning regression pipeline for retinal-layer segmentation in OCT and fundus images links layer-thickness changes to subsequent feature extraction, clustering, and classification, including for diabetic retinopathy.	The study focuses on estimating retinal layer attributes and clustering/segmenting image content from both modalities with regression-based learning, aiming to translate thickness variations into disease-specific classification signals that are clinically relevant for early diabetic retinopathy screening.

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Work	Year	Basis for inclusion	Principal contribution
[430]: An Open-Source Deep Learning Framework for Automated Corneal Segmentation in Anterior Segment Optical Coherence Tomography With Cross-Device External Validation	2026	In an anterior-segment OCT workflow, the study introduces CUNEX, a full-thickness corneal segmentation model based on nnU-Net. It validates the model without fine-tuning across four externally acquired OCT devices, then examines how the segmentations affect downstream classification of age, sex, and disease stage. Segmentation quality and downstream utility provide both internal and cross-device evidence.	CUNEX is released with pretrained weights and tested without fine-tuning across the included devices. The cross-device results support reproducibility and anatomically plausible full-thickness corneal segmentation relative to the stated comparators. Clinical integration remains a prospective question.
[442]: An Optical Coherence Tomography-Based Deep Learning Algorithm for Visual Acuity Prediction of Highly Myopic Eyes After Cataract Surgery	2021	The study develops an OCT-based deep learning ensemble that ingests preoperative macular OCT images to predict 4-week postoperative best-corrected visual acuity (BCVA) in highly myopic eyes after cataract surgery, with evaluation of prediction accuracy and error patterns using internal and external test datasets.	Multiple CNN backbones and their ensemble are trained internally and then tested externally for postoperative BCVA prediction. The reported errors indicate predictive feasibility. Any role in preoperative counselling or surgical planning requires prospective evaluation.
[155]: Analysis of early screening results for high-risk populations of retinal diseases based on artificial intelligence optical coherence tomography technology	2026	This diagnostic trial evaluates an ophthalmic deep learning AI-OCT system for early screening in a high-risk "Five-High" population, using wide-field macular and peripheral OCT imaging and measuring agreement with ophthalmologist review plus downstream referral and treatment outcomes.	The study quantifies the AI-OCT screening positive rate and lesion distribution in the "Five-High" cohort and shows high concordance with physician diagnosis while reporting referral and treatment follow-through after positive screening.
[297]: Anomaly Detection in Retinal OCT Images With Deep Learning-Based Knowledge Distillation	2025	The study develops an unsupervised anomaly-detection system for fovea-centred retinal OCT screening. A teacher–student knowledge-distillation framework, trained only on healthy scans, assigns anomaly scores at volume and B-scan levels and produces localised anomaly maps. Performance was assessed by receiver operating characteristic and precision–recall analyses on an internal test set covering several macular conditions and on external public datasets.	Training only on normal scans provides an annotation-free alternative to supervised pathology recognition, with feature-space deviations used for anomaly scores and localisation maps. The internal and public-dataset results establish benchmark performance across several disease patterns. Scalable screening remains to be prospectively assessed.

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Work	Year	Basis for inclusion	Principal contribution
[236]: Anti-VEGF treatment outcome prediction based on optical coherence tomography images in neovascular age-related macular degeneration using a deep neural network	2024	A DenseNet201 model uses SD-OCT acquired before and during anti-VEGF loading to predict anatomical response in neovascular AMD (nAMD). Its results are benchmarked against ophthalmologists.	Concatenating OCT sequences after the first and second injections improved reported treatment-response prediction relative to the participating human graders. This benchmark motivates longitudinal validation but does not itself support individualised treatment guidance.
[263]: Application of artificial intelligence-based dual-modality analysis combining fundus photography and optical coherence tomography in diabetic retinopathy screening in a community hospital	2022	This community-hospital study evaluates deep-learning AI for diabetic retinopathy screening by combining fundus photography with OCT. Findings graded by ophthalmologists using the International Clinical Diabetic Retinopathy severity scale form the reference standard. Evidence is reported as diagnostic accuracy and referral-rate comparisons in a real-world workflow.	The study extends AI-based screening beyond fundus-only grading by integrating OCT-derived macular oedema detection alongside AI DR classification, showing that dual-modality screening increases the number of referable cases flagged for ophthalmologist confirmation.
[364]: Application program interface for automatic segmentation of retinal layers and fluids in Optical Coherence Tomography - Neovascular Age related Macular degeneration retinal images using deep learning models	2024	Within an OCT-based ophthalmic imaging context for neovascular age-related macular degeneration, the study applies a deep learning segmentation approach (AR U-Net++) to delineate retinal layer boundaries and fluid compartments from retinal B-scans and reports standard overlap-metric comparisons against prior U-Net variants.	AR U-Net++ outputs spatially resolved retinal fluid locations and depths across multiple layers and automatically generates clinician-facing reports. The study evaluates segmentation, but not the reports' effect on AMD diagnosis or clinical workflow.
[202]: Applications of artificial intelligence in diagnosis of uncommon cystoid macular edema using optical coherence tomography imaging: A systematic review	2024	This systematic review evaluates AI approaches applied to ophthalmic OCT retinal imaging for diagnosing cystoid macular oedema (CME) in less common clinical contexts, covering the diagnostic tasks of detection, classification, and segmentation and summarising evidence from 23 included studies published between 2018 and November 2023, with CNN-based models reported as most consistently effective.	The review synthesises cross-database findings on how CNN-driven AI can automate identification and delineation of uncommon OCT-based CME phenotypes such as postoperative (Irvine Gass) CME and retinitis pigmentosa without evident vasculopathy, while also noting constraints related to dataset size and ethical considerations that affect clinical translation.

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Work	Year	Basis for inclusion	Principal contribution
[85]: Applications of Deep Learning algorithms for retinal diseases diagnosis based on Optical Coherence Tomography imaging	2023	Five convolutional neural network architectures are compared for OCT-based retinal disease detection and diagnosis using accuracy, AUC, F1-score, and loss.	The authors benchmark DenseNet121, DenseNet169, DenseNet201, InceptionResNet, and a 12-layer CNN on OCT images and report that DenseNet169 achieved the best overall metric values, supporting OCT image-based deep learning as a clinically relevant decision-support direction.
[174]: Artificial Intelligence Aided Analysis of Anterior Segment Optical Coherence Tomography Imaging to Monitor the Device-Cornea Joint After Synthetic Cornea Implantation	2025	In a rabbit keratoprosthesis translational study, the authors train a convolutional neural network to segment anterior lamellar tissue on serial anterior segment optical coherence tomography, then use quantified device-cornea joint tissue volume changes to flag subclinical peri-prosthetic thinning and retraction ahead of slit-lamp findings during 6 to 12 months of longitudinal imaging.	They introduce an AI-assisted, aligned AS-OCT analysis workflow that converts optic-cornea joint segmentation into absolute and percentage tissue-volume change estimates, identifying a practical change threshold that can detect tissue loss up to about 3 months before clinical detection, supporting objective postoperative surveillance after synthetic cornea implantation.
[161]: Artificial Intelligence applied to the classification of retinal diseases in Optical Coherence Tomography images	2022	This ophthalmology study uses k-nearest neighbours (KNN) and deep learning (two CNN architectures, including ResNet50) on greyscale OCT image data to perform retinal disease presence classification for CNV, DME, and drusen versus normal, with reported accuracy on a dataset of 83,484 images.	The authors evaluate a hybrid workflow that feeds KNN handcrafted histogram statistics and a CNN end-to-end approach, providing quantitative performance comparisons that may support AI-assisted diagnostic decision-making in OCT-based screening.
[386]: Artificial intelligence-based prediction of neurocardiovascular risk score from retinal swept-source optical coherence tomography-angiography	2024	The study applies ML and DL models to swept-source retinal OCTA microvasculature images to predict the CHA2DS2-VASc neurocardiovascular risk score from the publicly available RASTA dataset. It provides quantitative evidence of model performance for this ophthalmic imaging-to-cardiovascular risk-stratification task.	The authors compare conventional ML classifiers using clinical and OCTA-derived quantitative features with an image-only EfficientNetV2-B3 deep network trained on swept-source OCTA. They show how retinal microvascular imaging alone can estimate CHA2DS2-VASc risk categories in a clinically defined two-group scheme.
[189]: Artificial Intelligence-Based Quantification of Central Macular Fluid Volume and VA Prediction for Diabetic Macular Edema Using OCT Images	2023	In a retrospective study of treatment-naive diabetic macular oedema receiving anti-VEGF therapy, two deep-learning models quantify central macular fluid volume (CMFV) and central subfield thickness (CST) from OCT volume scans. Associations with one-month posttreatment BCVA are assessed by multivariable regression and ROC analysis.	The results indicate that automatically measured CMFV, rather than CST, was the only significant OCT biomarker for predicting one-month logarithm of the minimum angle of resolution (logMAR) BCVA. CMFV also showed slightly better discrimination for achieving BCVA of at least 20/40, supporting fluid-volume quantification as a more informative early outcome predictor.

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Work	Year	Basis for inclusion	Principal contribution
[240]: Artificial intelligence-based retinal disease classification using optical coherence tomography images	2022	The article describes a deep learning, CNN-based retinal disease classification method for ophthalmic optical coherence tomography images, targeting drusen, diabetic macular oedema, and choroidal neovascularisation and reporting test-set performance to support faster automated diagnostic screening and user/clinician alerting.	CNN and pretrained image features yielded the reported accuracy and F1 on the held-out set. The resulting suspicion alerts illustrate a screening-oriented output, but reduced manual review and clinical benefit were not assessed.
[345]: Artificial Intelligence-Driven Detection of LASIK Using Corneal Optical Coherence Tomography Maps	2025	In this ophthalmology study, a lightweight convolutional neural network was trained on corneal OCT parameter maps of pachymetry, epithelial thickness, posterior mean curvature, anterior axial power, and anterior stromal reflectance to classify previous laser-assisted in situ keratomileusis (LASIK). Repeated fivefold patient-stratified cross-validation and separate out-of-sample testing provided evidence of generalisation.	The model accurately discriminated myopic from hyperopic LASIK using down-sampled OCT maps and maintained its accuracy in an external post-LASIK test set. It therefore offers a practical imaging-based approach for identifying a history of LASIK when records may be incomplete.
[401]: ARTIFICIAL INTELLIGENCE FOR OPTICAL COHERENCE TOMOGRAPHY ANGIOGRAPHY-BASED DISEASE ACTIVITY PREDICTION IN AGE-RELATED MACULAR DEGENERATION	2024	In ophthalmology, this retrospective cross-instrument deep learning study evaluates whether a deep neural network trained on OCTA en-face vascular 2D projections can classify age-related macular degeneration disease activity into four clinically defined categories across Heidelberg Spectralis and Optovue Solix, using clinician-labelled fluid status from same-day OCT as the labelling evidence and reporting performance on held-out test sets.	Combining Heidelberg Spectralis and Optovue Solix OCTA data allowed the authors to quantify changes in AMD activity classification across instruments and imaging fields of view. The reported gains support cross-instrument training on the evaluated data, but do not establish manufacturer-independent performance.
[231]: Artificial intelligence for the detection of glaucoma with SD-OCT images: a systematic review and Meta-analysis	2024	This systematic review and meta-analysis evaluates ophthalmic AI for glaucoma detection using SD-OCT imaging, quantitatively synthesising diagnostic accuracy evidence across multiple included studies using pooled sensitivity, specificity, likelihood ratios, diagnostic odds ratio, and AUC.	By aggregating results from 20 studies and 51 AI models, it provides pooled performance estimates that help benchmark SD-OCT-based AI as an adjunct to clinical decision-making for glaucoma screening or diagnosis.
[163]: Artificial intelligence hybrid system for enhancing retinal diseases classification using automated deep features extracted from oct images	2021	The study evaluates a hybrid computer-aided diagnosis system for OCT-based retinal disease classification. AOCTNet extracts deep features from SD-OCT, which conventional supervised classifiers use to distinguish healthy eyes from AMD, CNV, DME, and drusen. Performance was assessed on an SD-OCT test set.	AOCTNet features are passed to eight machine-learning classifiers for multiclass SD-OCT categorisation. The best-performing models achieved strong test-stage results on the study data, illustrating the benchmark value of deep features without establishing clinical efficiency.

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Work	Year	Basis for inclusion	Principal contribution
[456]: Artificial intelligence in predicting macular hole surgery outcomes: a focus on optical coherence tomography parameters	2025	This retrospective ophthalmic study derives the macular hole index (MHI), tractional hole index (THI), hole form factor (HFF), basal hole diameter (BHD), and minimum hole diameter (MHD) from preoperative Spectralis OCT. These measures are used to train and test a generative pre-trained transformer (GPT) classifier that predicts anatomical macular hole closure after idiopathic vitrectomy. Performance is compared with logistic regression using AUC, predictive values, and kappa agreement.	A direct head-to-head evaluation compares OCT index-conditioned GPT predictions with a data-driven logistic regression model. MHI, THI, and HFF carried more prognostic information than simple diameter measures, while GPT tended to under-identify non-closure cases.
[458]: Artificial intelligence in retinal screening using OCT images: A review of the last decade (2013-2023)	2024	This PRISMA-structured systematic review covers machine learning and deep learning methods applied to OCT-derived retinal and choroidal images for screening and decision-support tasks in glaucoma, AMD, and related macular and vascular conditions, summarising the available evidence mainly from peer-reviewed journal studies.	The review synthesises what OCT-based ML versus DL approaches imply for feature handling and usability, highlighting how pretrained deep networks tend to support plug-and-play classification and pointing to continuous learning as a future direction for detecting subtle disease progression from OCT.
[360]: Artificial intelligence method based on multi-feature fusion for automatic macular edema (ME) classification on spectral-domain optical coherence tomography (SD-OCT) images	2023	This ophthalmology study develops a multiclass computer-aided diagnosis system that fuses handcrafted radiomics-style features with deep features extracted from SD-OCT (Heidelberg Spectralis) to classify macular oedema subtypes (DME, AMD, RVO, and central serous chorioretinopathy (CSC)), using standard test-set evaluation with accuracy and ROC/AUC plus model interpretability via Grad-CAM.	The reported results indicate that early multi-feature fusion followed by dimensionality reduction and least absolute shrinkage and selection operator (LASSO)-driven feature selection yields the best SVM performance on the held-out SD-OCT test set, while Grad-CAM heatmaps help indicate the retinal regions driving the subtype decisions.
[383]: Artificial Intelligence Versus Rules-Based Approach for Segmenting NonPerfusion Area in a DRCR Retina Network Optical Coherence Tomography Angiography Dataset	2025	In a multicentre diabetic retinopathy OCT angiography dataset, the study compares convolutional neural network-based versus rules-based nonperfusion area segmentation across four macular slabs, using manual grading and Early Treatment Diabetic Retinopathy Study (ETDRS) severity-linked diagnostics to quantify differences in segmentation agreement and clinical staging performance.	The authors show that the AI-derived nonperfusion outputs match manual segmentation more closely and yield higher DR-severity discrimination for referable status in most slabs, supporting more scalable OCTA quantification than the rules-based approach for screening and triage.

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Work	Year	Basis for inclusion	Principal contribution
[335]: Assessment of a Segmentation-Free Deep Learning Algorithm for Diagnosing Glaucoma From Optical Coherence Tomography Scans	2020	In a single-institution cross-sectional ophthalmic dataset, the study evaluates a segmentation-free convolutional neural network that classifies glaucoma versus normal eyes from full peripapillary SD-OCT circular B-scans and reports ROC AUC and sensitivity at fixed specificities.	On the single-institution test data, the raw-circle B-scan model achieved better reported glaucoma discrimination than conventional RNFL thickness metrics, especially for preperimetric and early perimetric disease. External and prospective evaluation would be needed to determine clinical advantage.
[71]: Attention Assisted Patch-Wise CNN for the Segmentation of Fluids from the Retinal Optical Coherence Tomography Images	2024	An attention-assisted patch-wise CNN with encoder-decoder and multiscale modules segments and quantifies three retinal fluid types on the public RETOUCH dataset. The study examines both qualitative and quantitative results.	The attention-enhanced framework predicts cyst subtype and estimates fluid volume. On the RETOUCH benchmark, the authors report improved precision, recall, and Dice coefficient over prior approaches. These results do not establish benefit in treatment-planning workflows.
[294]: Attention-Guided Neural Networks for Explainable Retinal Disease Detection in OCT Images	2025	The article describes an attention-guided convolutional neural network for automated retinal disease classification from OCT images, using Grad-CAM to generate explanation maps tied to clinically relevant regions and reporting category-level accuracy as evidence in an ophthalmic imaging task.	The study compares several attention-enhanced architectures and reports 98.34% category accuracy for the CBAM-augmented model, with Grad-CAM used to visualise highlighted regions. These results support benchmark classification and visual explanation, but do not establish the fidelity of the explanations or the effectiveness of screening-oriented decision support.
[434]: Augmenting Kalman Filter Machine Learning Models with Data from OCT to Predict Future Visual Field Loss: An Analysis Using Data from the African Descent and Glaucoma Evaluation Study and the Diagnostic Innovation in Glaucoma Study	2022	This retrospective longitudinal cohort study uses Kalman filter machine-learning models to forecast 36-month visual-field loss from perimetry and tonometry. It evaluates whether adding global OCT retinal nerve fibre layer data improves prediction and whether performance varies between race-matched and non-matched training and testing samples from the African Descent and Glaucoma Evaluation Study (ADAGES) and Diagnostic Innovation in Glaucoma Study (DIGS).	The results indicate that incorporating global RNFL from OCT into Kalman filter forecasting produces minimal gains in mean deviation prediction accuracy over tonometry-plus-perimetry models, while race alignment between training and testing yields only modest improvement in mean absolute prediction error.

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Work	Year	Basis for inclusion	Principal contribution
[83]: Automated Age-Related Macular Degeneration and Diabetic Macular Edema Detection on OCT Images using Deep Learning	2018	Deep learning is assessed for multiclass OCT classification of healthy retina, dry AMD, wet AMD, and DME, with a transfer-learning model as the comparator.	The study proposes an end-to-end OCT-based deep learning pipeline that improves over an earlier transfer learning approach for distinguishing healthy eyes, AMD subtypes, and diabetic macular oedema, supporting more accurate automated screening at the image level.
[426]: Automated assessment of the smoothness of retinal layers in optical coherence tomography images using a machine learning algorithm	2023	In this ophthalmology study, a support vector regression model with a wavelet kernel was used on spectral-domain OCT B-scans to automatically segment the inner plexiform layer (IPL) and outer plexiform layer (OPL) and compute a smoothness index, with agreement assessed against manual expert delineations on 50 slabs from patients with diabetic retinopathy using Bland-Altman and error statistics.	By replacing time-consuming manual layer tracing with an end-to-end OCT pipeline for IPL/OPL smoothness quantification, the authors show reproducible segmentation and smoothness-index agreement sufficient to support rapid, biomarker-style measurement in images spanning regular to disorganised layers.
[96]: Automated classification of age-related macular degeneration from optical coherence tomography images using deep learning approach	2023	The authors apply deep learning to ophthalmology by segmenting and classifying OCT macular findings related to AMD, using an attention-based nested U-Net with flower pollination hyperparameter optimisation for a drusen-versus-CNV-versus-normal workflow and reporting UCSD dataset-based evaluation with overall diagnostic performance metrics.	The study develops an OCT pre-processing and ANU-Net plus SqueezeNet classification pipeline, where retinal flattening and cropping precede hyperparameter-tuned learning to enable automated preliminary screening in hospital or eye-clinic settings.
[183]: Automated deep learning-based AMD detection and staging in real-world OCT datasets (PINNACLE study report 5)	2023	This clinical ophthalmology study develops a two-stage CNN for OCT-based AMD staging. Macula-centred 3D Topcon volumes are processed first by a B-scan ResNet50 and then by volume-level ResNets with Monte Carlo dropout. The model produces biomarker-based classifications of normal, intermediate AMD (iAMD), geographic atrophy (GA), and nAMD, and is evaluated on real-world retinal OCT test data using AUC.	The framework combines B-scan localisation, volume-level biomarker prediction, and uncertainty-aware inference to grade routine 3D OCT volumes retrospectively. Independent OCT datasets extend the assessment across normal, early/intermediate, and late AMD categories, but do not establish prospective staging utility.
[270]: Automated Deep Learning Framework for Retinal Disease Classification from OCT Images: A Large-Scale Study Across Four Categories	2024	The article describes an ophthalmology-focused deep learning framework using convolutional neural networks to classify high-resolution retinal OCT cross-sectional images into four retinal disease categories, supported by a large-scale dataset and reported classification accuracy evidence.	The study contributes an automated CNN-based retinal OCT image classification pipeline trained on 84,495 images, offering a practical route to reduce manual reading burden and speed diagnostic decision-making from OCT.

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Work	Year	Basis for inclusion	Principal contribution
[147]: Automated detection of a nonperfusion area caused by retinal vein occlusion in optical coherence tomography angiography images using deep learning	2019	In this ophthalmic deep-learning study, OCTA images of the superficial and deep capillary plexus are used to classify retinal vein occlusion eyes with nonperfusion area versus normal eyes, using a VGG-16-based deep neural network (DNN) compared with an SVM and evaluated with ROC-derived AUC plus sensitivity, specificity, and reading time against seven ophthalmologists.	The authors provide an OCTA-based automated pipeline for detecting nonperfusion area due to retinal vein occlusion and show DNN-based performance and Grad-CAM heat maps that emphasise the foveal avascular zone and nonperfusion regions, supported by head-to-head benchmarking of model outputs against clinician interpretation.
[201]: Automated Detection of Central Retinal Artery Occlusion Using OCT Imaging via Explainable Deep Learning	2025	In this retrospective external validation study from two German institutions, a transfer-learned explainable deep learning classifier analyses Spectralis OCT volumes and slides to triage an acute emergency 3-class task (central retinal artery occlusion (CRAO) versus control and differential diagnoses), with performance estimated using nested five-fold cross-validation and reported primarily via one-vs-all AUC.	The model attained high multiclass discrimination for CRAO and non-CRAO aetiologies on the retrospective external-validation data, while Grad-CAM maps highlighted inner-to-middle retinal regions associated with its decisions. These results motivate prospective urgent-care evaluation rather than establish practical decision-support use.
[287]: Automated detection of exudative age-related macular degeneration in spectral domain optical coherence tomography using deep learning	2018	The study reports a TensorFlow-based deep convolutional neural network (DCNN) that analyses spectral-domain optical coherence tomography cross-sectional images to detect exudative age-related macular degeneration, using held-out testing-stage evaluation on untrained scans as evidence of an image-based AI detection task in ophthalmology.	Training the pretrained DCNN and testing it on a separate set of cross-sections yielded the reported AMD detection score. This supports technical identification of exudative AMD on the evaluated images, but does not establish utility as clinical decision support.
[222]: Automated Detection of Glaucoma using OCT RNFLT Maps and Supervised Deep Learning Models: A Comparative Study of CNN, MobileNet, ResNet50, and VGG16 Architectures	2025	The authors test supervised deep learning classification of glaucoma using OCT retinal nerve fibre layer thickness (RNFLT) maps with several CNN families (CNN, MobileNet, ResNet50, and VGG16), reporting validation results with multiple diagnostic metrics (accuracy, sensitivity, specificity, and F1-score) to determine model effectiveness for automated early detection.	The study directly compares the diagnostic accuracy, error balance, and computational efficiency implications of four architectures on a dataset of 1000 balanced OCT RNFLT samples, highlighting VGG16's top precision and validation accuracy alongside MobileNet's best F1-score.

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Work	Year	Basis for inclusion	Principal contribution
[445]: Automated Detection of Macular Diseases by Optical Coherence Tomography and Artificial Intelligence Machine Learning of Optical Coherence Tomography Images	2019	Using Cirrus HD-OCT macular B-scans with masked ophthalmologist labels, this 2019 study trains a CNN with image augmentation and evaluates automated classification of multiple macular disorders in an ophthalmic diagnostic setting, reporting precision and recall on a held-out set of 100 OCT images.	By expanding limited training data through horizontal flipping, rotation, and translation, it demonstrates that the augmented CNN can generate clinically plausible top candidate disease predictions and quantifies detection accuracy across major classes including normal eyes, wet AMD, DR, and ERM.
[145]: Automated Detection of Unhealthy Retinal Optical Coherence Tomography Images Using Geometrical Features for Machine Learning Modeling	2026	The article describes an ophthalmic OCT imaging study using a supervised machine-learning classifier that inputs inner limiting membrane (ILM) and retinal pigment epithelium (RPE) geometry (symmetry and smoothness) to distinguish unhealthy from healthy retinal layer patterns, providing evidence from reported comparative results against a profile-feature baseline with emphasis on reduced false negatives.	The article specifies a feature-driven modelling strategy for OCT layer quality assessment that improves classification over a vertical profile baseline and highlights lower missed-unhealthy cases, supporting more efficient ophthalmologist prescreening workflows.
[41]: Automated Diagnosis of Eight Retinal Diseases from OCT Images Using Deep Learning with Attention Mechanisms	2025	The authors apply an attention-augmented ResNet deep learning model to classify eight retinal disease categories and normal retina from OCT images, using the RetinalOCT-C8 dataset and reporting validation results with per-class precision-recall and macro F1 to support diagnostic utility.	The study contributes an Attention Residual Classifiers design with a squeeze-and-excitation attention block and residual connections, achieving high validation accuracy and macro F1 while indicating strong class-level performance for several conditions relevant to OCT-based interpretation.
[185]: Automated Identification and Segmentation of Ellipsoid Zone At-Risk Using Deep Learning on SD-OCT for Predicting Progression in Dry AMD	2023	In patients with nonexudative dry age-related macular degeneration, the study uses a deep learning model on SD-OCT B-scans to automatically detect and pixel-wise segment ellipsoid zone at-risk regions, with performance benchmarked against expert ground truth and evaluated for prognostic relevance to geographic atrophy progression using an independent longitudinal test dataset.	The study provides an automated, high-accuracy EZ at-risk quantification pipeline and shows that baseline EZ at-risk area is positively associated with later geographic atrophy growth, supporting its use as a quantitative OCT biomarker for progression risk stratification.

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Work	Year	Basis for inclusion	Principal contribution
[278]: Automated inter-device 3D OCT image registration using deep learning and retinal layer segmentation	2023	This ophthalmic study develops a two-stage deep-learning registration pipeline for aligning 3D OCT volumes from different devices, using unsupervised multimodal en-face feature matching between polarisation-sensitive OCT (PS-OCT) and Spectralis scanning laser ophthalmoscopy (SLO) followed by retinal-layer segmentation-guided axial (Z-axis) alignment, with performance evaluated on paired clinical and experimental OCT data using landmark-based error and RPE-layer-derived axial metrics.	The study provides an automated way to transfer spatial correspondence across Heidelberg Spectralis and a custom PS-OCT system by coupling R2D2-style keypoint detection and random sample consensus (RANSAC)-based 2D alignment with ILM-referenced B-scan adjustment, enabling consistent fusion and cross-device mapping of signals such as PS-OCT degree of polarisation uniformity (DOPU) onto conventional OCT.
[241]: Automated machine learning-based classification of proliferative and non-proliferative diabetic retinopathy using optical coherence tomography angiography vascular density maps	2023	A supervised SVM, tuned by a genetic evolutionary algorithm, classifies proliferative diabetic retinopathy (PDR) versus NPDR from retinal-layer OCTA vascular-density maps. Evaluation includes 148 eyes from 78 patients.	By comparing vascular density maps at superficial, deep, and total retina levels, it identifies the deep retinal layer map as the most informative input, with the best reported accuracy reaching 90% for PDR versus NPDR discrimination and supporting layer-specific OCTA feature use for retinal disease classification.
[90]: AUTOMATED MACHINE LEARNING CLASSIFICATION OF OPTICAL COHERENCE TOMOGRAPHY IMAGES OF RETINAL CONDITIONS USING GOOGLE CLOUD VERTEX ARTIFICIAL INTELLIGENCE	2025	Using Google Cloud Vertex AI automated machine learning, the study evaluates single-label classification of optical coherence tomography images for five retinal categories, including a neovascular AMD subanalysis, with evidence from a validated publicly available dataset and reported diagnostic performance metrics.	The off-the-shelf Vertex AI pipeline produced the reported AUPRC, sensitivity, and specificity for the retinal categories studied. These benchmark results indicate technical feasibility, whereas scalability and diagnostic utility require external, workflow-level validation.
[254]: Automated Macular Disease Detection using Retinal Optical Coherence Tomography images by Fusion of Deep Learning Networks	2021	The authors apply deep learning to retinal OCT for automated macular disease image classification, using a two-network fusion strategy that concatenates learned feature vectors from pretrained DCNNs to support screening and clinician diagnosis-recommendation workflows, with comparative evidence reported as improved classification accuracy over single models on the same dataset.	The study introduces a feature-fusion DCNN framework for macular OCT classification in which concatenated representations from two pretrained networks yield better accuracy than each constituent model, offering a practical approach for OCT-based decision support.

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Work	Year	Basis for inclusion	Principal contribution
[127]: Automated Macular OCT Retinal Surface Segmentation in Cases of Severe Glaucoma Using Deep Learning	2022	In an ophthalmic glaucoma setting, the study describes an independently evaluated deep learning approach for automated macular OCT retinal surface segmentation that targets known failure modes in severe disease, using three-dimensional OCT imaging through data augmentation, contextual multi-slice input from adjacent B-scans, and topology-preserving fusion of predictions to produce more reliable surface delineations.	The authors test three design choices: vertical and jittered B-scan augmentation, contrast-enhanced six-channel stacking from adjacent slices, and topology-preserving fusion of horizontal and vertical surfaces. On the severe-glaucoma dataset, the reported mean absolute distance, Dice, and Hausdorff results favoured the resulting approach over the selected contemporary comparators.
[455]: Automated Quantification of Photoreceptor alteration in macular disease using Optical Coherence Tomography and Deep Learning	2020	In this 2020 study, an ensemble of U-shaped convolutional neural networks segments the photoreceptor layer in Spectralis OCT and produces en face thickness and uncertainty maps for DME and RVO cases. Quantitative benchmarking uses consensus expert annotations and measurements from a second retina specialist.	The ensemble averages model score maps to generate photoreceptor boundaries and disagreement maps. Comparison with consensus annotations and a second specialist supports thickness quantification on the evaluated Spectralis data, while the uncertainty output identifies regions for human review.
[186]: Automated Retinal Disease Detection: VGG16-Based Deep Learning Approach for OCT Image Classification	2025	Using a VGG16 deep learning approach on retinal OCT images, the authors perform four-class disease classification (CNV, DME, DRUSEN, NORMAL), reporting diagnostic evidence via standard evaluation metrics and confusion-matrix analysis on a train/validation/test split.	The article details an OCT preprocessing and augmentation pipeline with VGG16 and reports accuracy plus precision, recall, and F1 across all classes, supporting the feasibility of automated retinal diagnostic support for image-based screening.
[262]: Automated retinal disorders classification: leveraging digital image enhancement techniques and deep learning on OCT Images	2024	This ophthalmology study uses deep learning with transfer learning and data augmentation on retinal OCT images to classify CNV, DME, DRUSEN, and normal, reporting quantitative evidence via test-set classification accuracy.	The authors evaluate VGG16, ResNet18, and DenseNet with image preprocessing and enhancement, highlighting DenseNet's top test accuracy (92.41%) as support for automated OCT-based retinal disorder recognition.
[47]: Automated segmentation of choroidal neovascularization on optical coherence tomography angiography images of neovascular age-related macular degeneration patients based on deep learning	2023	The study develops a deep learning segmentation model to delineate choroidal neovascularisation regions from en face OCT angiography in neovascular age-related macular degeneration, using a modified U-net architecture with ResNeSt and spatial pyramid pooling to address CNV scale variability, and it provides clinical imaging evidence via testing on manually annotated OCTA images.	The ResNeSt-based U-net and multiscale spatial-pyramid pooling address CNV scale variation in OCTA data. On the annotated test images, the authors report better segmentation metrics than the standard saliency approach and baseline U-net. The comparison remains dataset-specific.

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Work	Year	Basis for inclusion	Principal contribution
[377]: Automated Segmentation of Retinal Fluid Volumes From Structural and Angiographic Optical Coherence Tomography Using Deep Learning	2020	In a diabetic macular oedema setting, the study evaluates a U-Net-like deep CNN (ReF-Net) for volumetric retinal fluid segmentation using structural OCT together with angiographic OCTA on simultaneously acquired 3D OCT/OCTA volumes, reporting agreement-based evidence from manual ground-truth labelling and quantitative segmentation metrics.	ReF-Net fuses simultaneously acquired structural OCT and OCTA to segment cystic fluid volumes. The reported comparisons favour fusion over structural OCT alone and examine typical shadow artefacts. The authors use these results to argue that volumetric assessment adds information beyond 2D projections.
[86]: Automated segmentation of the choroid in EDI-OCT images with retinal pathology using convolution neural networks	2017	In this 2017 study, a convolutional neural network with SegNet-style pixel-wise boundary probability maps and seam carving segments the choroid in enhanced depth imaging OCT (EDI-OCT) B-scans. Evaluation uses 62 manually delineated choroid segmentations from patients with age-related macular degeneration and does not require retinal-layer boundary preprocessing.	The pipeline estimates Bruch's membrane and the choroid-sclera interface directly, avoiding reliance on upstream retinal-boundary detection. Results on the annotated, disease-affected B-scans support feasibility for pathological EDI-OCT, while broader robustness across pathologies remains to be tested.
[340]: Automatic classification of retinal OCT images based on convolutional neural network	2020	The authors combine a convolutional neural network with OCT-focused preprocessing and augmentation to classify retinal images, reporting accuracy, sensitivity, and specificity from training and validation splits.	The article details an end-to-end CNN pipeline that couples pathological-area centering, noise-reducing cropping, and augmentation via random translation and horizontal flips, achieving reported performance of 93.43% accuracy, 91.38% sensitivity, and 95.88% specificity.
[230]: Automatic Detection and Classification of Diabetic Retinopathy from Optical Coherence Tomography Angiography Images using Deep Learning-A Review	2025	This review focuses on ophthalmic diabetic retinopathy assessment using OCTA microvascular imaging, surveying deep learning and machine learning approaches for DR detection and classification while summarising the reported OCTA datasets and evaluation approaches.	The review synthesises recent CNN-centred OCTA DR classification studies and contrasts a proposed deep learning workflow with traditional machine learning pipelines that rely on manually engineered features, underscoring clinical challenges such as data variability, interpretability, and workflow integration.
[298]: Automatic detection of microaneurysms in optical coherence tomography images of retina using convolutional neural networks and transfer learning	2022	This ophthalmic deep learning study targets microaneurysm detection from diabetic retinopathy OCT B-scans by converting them into labelled OCT strips using OCT-fluorescein angiography registration, then classifying strips with a stacked ensemble of transfer-learned CNNs for evidence from model evaluation on a hold-out test split.	The study provides a practical labelled OCT strip creation workflow for four classes (microaneurysm, normal, abnormal, and vessel) and uses stacking across fine-tuned ImageNet-pretrained networks to improve strip-level discrimination, supporting OCT-based microaneurysm localisation without needing fluorescein angiography at inference.

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Work	Year	Basis for inclusion	Principal contribution
[137]: Automatic diagnosis of macular diseases from OCT volume based on its two-dimensional feature map and convolutional neural network with attention mechanism	2020	This primary technical study develops an AI method for classifying macular disease from three-dimensional OCT volumes. A convolutional neural network extracts B-scan features to construct a two-dimensional feature map. At the volume level, support vector machines and convolutional neural networks with and without attention classify the map. Evaluation uses the public Duke AMD dataset and a private AMD/DME cohort.	The resulting volume-classification framework converts B-scan features into a two-dimensional map and compares attention-based with non-attention classifiers. Reported cross-validation results on the public and private cohorts exceeded the selected comparators, supporting technical feasibility for OCT-volume classification without establishing clinical screening utility.
[347]: AUTOMATIC METHOD OF MACULAR DISEASES DETECTION USING DEEP CNN-GRU NETWORK IN OCT IMAGES	2024	Using OCT B-scan images from the labelled OCT Retina dataset, this analysis develops a deep CNN with a gated recurrent unit (GRU) architecture that performs four-class macular disease recognition (drusen, choroidal neovascularisation, diabetic macular oedema, and normal), supported by reported accuracy, precision, recall, and F1 score with repeat experiments, leave-one-out cross-validation (LOOCV), and comparison to prior deep learning baselines.	The paper's main advance is the use of GRU units connected to a CNN feature extractor to improve OCT-based classification performance for early and late macular disease states, yielding strong results on the established dataset and positioning the method among the higher-performing approaches in the study's comparison with published methods.
[268]: Automatic Retinal and Choroidal Boundary Segmentation in OCT Images Using Patch-Based Supervised Machine Learning Methods	2019	In SD-OCT, this research evaluates patch-based supervised deep learning, using convolutional neural networks and recurrent neural networks (RNNs) to predict per-layer probability maps for retinal and choroidal boundary locations, then derives boundary positions through graph-search tracing and compares results with manual and prior image-analysis methods.	The study provides a practical framework for automated retinal and choroidal boundary localisation that reports how CNN versus RNN choice, patch size, and intensity compensation affect probability maps and the resulting thickness-related boundary tracing.
[352]: Automatic segmentation of multitype retinal fluid from optical coherence tomography images using semisupervised deep learning network	2023	The study addresses an ophthalmology imaging task by using a semisupervised deep learning network to segment subretinal and intraretinal fluid on OCT, with performance assessed using quantitative and qualitative analyses, component ablations, and cross-dataset testing on an unseen Kermany dataset.	The study reports a coarse-to-refine Ref-Net that can delineate multitype retinal fluid with limited labelled OCT images, providing evidence that expert-benchmarked segmentation and reasonable generalisation are achievable when training data are scarce.

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Work	Year	Basis for inclusion	Principal contribution
[432]: Automatic segmentation of nine retinal layer boundaries in OCT images of non-exudative AMD patients using deep learning and graph search	2017	CNN-guided graph search is tested for segmentation of nine retinal-layer boundaries in SD-OCT B-scans from non-exudative AMD eyes, using expert-corrected annotations in a held-out test set.	The article details an end-to-end pipeline that converts CNN-derived probability maps into refined layer boundaries via a modified graph-theory-and-dynamic-programming (GTDP) search, supporting practical AI-assisted OCT layer segmentation without relying on many ad hoc rules.
[295]: Automatic segmentation of OCT retinal boundaries using recurrent neural networks and graph search	2018	In an ophthalmology OCT setting, the analysis proposes a recurrent neural network patch classifier integrated with graph search (RNN-GS) to segment retinal layer boundaries in cross-sectional B-scan images and evaluates boundary-position errors against expert-corrected labels in healthy children and patients with AMD, with comparisons to CNN-based graph-search methods.	The authors replace a CNN with an RNN within a graph-search segmentation pipeline and show competitive, sometimes marginally improved boundary localisation accuracy and consistency, providing an evidence-backed approach for delineating multiple OCT retinal boundaries in both normal and AMD data.
[119]: Automatic Segmentation of Retinal Fluid and Photoreceptor Layer from Optical Coherence Tomography Images of Diabetic Macular Edema Patients Using Deep Learning and Associations with Visual Acuity	2022	This deep learning study in diabetic macular oedema uses a modified U-Net with an EfficientNet-B5 encoder to automatically segment OCT biomarkers: intraretinal cystoid fluid, subretinal fluid, and ellipsoid zone structure. It evaluates these outputs against manually labelled ground truth and uses ROC-based detection of ellipsoid zone disruption to support clinical interpretation.	The study further links AI-derived biomarker quantification to visual function by modelling associations with logMAR best-corrected visual acuity, showing that ellipsoid zone disruption has the strongest relationship and thereby supporting OCT-derived severity assessment beyond central subfield thickness.
[358]: Breaking the Mold: ViT-CNN Fusion for Enhanced Glaucoma Prediction in OCT Images	2025	The study targets an ophthalmic glaucoma prediction task using OCT imaging, and evaluates a ViT-CNN fusion model that combines localised CNN feature extraction with global Vision Transformer context followed by ML-based classification, with evidence reported from experiments across four datasets showing improved results versus single-model baselines.	Fusing localised CNN features with global Vision Transformer context is the study's central comparison. Across the evaluated datasets, the fused model improved the reported classification results relative to single-model baselines, while Grad-CAM visualisations describe attention rather than verify clinical reliability.

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Work	Year	Basis for inclusion	Principal contribution
[375]: CATALYZE: a deep learning approach for cataract assessment and grading on SS-OCT images	2025	This prospective single-centre study evaluates a deep learning cataract grading pipeline from Anterion swept-source OCT (SS-OCT) that outputs layerwise metrics and a clinical significance index, then compares its discrimination with an established optical ocular scatter index using correlation and ROC analyses across development and validation cohorts.	CATALYZE yields an SS-OCT-derived cataract clinical significance index that closely tracks the ocular scatter index and shows comparable validation AUC, supporting an objective, image-based approach to cataract assessment tied to optical scattering measures.
[443]: Circumpapillary OCT-focused hybrid learning for glaucoma grading using tailored prototypical neural networks	2021	The study develops a deep-learning model that grades glaucoma from raw circumpapillary SD-OCT B-scans. It fuses an RNFL descriptor with a residual-attention CNN and supervised few-shot or k-shot prototypical classification. Evaluation on Heidelberg Spectralis datasets covered categorical grading and localisation with class activation maps.	The study further refines glaucoma severity modelling by formulating static and dynamic tailored prototypical networks on top of an OCT-specific hybrid backbone that incorporates handcrafted RNFL thickness histogram features, enabling discrimination among healthy, early, and advanced glaucoma while avoiding reliance on a conventional projection-head classification stage.
[284]: Classification and Prediction Retinal Oct Images by CNN Algorithm	2022	The article describes a MATLAB-based convolutional neural network trained on well-lit ophthalmic fundus images to classify and predict diabetic retinopathy status for early detection and stage-related characterisation, reporting classification metrics from a single training/testing split.	The study provides an end-to-end CNN pipeline for automated diabetic retinopathy screening using extracted fundus image features and reports high diagnostic performance, offering quantitative support for image-driven triage in an ophthalmic monitoring context.
[353]: Classification and Quantification of Retinal Cysts in OCT B-Scans: Efficacy of Machine Learning Methods	2019	The study addresses an ophthalmic OCT task by using an end-to-end machine learning approach that combines a random forest classifier with a DeepLab-style segmentation algorithm to detect, label, and quantify retinal fluid compartments in B-scans of highly distorted florid disease, with evidence reported as agreement versus expert manual delineations using an average Dice score.	The study reports an automated segmentation and quantification pipeline for retinal fluid in OCT B-scans, highlighting clinically relevant measurement capability in challenging cases through Dice-based performance relative to trained expert labelling.
[260]: Classification of diabetes-related retinal diseases using a deep learning approach in optical coherence tomography	2019	This 2019 ophthalmology study uses an OCT-NET convolutional neural network to classify diabetes-related retinal disease categories from spectral-domain optical coherence tomography B-scan volumes, with evidence from evaluation on public SD-OCT datasets. The study reports quantitative performance and clinician-oriented class activation map (CAM) localisation for interpretability.	The study extends end-to-end SD-OCT volume classification by adding a CAM-based feedback stage that highlights diagnostically relevant regions within each scan, helping specialists understand and review the basis of model predictions beyond the disease label alone.

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Work	Year	Basis for inclusion	Principal contribution
[111]: Classification of Fluid Filled Abnormalities In Retinal Optical Coherence Tomography Images Using Chicken Swarm Optimized Deep Learning Classifier	2023	In an ophthalmic OCT context, the study evaluates a deep learning pipeline for classifying fluid-filled retinal abnormalities using contrast-limited adaptive histogram equalisation (CLAHE)-preprocessed OCT images, retinal-layer segmentation, Gabor-based feature extraction, and an RNN classifier optimised with chicken swarm optimisation (CSO), reporting accuracy and processing time from dataset-based MATLAB validation.	The study combines segmentation-guided feature selection with CSO-trained RNN classification and reports accuracy and processing time from dataset-based MATLAB validation. These results address technical performance and efficiency but do not establish clinical feasibility.
[304]: Classification of range of OCT-angiography capillary density using multichannel deep learning models in diabetic retinopathy, aging macular degeneration, and radiation retinopathy	2025	The article describes a nnU-Net-based deep learning approach for segmenting OCT-angiography capillary density into four severity categories in retinal vascular disease, using OCTA (alone and with FAZ and/or large vessel tree) and evaluating agreement with expert annotations via Dice coefficient comparisons.	The authors compare four multi-input configurations to determine how adding FAZ and large vessel tree information affects consistency of the categorised capillary-density maps, highlighting the three-channel OCTA plus FAZ plus large vessel tree model as a more objective option than OCTA-only labelling.
[315]: Classification of Retinal Diseases in Optical Coherence Tomography Images Using Artificial Intelligence and Firefly Algorithm	2023	A 2023 study combines retinal-layer extraction with handcrafted texture and shape descriptors and transfer-learned deep features for OCT disease classification. Firefly-algorithm feature selection precedes hierarchical multi-binary classification by SVM, logistic regression, and random forest on two public OCT datasets, with mean classifier accuracy reported.	The study adds an efficiency-oriented workflow that reduces the high-dimensional fused feature set via Firefly-based selection and improves training feasibility while maintaining diagnostic performance across the abnormal versus normal and AMD-related binary splits, highlighting how preprocessing, hybrid features, and hierarchical multi-binary decisions work together to support automated OCT-based screening.
[235]: Classification of Retinal OCT Images Using Deep Learning	2022	The authors apply a convolutional neural network to classify retinal disorder categories from OCT cross-sectional images, using the ophthalmic imaging modality to address automated diagnostic image classification with reported performance on a retinal OCT dataset.	The lightweight VGG16-based classifier assigns OCT images to eight disease states or Normal Eye. Its reported dataset performance demonstrates benchmark classification. Use for early identification of retinal pathology was not clinically evaluated.

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Work	Year	Basis for inclusion	Principal contribution
[191]: CNN-Hyperparameter Optimization for Diabetic Maculopathy Diagnosis in Optical Coherence Tomography and Fundus Retinography	2022	A CNN accept-or-reject classifier for diabetic maculopathy is tested on OCT and fundus retinography. Bayesian hyperparameter search is used to optimise performance within the end-to-end diagnostic pipeline.	The article details how Bayesian optimisation is used to tune CNN hyperparameters and design choices for diabetic maculopathy detection across OCT and fundus retinography, offering a practical optimisation strategy that can improve automated retinal screening performance.
[58]: CNN Layer-wise explanations for the prediction of Diabetic Retinopathy from OCTA images	2025	An explainable CNN classifies diabetic retinopathy severity from OCTA images, while Signature Activation generates holistic, class-agnostic explanations of its predictions.	The authors apply a layer-wise explanation method to clarify what the CNN attends to when grading OCTA-derived DR severity, supporting clinician interpretability needs for AI-driven retinal screening.
[429]: Comparative Analysis of Deep Learning Architectures for Macular Hole Segmentation in OCT Images: A Performance Evaluation of U-Net Variants	2025	This OCT macular hole segmentation study evaluates multiple U-Net variants that swap in different CNN and Transformer-style encoders, using the OIMHS dataset and reporting overlap and boundary evidence via Dice coefficient and 95th-percentile Hausdorff distance (HD95) to compare architectural families on the macular hole (MH) task.	The metric-based comparison identifies InceptionNetV4 plus U-Net as yielding the highest reported Dice values among the tested variants. The authors also document unstable HD95 behaviour for small MH regions and weaker Transformer results for MH and intraretinal cyst segmentation, limiting any single ranking across metrics.
[177]: Comparing code-free deep learning models to expert-designed models for detecting retinal diseases from optical coherence tomography	2024	This ophthalmology study compares code-free deep learning and clinician-built AutoML models with expert-designed bespoke deep learning models for multiclass retinal pathology detection from OCT macular videos and fovea-centred images, using an internal dataset and reporting point-estimate discriminative performance metrics.	On the same internal OCT-classification task, trainee-developed code-free models achieved point estimates comparable to those of the expert-designed models. This comparison concerns model-development approaches, not performance relative to clinical experts or screening effectiveness.
[392]: Comparison of Autonomous AS-OCT Deep Learning Algorithm and Clinical Dry Eye Tests in Diagnosis of Dry Eye Disease	2021	In this ophthalmology study, a deep-learning classifier autonomously detects dry eye disease (DED) from anterior-segment OCT images. Diagnostic agreement is assessed against masked cornea specialist diagnoses and compared with the Ocular Surface Disease Index (OSDI), tear break-up time (TBUT), corneal and conjunctival staining, and Schirmer's test.	The authors provide a head-to-head evaluation showing that AS-OCT deep learning performs significantly better than several conventional tests and is comparable to others, supporting the feasibility of autonomous imaging-based DED diagnosis for clinic-relevant screening.

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Work	Year	Basis for inclusion	Principal contribution
[193]: Comparison of Different Machine Learning Classifiers for Glaucoma Diagnosis Based on Spectralis OCT	2021	This glaucoma diagnostic study evaluates several classical machine-learning classifier families using Heidelberg Spectralis spectral-domain OCT measurements. Inputs include circumpapillary retinal nerve fibre layer (cRNFL) thickness, BMO-MRW, ETDRS macular thickness, and posterior pole asymmetry analysis (PPAA). The models distinguish normal from glaucomatous eyes and stratify early, moderate, and severe disease. Evidence comes from cross-validation, bootstrap analysis, and comparison with logistic regression.	The study identifies random forest as the top-performing approach across glaucoma subsets and uses model-derived variable-importance rankings to pinpoint which OCT-derived ganglion cell layer, cRNFL, and BMO-MRW features and locations matter most as disease severity increases.
[331]: Context-aware diagnosis of age-related macular degeneration (AMD) stages: a unified approach using self-supervised U-Net and graph neural networks on OCT imaging	2025	The study targets context-aware staging of age-related macular degeneration from optical coherence tomography using a self-supervised U-Net to segment superpixels and graph neural networks to model spatial and structural relationships, providing quantitative diagnostic evidence via reported accuracy, F1-score, and AUC.	The study introduces a unified, self-supervised pretext inpainting training scheme with a context-aware loss that links superpixel interdependencies, and reports improved AMD stage classification over baselines, supporting the clinical relevance of OCT-based, relation-informed modelling.
[118]: Convolution neural network model for predicting various lesion-based diseases in diabetic macula edema in optical coherence tomography images	2023	The article describes a lesion-based convolutional neural network that uses OCT retinal images to detect and distinguish DME-related lesions (including haemorrhages, microaneurysms, and exudate), evaluated against established CNN architectures as evidence from an ophthalmic imaging diagnosis task.	The study introduces an OCT lesion-focused CNN pipeline aimed at improving DME lesion prediction and reports higher accuracy than standard CNN baselines such as ResNet, VGG16, Inception, and AlexNet, supporting more effective automated OCT interpretation.
[122]: CorneaNet: fast segmentation of cornea OCT scans of healthy and keratoconic eyes using deep learning	2019	The 2019 study develops CorneaNet, a supervised fully convolutional neural network that segments corneal OCT B-scans into epithelium, Bowman's layer, stroma, and background in healthy and keratoconic eyes. Validation uses six-fold cross-validation and produces layer-wise segmentations and thickness maps.	CorneaNet generates layer segmentations and thickness maps with substantially shorter reported runtime than the authors' earlier iterative robust-fitting algorithm. The study establishes computational efficiency on its cross-validation data. Effects on keratoconus assessment or early detection were not evaluated directly.

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Work	Year	Basis for inclusion	Principal contribution
[414]: Cross-instrument optical coherence tomography-angiography (OCTA)-based prediction of age-related macular degeneration (AMD) disease activity using artificial intelligence	2024	This retrospective ophthalmic study evaluates DNN-based multiclass prediction of AMD disease activity from 2D vascular en-face OCTA images using cross-instrument training data from Heidelberg Spectralis and Optovue Solix, with labels derived from clinical diagnoses and contemporaneous B-scan OCT fluid to test generalisability across devices via separate test sets.	By training EfficientNet-b5-based classifiers on combined Heidelberg and Optovue OCTA vascular morphology, the authors demonstrate cross-manufacturer feasibility for activity classification and quantify how performance varies by training source, supporting practical deployment considerations for OCTA-based AMD activity grading.
[80]: CSANet: a lightweight channel and spatial attention neural network for grading diabetic retinopathy with optical coherence tomography angiography	2024	This ophthalmology study proposes CSANet, an attention-based CNN for fine-grained diabetic retinopathy grading using OCTA en-face images, tackling small-lesion visibility and adjacent-grade confusion, and evaluates the approach with benchmark experiments on OCTA-DR and DRAC 2022 classification evidence.	The study introduces a lightweight hybrid attention module combining channel and spatial attention to better capture lesion-relevant OCTA features and reduce missed small findings, supported by strong grading results across the reported datasets and module ablations.
[457]: Cynomolgus monkey's retina volume reference database based on hybrid deep learning optical coherence tomography segmentation	2023	Using spectral-domain OCT from healthy cynomolgus monkeys, the study applies deep learning semantic retinal segmentation plus classical computer-vision foveolar landmark detection to compute retinal volumes across predefined zones, quadrants, and slices, providing evidence for natural variation and analysing effects of sex, geographic origin, and eye laterality using a large retrospective dataset of 374 eyes from 203 animals.	The study establishes an OCT-derived macaque retinal volume reference database and quantifies how origin and sex systematically shift foveolar subfield volume estimates, which is critical for interpreting future OCT-based preclinical measurements in light of biologic baseline differences.
[115]: DAISY Descriptors Combined with Deep Learning to Diagnose Retinal Disease from High Resolution 2D OCT Images	2020	In this ophthalmic study, deep learning is used to classify high-resolution 2D spectral-domain OCT B-scans for four retinal conditions using DAISY-descriptor-driven intelligent downsampling followed by a GRU, with evaluation on a publicly available dataset that includes independent testing with bootstrapped confidence intervals.	The authors show that replacing random downsampling with DAISY-based feature-preserving reduction enables effective OCT disease classification from descriptors, improving test accuracy and AUC over an InceptionV3 baseline.
[318]: Deep CNN framework for retinal disease diagnosis using optical coherence tomography images	2021	The now-retracted report combines speckle denoising, hyperparameter tuning, and a deep convolutional neural network to classify labelled OCT images as normal, drusen macular degeneration, or DME, evaluating performance with K-fold validation.	The end-to-end workflow combines Kuan-filter despeckling with a tuned CNN and comparison against pretrained models, attaining a reported classification accuracy of 95.7%. The publisher later retracted the article after finding a compromised editorial process and manipulated peer review. Accordingly, this accuracy should not be treated as reliable evidence.

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Work	Year	Basis for inclusion	Principal contribution
[292]: DEEP CONVOLUTIONAL NEURAL NETWORK PREDICTION FOR GLAUCOMA DETECTION USING OCT AND OCT-ANGIOGRAPHY DISC- AND MACULA-CENTERED IMAGES AND THEIR COMBINED POWER	2024	This cross-sectional ophthalmic deep learning study used a deep CNN intermediate fusion framework on en-face OCT and OCTA disc- and macula-centred images to classify eyes as healthy versus glaucomatous, evaluating discriminative evidence using standard accuracy and AUC metrics (with additional sensitivity and specificity).	The study reports that combining disc and macula regions improves binary glaucoma detection across modality pairings, while OCTA alone shows consistently superior sensitivity on disc-and-macula inputs, offering practical guidance on which OCT/OCTA image sets to prioritise for early diagnosis.
[210]: Deep learning algorithms for detection of diabetic macular edema in OCT images: A systematic review and meta-analysis	2023	This systematic review and diagnostic test accuracy meta-analysis evaluates deep learning neural networks, predominantly convolutional architectures, using OCT and fundus retinal imaging to detect diabetic macular oedema, synthesising pooled evidence on diagnostic performance and heterogeneity across included studies.	By aggregating results from 53 studies, the meta-analysis estimates sensitivity and compares OCT with fundus imaging. The pooled results favour OCT in the reported analysis, while between-study heterogeneity limits direct clinical inference.
[228]: Deep Learning and OCT Imaging: A Novel Ensemble Approach for Eye Disease Diagnosis	2026	Using retinal OCT imaging, the study describes a deep learning ensemble of three CNNs (VGG16, InceptionV3, and InceptionResNetV2) for automated ocular disease classification and reports high classification accuracy.	The study proposes an OCT-focused inference approach that combines heterogeneous CNNs and reuses preexisting architectures to improve disease identification performance on an OCT dataset.
[418]: Deep learning and optical coherence tomography in glaucoma: Bridging the diagnostic gap on structural imaging	2022	This 2022 systematic review covers ophthalmic deep learning approaches for glaucoma diagnosis and progression, focusing on models trained on structural OCT (including optic nerve and macular scans and derived OCT maps) and extending OCT-trained learning to fundus photography to support detection in broader clinical screening contexts.	The review synthesises a decade of OCT- and fundus-based deep learning strategies, detailing how OCT-derived ground truth supports automated glaucoma assessment and highlighting practical evidence-based gaps and future directions for translation and research use.
[343]: Deep Learning Approaches for Retinal Disease Diagnosis: Insights from Fundus and OCT Analysis	2025	The authors benchmark deep learning for glaucoma and diabetic retinopathy on clinically relevant retinal imaging, using fundus photographs alongside 2D and 3D OCT inputs and reporting task-based diagnostic classification evidence with standard metrics (accuracy, precision, recall, specificity, and F1-score).	The study provides a modality-specific comparison of prominent architectures, highlighting transformer family models (and a ViT-large plus GRU hybrid for 3D OCT) together with EfficientNet results for 2D OCT, to inform which model families may be better suited for retinal disease screening across imaging types.

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Work	Year	Basis for inclusion	Principal contribution
[200]: Deep Learning Approaches Predict Glaucomatous Visual Field Damage from OCT Optic Nerve Head En Face Images and Retinal Nerve Fiber Layer Thickness Maps	2020	This evaluation tests deep learning on spectral-domain OCT optic nerve head en face images and RNFL thickness maps to classify eyes with and without glaucomatous visual field damage and to predict quantitative functional outcomes from visual field testing, providing independent-test diagnostic and regression evidence (AUC, sensitivity/specificity, R2, MAE) in an ophthalmic glaucoma setting.	On the independent test set, RNFL en face models yielded better reported detection and functional-estimation metrics than mean RNFL thickness. The work links SD-OCT structure to visual-field mean deviation, pattern standard deviation, and sectoral pattern deviation, but does not establish functional risk stratification prospectively.
[179]: Deep learning approaches to predict 10-2 visual field from wide-field swept-source optical coherence tomography en face images in glaucoma	2022	This glaucoma study uses a convolutional deep learning approach based on Inception-ResNet-V2 to predict 10-2 visual field (VF) threshold values from wide-field swept-source OCT en face-derived thickness-map inputs, with performance evaluated against ground-truth 10-2 perimetry in a held-out test set.	By contrasting an en face-based input pipeline with an RNFL thickness-based alternative (both combined with macular ganglion cell/inner plexiform layer thickness (mGC/IPLT) maps), the authors show that the en face model yields lower sector-specific prediction errors and supports clinically relevant central VF estimation from SS-OCT imaging.
[204]: Deep learning architectures analysis for age-related macular degeneration segmentation on optical coherence tomography scans	2020	Using deep learning CNN-based semantic segmentation on OCT scans, the authors delineate three AMD-associated retinal fluid types, applying patch-based training on the RETOUCH challenge data and testing generalisation via OPTIMA, with evidence reported using Dice similarity metrics versus human performance.	The authors compare four widely used segmentation architectures and show that patching plus fine-tuning improves dense fluid delineation, yielding high mean Dice scores (up to 0.85) that support more accurate OCT-based AMD fluid segmentation.
[361]: Deep Learning Architectures for OCT Images Retinal Disease Classification	2025	Architectural and training choices, including layer depth, kernel size, and pooling strategy, are compared in deep learning classifiers for retinal disease on OCT. Quantitative metrics assess the resulting models.	The study identifies OCTDeepNet2 as the best-performing configuration under the tested layer and training conditions, offering practical guidance on which architecture and optimisation settings may improve retinal disease classification accuracy from OCT scans.
[171]: Deep Learning-Assisted Detection of Glaucoma Progression in Spectral-Domain OCT	2023	This retrospective cohort ophthalmic study trains a convolutional neural network to estimate the probability of glaucoma structural progression from baseline and follow-up spectral-domain OCT RNFL thickness, with performance reported using AUC, sensitivity, and specificity and benchmarked against conventional trend-based change methods.	The reported results indicate that incorporating time between OCT visits into the DL framework yields better diagnostic discrimination than trend-based analyses and provides likelihood ratios that can substantially shift post-test progression probability.

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Work	Year	Basis for inclusion	Principal contribution
[433]: Deep learning-based algorithm for the detection of idiopathic full thickness macular holes in spectral domain optical coherence tomography	2024	This cross-sectional study evaluates a Zeiss "B-scans of interest" model for detecting idiopathic full-thickness macular hole (IFTMH) features in SD-OCT B-scans. Two graders provided B-scan-level reference labels, and severity was adjudicated according to the International Vitreomacular Traction Study Group. Internal test-set performance and the correlation between model probability and severity stage were assessed.	The authors validate automated abnormal B-scan detection for IFTMH across two test sets derived from different grader-agreement labelling rules, showing high accuracy for identifying abnormal scans while demonstrating only a weak relationship between the model's probability scores and cube-level severity stages, highlighting the need for task-specific training to support severity stratification.
[293]: Deep learning-based automated detection of retinal diseases using optical coherence tomography images	2019	In this 2019 study, an ensemble of four modified ResNet50 models performs multiclass retinal disease recognition on OCT B-scans. Evaluation comprises patient-level 10-fold cross-validation, comparison with ophthalmologists, occlusion testing, and qualitative analysis of misclassifications.	The study provides four-class classification of CNV, DME, drusen, and normal from OCT with validation on an independent testing set and additional external dataset checks, while also examining how integrating OCT with patient medical history affects sensitivity and specificity and showing occlusion-derived regions consistent with clinically relevant pathology.
[77]: Deep Learning-Based Automatic Detection of Ellipsoid Zone Loss in Spectral-Domain OCT for Hydroxychloroquine Retinal Toxicity Screening	2021	In a retrospective, single-centre case-control ophthalmic study for hydroxychloroquine retinal toxicity screening, the authors used a dual mask region-based convolutional neural network on spectral-domain OCT to detect and quantify ellipsoid zone loss from B-scans and reconstruct an en face EZ-loss map, with 10-fold cross-validation performance evaluated against human ground-truth annotations and American Academy of Ophthalmology (AAO) guideline-based toxicity determinations.	They introduce a combined projection architecture that aggregates parallel horizontal and vertical scan detections to produce more robust en face EZ-loss quantification, enabling fast objective metrics that closely match human grading and support AAO-recommended toxicity screening.
[95]: Deep learning-based classification and segmentation of retinal cavitations on optical coherence tomography images of macular telangiectasia type 2	2022	The study in ophthalmology builds a two-stage convolutional neural network to classify and segment retinal cavitations on spectral-domain OCT B-scans from MacTel2 eyes, using manual segmentations as the reference standard and evaluating the method with reported detection, overlap, and volume-change agreement across longitudinal time points in a clinical trial dataset.	The study advances automated cavitation quantification for MacTel2 by adding a B-scan prescreening classifier before U-Net-based segmentation, improving segmentation accuracy and enabling highly correlated retinal cavitation volume and longitudinal change estimates relative to reader measurements for clinical research use.

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Work	Year	Basis for inclusion	Principal contribution
[374]: Deep learning-based classification of eye diseases using Convolutional Neural Network for OCT images	2023	The authors apply a CNN with transfer learning and fine-tuning to publicly available OCT retinal images for multi-class ophthalmic disease classification, targeting early detection of diabetic macular oedema, choroidal neovascular membranes, and age-related macular degeneration versus normal retina using diagnostic evidence reported from model testing.	The study provides a four-class OCT grading pipeline that couples OCT-specific preprocessing (brightness/-contrast adjustment, multiscale retinex enhancement, and Gaussian blur) with a VGG16-based CNN to improve discrimination among the three disease categories and normal, supported by reported classification performance before and after fine-tuning.
[121]: Deep learning-based detection of diabetic macular edema using optical coherence tomography and fundus images: A meta-analysis	2023	This systematic review and meta-analysis evaluates deep learning-based detection of diabetic macular oedema in ophthalmology using optical coherence tomography and fundus imaging, synthesising sensitivity evidence from 53 studies and summarising model performance reported with established diagnostic metrics (e.g., AUC and sensitivity).	Across large volumes of OCT B-scans and fundus images, it pools results to quantify overall detection sensitivity for OCT versus fundus and discusses risk-of-bias patterns, helping readers interpret evidence strength and relative modality utility for AI-enabled DME screening and referral.
[175]: Deep Learning-Based Detection of Reticular Pseudodrusen in Age-Related Macular Degeneration on Optical Coherence Tomography	2024	The study develops a deep learning model for OCT B-scan-based segmentation of reticular pseudodrusen (RPD) in age-related macular degeneration, with evaluation that includes expert comparisons and external testing on multiple held-out datasets to substantiate segmentation accuracy and clinical usability.	The study presents an automated RPD-segmentation pipeline and releases the code publicly. On the evaluated datasets, the authors report greater agreement with the reference standard than the participating specialists. Downstream clinical reliability was not established.
[108]: Deep Learning-Based Diagnosis of Corneal Condition by Using Raw Optical Coherence Tomography Data	2025	This ophthalmic study applies modified convolutional neural networks to raw anterior-segment OCT corneal images from the Casia2 device to classify each eye into normal, ectasia (keratoconus), or other corneal disease, with diagnostic performance reported on a labelled test set and supportive evidence from Grad-CAM visualisations.	Stacking sixteen unprocessed meridional Casia2 OCT slices supplies the multichannel input for three-class classification. The test-set metrics characterise model performance, and the GUI demonstrates an interface for raw files with the 3dv extension. Neither result alone establishes clinical usability.
[367]: Deep Learning-based Diagnosis of Glaucoma Using Wide-field Optical Coherence Tomography Images	2021	This retrospective ophthalmic deep-learning study evaluates convolutional neural network-based detection and five-stage classification of glaucoma from wide-field swept-source OCT imaging, comparing DL fusion strategies against conventional retinal thickness parameters using reported diagnostic performance measures.	The results indicate that the study's fusion by convolution network and fusion by fully connected network architectures applied to multiple wide-field SS-OCT image types can improve glaucoma discrimination, including early disease, beyond thickness-based conventional metrics, while also providing stage-specific confusion-matrix results.

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Work	Year	Basis for inclusion	Principal contribution
[404]: Deep learning based eye disease classification using Optical Coherence Tomography (OCT) images	2026	Using OCT imaging, the study applies and compares multiple CNN model families (VGG16, VGG19, ResNet50, InceptionV3) for four-class retinal disease classification (CNV, DME, drusen, normal) and evaluates them with internal testing plus an independent external validation set, providing evidence on generalisability for an ophthalmic diagnostic task.	The study identifies VGG19 as the most stable performer across preprocessing and the Synthetic Minority Oversampling Technique (SMOTE) applied to latent features for class imbalance, and it quantifies how performance drops on external data, which is useful for judging readiness and dataset-diversity needs before clinical deployment.
[279]: Deep Learning Based Fluid Segmentation in Retinal Optical Coherence Tomography Images	2019	Using deep learning with an attention mechanism and dense skip connections, the authors segment retinal fluid in OCT imaging for macular oedema, targeting an ophthalmic imaging task that supports quantitative disease monitoring using results reported on a public dataset and described as adaptable across different scanning devices.	The study introduces a single-stage, parameter-efficient attention-based segmentation framework with joint losses to automatically delineate macular oedema fluid regions on OCT, emphasising improved segmentation accuracy and cross-device adaptability for reducing the burden of manual labelling.
[208]: Deep learning-based image quality assessment for optical coherence tomography macular scans: a multicentre study	2024	This multicentre, externally tested deep learning study evaluates imaging-quality assessment for 3D OCT macular scans acquired on Cirrus and Spectralis platforms, using ResNet-18-based architectures to classify scans as gradable versus ungradable across internal and multiple unseen sites, providing evidence to support inclusion of an image-quality gating step prior to downstream automated retinal disease detection.	The study provides a robust, device-specific gradability classifier for filtering ungradable 3D OCT macular volumes and demonstrates how this quality screen can be integrated into an end-to-end automated workflow for more dependable ophthalmic analysis.
[212]: Deep learning based joint segmentation and characterization of multi-class retinal fluid lesions on OCT scans for clinical use in anti-VEGF therapy	2021	This ophthalmic OCT study develops and evaluates an end-to-end deep learning model (RFS-Net) that jointly segments and recognises three retinal fluid lesion types (IRF, SRF, PED) from multi-vendor OCT images for anti-VEGF therapy dosing and activity assessment, with evidence coming from training and validation using publicly available datasets and performance reported with quantitative metrics.	The article specifies an automated, multi-class retinal fluid segmentation approach trained on OCT from Topcon, Cirrus, and Spectralis to reduce reliance on labour-intensive manual delineation and thereby support more consistent lesion characterisation relevant to treatment planning.

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Work	Year	Basis for inclusion	Principal contribution
[289]: Deep-learning based multiclass retinal fluid segmentation and detection in optical coherence tomography images using a fully convolutional neural network	2019	In retinal OCT, the study applies a fully convolutional neural network to multiclass fluid segmentation and detection by using OCT intensity together with retinal layer segmentations from a graph-cut step, with evidence from MICCAI RETOUCH challenge performance reported for both segmentation and detection tasks.	The study introduces an end-to-end OCT fluid labelling pipeline that adds random-forest classification to identify and reject falsely labelled fluid regions, and this design is supported by challenge results that quantify both segmentation accuracy and detection discrimination.
[330]: Deep Learning-Based Neuro Computing to Classify and Diagnosis of Ophthalmology by OCT	2025	Deep learning models are tested for retinal disease screening, lesion classification, and lesion localisation on labelled OCT datasets, including evaluation in an independent patient group.	The study reports model performance for binary lesion presence detection and multiclass multilabel identification of age-related macular degeneration and vitreomacular traction syndrome, with additional lesion-localisation results that support automated screening and diagnostic assistance from OCT.
[243]: Deep Learning Based on Ensemble to Diagnose of Retinal Disease using Optical Coherence Tomography	2021	Using retinal cross-sectional optical coherence tomography images, the study describes an ensemble transfer-learning approach that fuses class-probability outputs from MobileNetV3Large and ResNet50 (with CLAHE pre-processing) to classify five retinal diseases, supported by 5-fold cross-validation and early stopping.	The article details an end-to-end OCT diagnostic pipeline that enhances image contrast and improves prediction stability through probability-level ensemble fusion, offering a practical deep learning strategy for automatic retinal disease classification across five conditions.
[258]: Deep learning-based optical coherence tomography and retinal images for detection of diabetic retinopathy: a systematic and meta analysis	2025	This systematic review and meta-analysis evaluates ophthalmic deep learning models for diabetic retinopathy detection using OCT and colour retinal imaging, pooling diagnostic accuracy evidence (from 47 studies, with 10 contributing to meta-analysis) to quantify performance relevant to clinical screening decisions.	The review synthesises results across large numbers of retinal images and OCT scans to produce pooled diagnostic odds ratios and summary sensitivity and specificity estimates. It thereby provides a quantitative synthesis of the reported diagnostic performance of AI-assisted interpretation of OCT and retinal imaging.
[425]: Deep Learning-Based Prediction of Visual Field Mean Deviation from Numeric OCT Data in Glaucoma	2026	This retrospective glaucoma study trained deep learning models on numeric macular ganglion cell complex (GCC) thickness exported from spectral-domain OCT to predict Humphrey Field Analyzer (HFA) 30-2 mean deviation (MD) in a regression setting and to classify visual field deterioration at predefined MD thresholds using fivefold cross-validation.	The study reports a clinically oriented approach that avoids image rendering by feeding raw 512×128 OCT thickness arrays into multiple CNN and vision-transformer architectures, yielding robust mean deviation estimation and threshold-based discrimination with Grad-CAM-based qualitative explanations.

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Work	Year	Basis for inclusion	Principal contribution
[393]: Deep learning based retinal hard exudates quantification of optical coherence tomography	2025	In this 2025 study, a modified U-Net (DUCK-Net) segments retinal hard exudates on OCT B-scans from diabetic retinopathy and branch retinal vein occlusion. Evaluation includes test-set segmentation accuracy, comparison of OCT-derived exudate area and volume with independent manual measurements, and a 2D projection for visualisation.	Pixel-wise segmentation is combined with volumetric area and volume readouts for automated hard-exudate quantification from OCT. The method reduces manual measurement effort and depicts exudate structure more directly than fundus photography, although cross-device performance was not evaluated.
[378]: Deep Learning-Based Retinal Layer Segmentation in Optical Coherence Tomography Scans of Patients with Inherited Retinal Diseases	2025	Using a U-Net-based deep learning model with adaptive batch normalisation domain adaptation, the authors segment the outer nuclear layer on spectral-domain OCT B-scans from patients with inherited retinal diseases (IRDs), producing retinal and ONL thickness maps with performance evaluated on a test dataset using Dice score metrics.	The study reports reliable ONL segmentation in OCT from pathologically altered eyes with IRDs where existing tools fail and supplies automatically generated full-retina and ONL thickness maps that can support structural assessment for structure-function and treatment-outcome studies.
[286]: Deep learning based retinal OCT segmentation	2019	A fully convolutional network (FCN) with Gaussian-process post-processing segments SD-OCT images from mild non-proliferative diabetic retinopathy cases. Segmentation accuracy is compared with clinician results and other machine-learning baselines.	The study presents a tailored deep learning plus Gaussian-process pipeline for OCT retinal segmentation on a University of Miami dataset and highlights near-human performance with improved mean unsigned error relative to graph-based and other ML methods.
[307]: Deep Learning-Based Retinal OCT Segmentation: A Comparative Study	2025	The study addresses ophthalmic OCT image analysis by performing retinal layer segmentation using CNN, hybrid CNN-graph cut, and Transformer U-Net variants on normal and AMD scans, with evidence based on the reported Dice similarity coefficient (DSC), intersection over union (IoU), precision, and recall on the DUKE OCT dataset.	The study quantifies how Transformer U-Net compares with U-Net and CNN-graph cut approaches for segmentation accuracy and boundary stability across normal versus AMD subsets, providing practical guidance on architecture trade-offs for retinal OCT layer measurement.
[151]: Deep Learning-Based SD-OCT Layer Segmentation Quantifies Outer Retina Changes in Patients With Biallelic RPE65 Mutations Undergoing Gene Therapy	2025	Using deep-learning-based automated segmentation of SD-OCT retinal layers in patients with biallelic RPE65-IRD, the authors quantify outer-retina ellipsoid zone structural biomarkers, including baseline group differences and longitudinal treatment-related changes around voretigene neparvovec, providing evidence from retrospective in-hospital imaging analysis with control comparisons.	The study develops and applies a modified U-Net approach to derive five ellipsoid zone biomarkers (including thickness, granularity, reflectivity/intensity, and EZ area-related measures) with outer-layer distance metrics, enabling objective, repeatable OCT-derived quantification that supports monitoring of disease and treatment response beyond manual segmentation.

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Work	Year	Basis for inclusion	Principal contribution
[283]: Deep learning-based segmentation and quantitative analysis of retinal microstructures in optical coherence tomography angiography images using RS_Unet3+	2026	The study develops a deep learning segmentation framework (RS_Unet3+) for OCTA en face images to quantify retinal vascular network (RVN) and foveal avascular zone metrics, testing the approach on public OCTA vessel segmentation datasets and examining clinically consistent parameter differences in diabetic retinopathy using additional clinical cohorts.	The study reports multiclass RVN and FAZ segmentation with automated extraction of vessel perfusion density (VPD), vessel length density (VLD), fractal dimension (FD), and other vascular parameters and demonstrates that FAZ morphology and vascular metrics vary in a DR cohort in a way that can support diagnosis-related trend analyses.
[354]: Deep learning-based signal-independent assessment of macular avascular area on 6×6 mm optical coherence tomography angiogram in diabetic retinopathy: a comparison to instrument-embedded software	2023	This clinical study evaluates a deep-learning method for signal-independent quantification of macular extrafoveal avascular area from 6×6 mm OCT angiography in diabetic retinopathy, comparing it with instrument-embedded vessel density software to predict DR severity using evidence from a cross-sectional dataset with diagnostic accuracy and correlation analyses.	The study reports that MEDnet-based extrafoveal avascular area measured on the superficial vascular complex yields higher sensitivity for distinguishing diabetes and DR stages from controls at fixed specificity and shows minimal dependence on signal strength and shadow artefacts compared with commercial vessel density, supporting more robust OCTA-based DR assessment.
[48]: Deep Learning Based Sub-Retinal Fluid Segmentation in Central Serous Chorioretinopathy Optical Coherence Tomography Scans	2019	The article describes a fully convolutional neural network that performs automated sub-retinal fluid segmentation on OCT volumes from patients with central serous chorioretinopathy, tackling noise and motion artefacts and spatial variability typical of this imaging-based task, with evidence provided as reported test-set segmentation accuracy via Dice, precision, and recall.	The study proposes an end-to-end learning approach tailored to differentiate subtle sub-retinal fluid patterns in CSC OCT and reports quantitative test-set results on 15 OCT volumes, indicating practical effectiveness for pixel-level fluid delineation despite imaging variability.
[182]: Deep Learning Classification of Drusen, Choroidal Neovascularization, and Diabetic Macular Edema in Optical Coherence Tomography (OCT) Images	2023	A 2023 deep learning study classifies drusen, choroidal neovascularisation, and diabetic macular oedema from OCT images and reports performance on a labelled dataset of diseased and healthy scans.	The study trains a CNN-based supervised model on a multiclass OCT dataset and reports test-set precision, recall, sensitivity, specificity, F1 score, and accuracy to support automated retinal pathology detection, while noting the need for external validation and clinical workflow evaluation.

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Work	Year	Basis for inclusion	Principal contribution
[344]: Deep Learning Classification on Optical Coherence Tomography Retina Images	2020	A VGG16 transfer-learning network classifies greyscale OCT scans, converted to red–green–blue (RGB) images with colour maps, into three retinal-disease classes and a normal class. Test accuracy provides the reported evidence.	The article details an end-to-end OCT retinal image classification pipeline with data augmentation to mitigate limited training data and evaluates a VGG16-based model to achieve 99.48% testing accuracy across all four classes, highlighting feasibility for automated diagnostic categorisation.
[416]: Deep learning detection of retinal detachment: Optical coherence tomography staging and estimation of duration of macular detachment	2025	This retrospective deep learning study uses transfer-learned 2D and 3D OCT imaging to classify rhegmatogenous retinal detachment (RRD), including macula status, six-stage OCT morphology, and a surrogate estimate of macula-off duration, with evaluation based on ROC and precision-recall performance on a single-centre dataset of RRD eyes versus controls.	The results indicate that a 3D OCT model can stratify macula-on versus macula-off RRD and discriminate short versus longer macula-off duration, while a 2D model supports automated stage classification and provides Grad-CAM feature maps to visualise model attention linked to clinically relevant staging associations.
[76]: Deep learning-driven automated quality assessment of ultra-widefield optical coherence tomography angiography images for diabetic retinopathy	2025	A ViT-based transfer learning framework was developed for automated image quality assessment of ultra-widefield OCTA in diabetic retinopathy, using UW-OCTA images (12×12 mm) fine-tuned from pre-training on $6{\times}6$ mm OCTA and evaluated with AUC and quadratic weighted Kappa plus ablations on a labelled DRAC2022 test set.	Pre-training on narrower-field OCTA and transformer attention enabled quality grading with scarce UW-OCTA data. On the held-out test set, the model exceeded the selected ResNet baselines in AUC and Kappa, while heatmaps and ablations characterise model behaviour but do not establish clinical usability.
[362]: Deep learning for objective OCTA detection of diabetic retinopathy	2020	The authors apply a transfer-learning convolutional neural network using VGG16 to perform OCTA-based diabetic retinopathy detection and multi-class classification (control, no DR, and nonproliferative DR), with augmentation and 5-fold cross-validation as evidence from preliminary dataset evaluation.	The study reports how fine-tuning nine layers of the VGG16 model improves OCTA image classification for DR staging, providing early evidence for objective, OCTA-driven deep learning in ophthalmic screening.
[336]: Deep learning for quality assessment of retinal OCT images	2019	The study develops a deep-learning OCT image quality assessment (OCT-IQA) pipeline for ophthalmic retinal B-scan images, using transfer-learned CNNs (VGG-16, Inception-V3, ResNet-18, ResNet-50) to classify scan quality based on signal completeness, location, and effectiveness, with evidence from five-fold cross-validation plus downstream retinopathy detection testing showing utility of the quality pre-filtering step.	Using the best-performing ResNet-50 OCT-IQA model, the authors compared a filtered “pure” high-quality set with a “mixed” set containing poor-quality scans. The filtered data yielded higher reported retinopathy-detection accuracy and AUC.

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Work	Year	Basis for inclusion	Principal contribution
[112]: Deep Learning for Retina Structural Biomarker Classification Using OCT Images	2024	A multimodal, multitask framework combines 3D OCT with one-dimensional clinical data and supervised and semi-supervised learning to predict six retinal structural biomarkers. Evaluation uses macro F1-score and a baseline comparison.	The study introduces a custom post-processing step that improves biomarker detection precision and demonstrates a 14.48% macro F1-score gain over baseline for six retinal structural biomarkers, supporting the utility of post-processing-enhanced OCT-based deep learning for diagnostic decision support.
[272]: Deep Learning for Retinal OCT Disease Classification using EfficientNetB0: A Transfer Learning Approach	2025	Using transfer learning with EfficientNet (EfficientNetB7), the authors classify retinal OCT images into CNV, DME, drusen, and normal, targeting early and more consistent ocular diagnosis despite observer variability, with evidence reported from experimental evaluation on the OCT2017 dataset.	The article outlines an end-to-end automated OCT disease classifier that incorporates preprocessing to handle scan-quality and demographic variation and reports balanced cross-class performance using confusion-matrix analysis, which is valuable for dependable clinical triage across visually similar findings.
[253]: Deep Learning for Segmentation of Optic Disc and Retinal Layers in Peripapillary Optical Coherence Tomography Images	2023	Using peripapillary spectral-domain OCT imaging, the study applies a deep-learning approach with transfer learning to automatically segment nine retinal layers and the optic disc centred in the peripapillary region, then derives layer thickness measurements along B-scans, with evidence reported as Dice overlap in both a public dataset and a depth-enhanced dataset.	The study reports an end-to-end optic-disc-centred OCT segmentation and thickness quantification pipeline with visualisation outputs, achieving a Dice score of 83.6% for nine-layer and optic-disc delineation across both image sets, which supports practical automated analysis of retinal layer structure.
[196]: Deep learning for the segmentation of preserved photoreceptors on en face optical coherence tomography in two inherited retinal diseases	2018	A patch-based convolutional neural network segments preserved ellipsoid zone regions in en face OCT images from patients with choroideremia or retinitis pigmentosa, producing pixelwise masks that are compared with expert manual grading.	The study provides a versatile CNN segmentation workflow that converts B-scan patch classifications into en face preserved photoreceptor maps and removes vessel-shadow artefacts, demonstrating automated quantification of preserved photoreceptors with agreement to manual labels in both included IRDs.
[249]: Deep Learning Image Analysis of Macular Optical Coherence Tomography Angiography Images for Detection of Progression in Glaucoma	2024	The authors apply a CNN-based longitudinal OCTA image analysis in glaucoma, using macular microvascular density derived from thick-vessel segmentation, thin-vessel masking, Otsu thresholding, and superpixel measurement to distinguish progressive versus stable eyes on baseline and follow-up scans, with evaluation framed by patient-level progression detection evidence.	The study provides a five-stage processing pipeline that converts serial macular OCTA into superpixelised thin-vessel density measurements and uses t-value patterning to separate progressive from stable glaucoma, offering a practical framework for tracking disease-related vascular dropout over time.

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Work	Year	Basis for inclusion	Principal contribution
[220]: Deep Learning Image Analysis of Optical Coherence Tomography Angiography Measured Vessel Density Improves Classification of Healthy and Glaucoma Eyes	2022	Using a VGG16 convolutional neural network on en face radial peripapillary capillary OCTA vessel density images, and five-fold cross-validation against feature-based gradient-boosting classifiers derived from instrument OCTA/OCT vessel density and OCT RNFL thickness, this ophthalmic diagnostic study evaluates healthy versus glaucomatous eyes with adjusted precision-recall evidence to compare diagnostic accuracy across method families.	The results indicate that end-to-end deep learning on 4.5 mm by 4.5 mm OCTA en face vessel density maps improves glaucoma classification over gradient-boosting models built from standard OCTA and OCT measurements, including comparisons that remain consistent in an early glaucoma subgroup.
[94]: Deep learning is effective for the classification of OCT images of normal versus Age-related Macular Degeneration	2017	This electronic medical record (EMR)-linked deep learning study in ophthalmology classifies central macular spectral-domain OCT images into normal versus age-related macular degeneration, evaluating performance with ROC analysis at image, macular scan, and patient levels to provide evidence for automated OCT screening and computer-aided diagnosis workflows.	The study trains a modified VGG16 convolutional neural network on a large OCT dataset extracted from Epic, then reports how aggregating predictions at the scan and patient levels improves discrimination and uses an occlusion test to identify regions driving the AMD calls.
[53]: Deep learning method for semi-automated segmentation of optic nerve head tissues in optical coherence tomography images	2025	For OCT imaging of optic nerve head (ONH) tissues, the study evaluates a CNN pipeline that segments the anterior lamina cribrosa surface, Bruch's membrane (BM), and choroid–scleral interface (CS) from SD-OCT radial scans. The user-in-the-loop workflow allows correction and iterative model updating. On the held-out test set, boundaries were compared with manual markings using RMSE and, for Bruch's membrane opening (BMO), Euclidean distance.	The study provides a semi-automated, interactive workflow that lets users accept or adjust predicted boundary markers and then uses the corrected masks to continue training the model, yielding improved BM and CS predictions in the reported test set and enabling boundary-level error quantification with time savings versus fully manual segmentation.
[173]: Deep Learning Model for the Detection of Corneal Edema Before Descemet Membrane Endothelial Keratoplasty on Optical Coherence Tomography Images	2022	This retrospective ophthalmology study evaluates an OCT-based deep-learning pipeline that segments corneal scans at pixel level and classifies normal tissue versus oedema. It used oedema fraction from 1,992 preoperative corneal OCT images to detect minimal corneal swelling before Descemet membrane endothelial keratoplasty (DMEK) in patients with and without Fuchs endothelial corneal dystrophy (FECD). Performance was evaluated against objective differential central corneal thickness (DCCT) thresholds using the AUC.	The study provides an automated, quantitative decision-support measure for pre-DMEK assessment by mapping epithelial and stromal oedema regions and showing strong discrimination of clinically meaningful oedema (DCCT of 20 µm) and consistent thresholds across patient groups, supporting objective detection of subclinical corneal oedema.

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Work	Year	Basis for inclusion	Principal contribution
[246]: Deep learning model to predict visual field in central 10° from optical coherence tomography measurement in glaucoma	2021	In open-angle glaucoma care, this research uses a convolutional neural network to predict HFA 10-2 threshold sensitivities within the central 10° visual field from spectral-domain OCT macular layer thickness. Accuracy is evaluated on an external test set against support vector machine and multiple linear regression baselines.	An externally tested CNN estimates whole-field and point-wise 10-2 threshold sensitivities from SD-OCT-derived layer thickness, with lower reported error than the conventional machine-learning and regression comparators.
[129]: Deep learning network (DL-Net) based classification and segmentation of multi-class retinal diseases using OCT scans	2024	The authors apply ophthalmology-oriented deep learning to volumetric retinal OCT scans to automate multi-class recognition and lesion delineation across common macular disease categories (normal, DME, CNV, AMD, and drusen), providing evidence of effectiveness via reported classification and segmentation performance compared with other OCT frameworks using publicly available datasets.	The study proposes a multi-scale DL-Net that modifies ResNet-50 and SqueezeNet for OCT-based multi-class classification and couples convolutional feature aggregation with a Bi-LSTM-based recurrent convolutional segmentation approach, offering an end-to-end diagnostic pipeline intended to improve OCT interpretation for clinical decision support.
[78]: Deep learning network for parallel self-denoising and segmentation in visible light optical coherence tomography of the human retina	2023	For visible-light optical coherence tomography (VIS-OCT) of the retina, the study evaluates a co-learning deep network that denoises speckle-laden B-scans and segments 10 retinal layers from the same input. It uses paired noisy and clean data with boundary annotations and reports evidence from repeated experiments and cross-protocol generalisation.	The study introduces a VIS-OCT dataset with 105 paired noisy and clean B-scans and 10 manually delineated layer boundaries. Parallel self-denoising improves layer segmentation, particularly when only 25% of segmentation labels are available, while remaining robust when applied to a different scanning protocol.
[227]: Deep learning network with differentiable dynamic programming for retina OCT surface segmentation	2023	In this 2023 study, a U-Net feature learner is coupled to constrained differentiable dynamic programming (DDP) to enforce smooth surface transitions across B-scan columns in retinal OCT data. External evaluation on the Duke AMD and Johns Hopkins University (JHU) multiple sclerosis (MS) datasets assesses subvoxel segmentation accuracy.	The study integrates DDP into segmentation-network training to regularise global surface smoothness. The DDP-constrained model improves segmentation accuracy over a U-Net baseline and an ablation without DDP on both AMD and MS retinal-layer surfaces.

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Work	Year	Basis for inclusion	Principal contribution
[59]: Deep learning of fundus images and optical coherence tomography images for ocular disease detection – a review	2024	This review is relevant to ophthalmic practice by synthesising deep learning methods for ocular disease detection and region delineation using fundus and optical coherence tomography imaging, addressing tasks such as diabetic retinopathy classification/grading and glaucoma detection as well as vessel, optic disc, and optic cup segmentation, with evidence drawn from reported literature rather than new experimental results.	The review synthesises convolutional, recurrent, generative adversarial, U-Net and Y-Net, and transformer-based approaches across multiple imaging types and task settings, highlighting architectural variants that improved on baselines and offering a unified reference that spans beyond single-disease or single-task reviews.
[461]: Deep Learning Techniques for Diabetic Retinopathy Diagnosis using Optical Coherence Tomography: A Review	2022	This review synthesises deep-learning techniques for diabetic retinopathy from optical coherence tomography imaging, covering OCT-based detection and classification as well as retinal-layer segmentation, with evidence drawn from method reports published over the prior four years.	The study gathers and summarises OCT-oriented model approaches and the diagnostic features used for diabetic retinopathy, clarifying how segmentation of retinal layers is leveraged within classification-oriented workflows.
[356]: Deep learning to detect macular atrophy in wet age-related macular degeneration using optical coherence tomography	2023	This ophthalmic deep learning study uses a U-Net-based CNN to segment and quantify optical coherence tomography B-scans from patients with wet age-related macular degeneration for automatic detection of six OCT features of macular atrophy defined by CAM criteria, with model performance reported using standard validation metrics against expert manual annotations.	The fully automated, one-vs-all pipeline produces pixel-level measurements for interrupted outer retina, interrupted RPE, absence of outer retina, absence of RPE, and hypertransmission thresholds. It offers a reproducible OCT workflow for lesion identification in wet AMD, although validation was internal only.
[310]: Deep Learning to Detect OCT-derived Diabetic Macular Edema from Color Retinal Photographs: A Multicenter Validation Study	2022	This retrospective multicentre study validates a deep learning system for detecting centre-involving diabetic macular oedema from 2D colour fundus photographs, using OCT-derived retinal thickening and intraretinal fluid as the reference standard and expert grading as the comparator across international DR screening and clinic datasets.	The study provides OCT-labelled, device-thresholded evidence that the model achieves higher specificity with noninferior sensitivity than expert graders across multiple populations, supporting potential use to reduce false-positive referrals pending prospective evaluation.
[273]: Deep Learning Using Preoperative Optical Coherence Tomography Images to Predict Visual Acuity Following Surgery for Idiopathic Full-Thickness Macular Holes	2024	Using fully automated deep learning on preoperative 3D SD-OCT imaging in an ophthalmic surgical setting, the authors predict postoperative visual acuity for idiopathic full-thickness macular holes, with quantitative evaluation of multiple model variants after preprocessing, quality assurance, and anomaly detection to support robustness to image-quality variability.	The quality-aware workflow ingests seven central SD-OCT images per patient and compares nine contemporary networks for estimating visual acuity in ETDRS letters. The reported prediction errors quantify benchmark feasibility. Automated postoperative prognostication requires prospective clinical validation.

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Work	Year	Basis for inclusion	Principal contribution
[277]: Deep learning visual field global index prediction with optical coherence tomography parameters in glaucoma patients	2023	This retrospective cross-sectional glaucoma study uses a deep neural network to estimate three global visual-field indices from spectral-domain OCT. These indices are MD, pattern standard deviation (PSD), and visual field index (VFI). Inputs comprise Bruch's membrane opening–minimum rim width and retinal nerve fibre layer parameters. Fivefold cross-validation assesses prediction error and correlation. The stated aim is to support care when visual-field testing is delayed.	Joint modelling yields estimates for all three VF global indices from OCT structural parameters. Cross-validated errors and correlations quantify structure-function prediction on the study data. Use as a surrogate for standard automated perimetry was not tested.
[133]: Deep learning with adaptive convolutions for classification of retinal diseases via optical coherence tomography	2024	In this 2024 study, a CNN classification pipeline combines Zeckendorf-theorem-based indirect noise reduction with a texture-sensitive adaptive convolution layer. Accuracy is compared across seven OCTNet-style architectures on the UCSD OCT dataset.	The study introduces a preprocessing-plus-adaptive-convolution substitute for standard convolution in OCT classification and quantifies modest but consistent accuracy improvements (0.44% to 2.44%) to support the practical value of the denoised base/fine representation and feature-content-driven adaptive filtering.
[448]: Deep Learning with Disc Photos or OCT Scans in Glaucoma Detection	2025	Using a retrospective tertiary-glaucoma dataset, the study compares deep learning models trained on OCT RNFLT maps versus disc photos to detect glaucoma defined by functional visual-field impairment, evaluated with AUC (including performance across demographic groups), providing diagnostic evidence relevant to ophthalmic screening workflows.	The reported results indicate that OCT RNFLT-based deep learning achieves higher glaucoma detection accuracy than disc photo-based modelling while maintaining broadly consistent AUC across demographic groups, informing modality choice and highlighting where dataset equity may be necessary.
[366]: Deep Neural Network for Scleral Spur Detection in Anterior Segment OCT Images: The Chinese American Eye Study	2020	This population-based anterior segment OCT study uses a ResNet-18 CNN to localise scleral spurs by predicting X- and Y-coordinate positions in AS-OCT images, and it validates performance against a human expert reference using prediction error distributions compared with intragrader variability.	By showing that CNN localisation errors were statistically similar to the reference grader's repeatability (and lower than intergrader variability), the study provides automated scleral spur detection that can remove manual, grader-dependent steps for AS-OCT analysis in angle-closure assessment.
[73]: Deep Neural Network Regression for Automated Retinal Layer Segmentation in Optical Coherence Tomography Images	2020	The study proposes an OCT-specific deep neural network regression approach that uses intensity, gradient, and adaptive normalised intensity score features to predict retinal boundary pixels for layer segmentation, with evidence from quantitative comparison to manual references on 114 images using Dice and pixel-distance metrics.	Boundary-wise regression avoids classification-style labelling and uses intensity-derived features to estimate retinal surfaces. The authors report results under low contrast, speckle noise, intensity variation, and vessel interference together with training and processing times. These controlled tests do not establish general robustness.

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Work	Year	Basis for inclusion	Principal contribution
[100]: Deep Neural Networks for Automated Outer Plexiform Layer Subsidence Detection on Retinal OCT of Patients With Intermediate AMD	2024	A deep-learning system localises and scores OPL subsidence on Spectralis retinal OCT volumes. Development data and an independent external dataset support evaluation by cross-validation and ROC, precision–recall (PR), and free-response receiver operating characteristic (FROC) analyses.	The study advances an object-detection and B-scan presence-scoring pipeline that merges bounding boxes across neighbouring B-scans into volume-level predictions. It enables rapid, screening-oriented assessment of OPL subsidence, with validated external generalisation and a quantified low false-positive burden.
[130]: DeepRetina: Layer Segmentation of Retina in OCT Images Using Deep Learning	2020	In an ophthalmic optical coherence tomography setting, this paper introduces an encoder-decoder deep learning segmentation method with an improved Xception65 backbone and atrous spatial pyramid pooling to automatically delineate a 10-layer retinal structure, with quantitative evidence from evaluation on a retinal OCT database including both healthy and disease images.	DeepRetina combines an Xception-based encoder-decoder with atrous spatial pyramid pooling for retinal-layer segmentation. The database results quantify per-layer accuracy and processing efficiency. Utility in clinical workflows was not evaluated.
[223]: Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI	2021	The study addresses retinal OCT disease classification in four diagnostic categories using a compact 6-layer CNN with LIME- and Grad-CAM-based explainability, providing clinician-facing interpretability alongside an evaluation protocol and reported classification outcomes.	The compact CNN reduces model size and is accompanied by LIME attributions and Grad-CAM visualisations. These methods expose image regions associated with predictions. They do not demonstrate real-time deployment, clinician comprehension, or clinical usefulness.
[178]: Detecting glaucoma based on spectral domain optical coherence tomography imaging of peripapillary retinal nerve fiber layer: a comparison study between hand-crafted features and deep learning model	2020	A pretrained CNN detects glaucoma from SD-OCT peripapillary RNFL images. Held-out AUC is compared with handcrafted sectoral and mean pRNFL thickness measures.	Direct comparison with handcrafted pRNFL features showed higher held-out ROC performance for the deep learning model than for conventional thickness parameters. This is a benchmark advantage. Practical screening value was not evaluated.
[165]: Detecting Retinal Nerve Fibre Layer Segmentation Errors on Spectral Domain-Optical Coherence Tomography with a Deep Learning Algorithm	2019	This ophthalmic study develops a supervised deep learning classifier that analyses spectral-domain OCT peripapillary circle B-scans to identify retinal nerve fibre layer segmentation errors, using human grader labels as the reference standard and reporting diagnostic evidence via ROC analysis and accuracy against the test set.	Trained on 25,250 SD-OCT B-scans, the probability-based detector flags mild and severe RNFL segmentation errors against human grading. This provides a testable quality-assurance signal. Whether its use reduces unreliable measurements in practice was not evaluated.

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Work	Year	Basis for inclusion	Principal contribution
[72]: Detection of Choroidal Neovascularization (CNV) in Retina OCT Images Using VGG16 and DenseNet CNN	2022	VGG16 and DenseNet are compared for choroidal neovascularisation detection from retinal OCT/OCTA, followed by segmentation of regions of interest. Reported accuracy assesses the classification stage.	The study refines an end-to-end OCT/OCTA CNV workflow by comparing VGG16 versus DenseNet and then applying region segmentation via OpenCV to distinguish diseased images from background, reporting higher performance for VGG16.
[104]: Detection of diabetic macular edema in optical coherence tomography scans using patch based deep learning	2018	The study targets ophthalmic screening of diabetic macular oedema by analysing optical coherence tomography scans using a patch-based deep learning pipeline that integrates image-processing candidate detection with CNN patch labelling and rule-based confidence aggregation to produce scan-level DME predictions.	The article specifies a two-stage OCT framework that first localises potential fluid and hard exudate regions and then uses a deep convolutional neural network to label patches, enabling automated DME detection from scans without relying on manual annotation at inference.
[351]: Detection of glaucoma progression on longitudinal series of en-face macular optical coherence tomography angiography images with a deep learning model	2024	In glaucoma monitoring, the research evaluates a custom convolutional neural network that uses longitudinal superficial en-face macular OCTA vessel-density patterns (baseline versus final scans aligned to the foveal avascular zone) to classify statistically defined 24-2 visual field mean deviation progression, reporting test-set discrimination via AUC, sensitivity, specificity, and accuracy with a logistic regression comparator and validation using confusion-matrix-based metrics.	Spatiotemporal learning from sequential macular OCTA images yielded better reported progression discrimination than logistic regression without explicit vessel-density rate features. Gradient-based maps localised model attention around microvascular loss and foveal avascular-zone changes, offering a plausibility check rather than biological validation.
[237]: Detection of Macular Neovascularization in Eyes Presenting with Macular Edema using OCT Angiography and a Deep Learning Model	2025	In this prospective cross-sectional ophthalmic study, a hybrid multitask convolutional neural network was evaluated for macular neovascularisation (MNV) detection and segmentation using OCTA and structural OCT. The cohort included eyes with macular oedema due to exudative AMD, diabetic macular oedema, or retinal vein occlusion. Diagnostic classification and segmentation quality were reported using standard sensitivity, specificity, IoU, and F1 metrics.	The authors report the best aiMNV performance on denser 3x3-mm OCTA acquisitions, together with automated MNV membrane-area segmentation in the macular-oedema cohort. Real-world sampling broadens the evaluation but does not establish clinical utility for distinguishing exudative AMD MNV with coexisting IRF or SRF.

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Work	Year	Basis for inclusion	Principal contribution
[132]: DETECTION OF MORPHOLOGIC PATTERNS OF DIABETIC MACULAR EDEMA USING A DEEP LEARNING APPROACH BASED ON OPTICAL COHERENCE TOMOGRAPHY IMAGES	2021	For pattern-level DME classification, the authors develop a deep learning model for ophthalmic OCT image analysis that classifies three morphologic diabetic macular oedema patterns (diffused retinal thickening, cystoid macular oedema, serous retinal detachment), using internal 5-fold cross-validation and external validation plus occlusion-based testing to provide evidence of model performance and interpretability.	The framework provides automated, pattern-level OCT detection with multilabel grader-derived references and uses occlusion maps to identify the image regions most critical for accurate recognition, which may help standardise DME assessment and support treatment planning.
[285]: Detection of Nonexudative Macular Neovascularization on Structural OCT Images Using Vision Transformers	2022	A vision-transformer framework is evaluated on structural OCT B-scans, segmenting the double-layer sign (DLS) and drusen and then producing an eye-level classification of nonexudative macular neovascularisation, with performance validated against human-graded labels from a held-out set and reported using diagnostic metrics and intergrader agreement.	The segmentation-to-projection-to-classification pipeline derives the neMNV decision from structural OCT alone and reports diagnostic detection performance and kappa-based concordance with a senior human grader, supporting transformer-based detection of DLS-associated neMNV in AMD.
[303]: Detection of Retinal Disease from Optical Coherence Tomography Images Using CNN Models	2024	The evaluation applies convolutional neural networks on optical coherence tomography to perform four-class classification of retinal disease states (DME, drusen, CNV, and normal), providing evidence from an imaging-based, ophthalmology-relevant diagnostic task with comparative AI model evaluation.	The authors present a proposed LayerV variant of VGG16 and report higher accuracy (97.45%) relative to other tested CNN architectures, indicating potential as an OCT-based automated screening tool for differentiating major retinal conditions.
[164]: Detection of Retinal Disorders in Optical Coherence Tomography using Deep Learning	2019	In this 2019 four-class macular screening study, the authors apply DenseNet deep learning to macular optical coherence tomography for screening and detecting normal retina versus CNV, DME, and drusen, with evidence reported via training, validation, and test classification metrics and AUC.	The reported experiments show how DenseNet can be parameter-tuned and evaluated for multi-class disorder recognition on OCT, providing quantitative performance results that support further evaluation as an automated screening aid for treatable retinal conditions.
[382]: Development and validation of a 3-D deep learning system for diabetic macular oedema classification on optical coherence tomography images	2025	Across multiple centres and OCT platforms, the authors build and externally validate a 3D OCT deep learning classifier that grades diabetic macular oedema into no DME, non-centre-involved DME, and centre-involved DME, using retrospective, cross-sectional annotated volumes and evaluating diagnostic performance against retina experts in a screening-relevant ophthalmic setting.	Training on 3D CNN features from multiple OCT device brands, followed by centre-split external testing, provides automated DME localisation with human-comparable agreement and quantitative ROC metrics, informing potential deployment for population-based DME screening.

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Work	Year	Basis for inclusion	Principal contribution
[394]: Development and validation of a pixel wise deep learning model to detect cataract on swept-source optical coherence tomography images	2022	In a retrospective SS-OCT cohort, the authors use a U-Net convolutional segmentation model on swept-source OCT to perform pixel-wise cataract detection and localisation, outputting per-pixel cataract probability maps summarised by Cataract Fraction and evaluated with ROC AUC, sensitivity, and specificity on a held-out validation set.	The principal methodological contribution is an SS-OCT-based, automated, interpretable probability mapping framework that converts lens imaging into quantitative Cataract Fraction to support objective cataract assessment and help guide clinical diagnosis and surgical planning.
[117]: DEVELOPMENT AND VALIDATION OF AN EXPLAINABLE ARTIFICIAL INTELLIGENCE FRAMEWORK FOR MACULAR DISEASE DIAGNOSIS BASED ON OPTICAL COHERENCE TOMOGRAPHY IMAGES	2022	A two-stage AI pipeline is evaluated on optical coherence tomography images to classify nine retinal lesions at the image level and produce an explainable eye-level diagnosis of three macular diseases, validating both levels against human expert readers.	The authors develop an integrated deep learning plus random-forest framework that links detected central macular lesions to interpretable disease predictions, demonstrating test performance on lesion labelling and eye-level accuracy relative to four ophthalmic experts.
[134]: Development of a deep learning system to detect glaucoma using macular vertical optical coherence tomography scans of myopic eyes	2023	For glaucoma detection in myopic eyes, the authors developed and validated a CNN-based deep learning glaucoma classifier using macular vertical SD-OCT B-scans from myopic eyes, and compared its discrimination against circumpapillary OCT while including an external test set, showing strong performance especially in eyes with large myopic parapapillary atrophy.	Comparison of macular vertical and circumpapillary OCT inputs, with demographic and ophthalmic features added for refinement, indicates an evidence-based OCT imaging approach that can better support glaucoma detection in challenging high-myopia anatomy.
[44]: Diagnosis of Retinal Diseases Based on Bayesian Optimization Deep Learning Network Using Optical Coherence Tomography Images	2022	The authors investigate ophthalmic OCT image-based classification of retinal diseases using transfer learning with pretrained CNNs (VGG16, DenseNet201, InceptionV3, Xception), paired with image augmentation and Bayesian hyperparameter optimisation, and report diagnostic performance across an eight-category setup (seven diseases plus normal) using validation and test results derived from the model evaluation metrics.	The evaluation compares feature-extractor and fine-tuning strategies for multiple CNN backbones while tuning classifier hyperparameters via Bayesian optimisation, providing test accuracies and class-wise precision, recall, and F1-scores and reporting that DenseNet201 fine-tuning achieves the highest reported performance for the seven-disease versus normal OCT diagnosis setting.

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Work	Year	Basis for inclusion	Principal contribution
[138]: Diagnosis of retinal diseases using the vision transformer model based on optical coherence tomography images	2023	For four-class retinal diagnosis, the authors use a vision transformer to classify retinal conditions from OCT images into normal and three abnormal classes (CNV, drusen, and diabetic macular oedema) and report diagnostic evidence via accuracy, sensitivity, and specificity.	The reported results indicate that the proposed transformer-based multi-class OCT classifier achieves high reported diagnostic metric values for CNV, DME, and drusen and compares favourably against traditional CNN approaches, supporting further evaluation of transformers for retinal disease diagnosis.
[148]: Diagnosis of Retinal Disorders from Optical Coherence Tomography Images using CNN	2025	The authors assess a lightweight CNN for automated retinal disorder diagnosis from optical coherence tomography imaging, performing multiclass classification of choroidal neovascularisation, diabetic macular oedema, drusen, and normal eyes with reported diagnostic performance metrics.	OpticNet-71 uses depthwise separable convolutions, residual blocks, and an encoder-decoder design to reduce computational cost and reduce inference demands for potential real-time use while delivering high accuracy on the specified retinal categories.
[449]: Diagnostic ability of macular microvasculature with swept-source OCT angiography for highly myopic glaucoma using deep learning	2023	For highly myopic glaucoma, the authors evaluate swept-source OCT angiography and swept-source OCT macular imaging to distinguish highly myopic glaucoma from healthy high myopia, using ViT-based classification with ROC analysis across superficial capillary plexus, deep capillary plexus, and ganglion cell layer thickness parameter sets, with both internal testing and external validation reported via AUC.	The ViT model trained on macular OCTA superficial capillary plexus images achieved glaucoma discrimination comparable to macular ganglion cell layer thickness measures and better than deep capillary plexus, supporting macular OCTA microvasculature as a potentially useful biomarker for glaucoma detection in high myopia.
[68]: Diagnostic performance and generalizability of deep learning for multiple retinal diseases using bimodal imaging of fundus photography and optical coherence tomography	2025	Across multiple centres, the study evaluates deep learning for detecting and grading seven retinal conditions using bimodal colour fundus photography and optical coherence tomography, employing multimodal multi-instance learning (MIL) fusion and reporting AUC-based diagnostic performance with external testing across different devices and scanning protocols in separate hospitals and dataset splits.	The comparative evaluation selects a bimodal Fusion-MIL model that improves multi-disease discrimination over single-modality baselines and extends to detailed pathologic myopia atrophy, traction, and neovascularisation (ATN) component grading, reporting cross-setting performance and heatmap-based localisation across heterogeneous imaging settings.
[52]: DRUNET: a dilated-residual U-Net deep learning network to segment optic nerve head tissues in optical coherence tomography images	2018	For glaucoma-focused OCT analysis, DRUNET, a dilated-residual U-Net, segments multiple optic nerve head tissue layers from Spectralis B-scan images, enabling automatic extraction of clinically relevant structural measures, with performance evaluated against expert manual segmentations using overlap and classification metrics.	The end-to-end framework isolates six neural and connective ONH tissues and converts the segmentations into six structural parameters. It reduced parameter-extraction error relative to a prior patch-based approach on the evaluated data.

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Work	Year	Basis for inclusion	Principal contribution
[368]: Dry age-related macular degeneration classification from optical coherence tomography images based on ensemble deep learning architecture	2024	For dry AMD staging, the authors develop an ensemble deep learning classifier that combines multiple pre-trained CNNs to categorise four OCT-derived stages (normal, drusen, nascent geographic atrophy, and geographic atrophy), using three-fold cross-validation to evaluate early-stage recognition, particularly nascent geographic atrophy, in retinal B-scan OCT images from a hospital dataset.	The weighted ensemble combines ResNet50, EfficientNetB4, MobileNetV3, and Xception on non-segmented OCT. In three-fold cross-validation, it improved discrimination of nascent geographic atrophy relative to the constituent models, while Grad-CAM visualised highlighted features. These results support dataset-specific staging performance but do not establish clinical feasibility.
[405]: Dry and Wet Age-Related Macular Degeneration Classification Using OCT Images and Deep Learning	2019	The authors evaluate deep convolutional neural networks, including ResNet, on OCT imaging to classify dry versus wet age-related macular degeneration and to support evidence in the form of reported stage-wise ROC-based performance from model evaluation.	The reported comparison indicates that an 18-layer ResNet improves classification relative to AlexNet for dry and wet AMD and provides stage-specific performance values, which matters for automated, time-sensitive triage towards appropriate treatment selection.
[371]: DTG: Dual transformers-based generative adversarial networks for retinal 2D/3D OCT image classification	2026	The evaluation examines a dual-transformer, GAN-based deep learning approach for classifying retinal disorders from 2D and 3D OCT images, using patient instance-based data augmentation and reporting results on real-world datasets with multiple classification metrics.	The integration in DTG of Vision Transformer and multiscale Vision Transformer encoders with a GAN-driven semantic representation step, followed by weighted classification, targets data scarcity in OCT and is shown to outperform prior 2D/3D CNN and transformer baselines on retinal disease classification.
[168]: Dual-input convolutional neural network for glaucoma diagnosis using spectral-domain optical coherence tomography	2021	The authors evaluate a VGG16-based dual-input convolutional neural network that ingests SD-OCT RNFL and ganglion cell–inner plexiform layer (GCIPL) inputs to perform glaucoma diagnosis, reporting test-set evidence for separating normal versus glaucoma and early-stage versus normal using accuracy and AUC.	The comparison indicates that jointly using RNFL and GCIPL thickness information within a dual-input architecture improves classification of early-stage primary open-angle glaucoma versus normal compared with single-input approaches, providing evidence for automated SD-OCT discrimination.
[216]: Dual-Stream CNN-ViT Framework for Interpretable Glaucoma Detection from OCT and Fundus Images	2025	A multimodal, interpretable model is developed for glaucoma stage classification by fusing OCT and fundus images in a dual-stream CNN-ViT architecture and is evaluated on public datasets while providing SHAP-based regional explanations that are reported to correlate with established retinal markers.	The OCT-fundus fusion model outputs Healthy, Mild, and Severe glaucoma stage predictions together with SHAP attributions linked to clinically meaningful features such as RNFL thinning and optic disc cupping, supporting transparent use for screening-oriented workflows.

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Work	Year	Basis for inclusion	Principal contribution
[74]: Early Detection of Cystoid Macular Edema in Retinitis Pigmentosa Using Longitudinal Deep Learning Analysis of OCT Scans	2025	In a retrospective RP cohort, the authors use deep learning on longitudinal spectral-domain OCT macular B-scan images to predict early cystoid macular oedema onset in retinitis pigmentosa, using a patient-wise split for evidence from internal validation/testing rather than external cohorts.	ResNet-based models trained on early-visit OCTs provide an OCT-only approach to CME-risk prediction in RP. ResNet-34 was the best-performing model in the internal evaluation. Whether it enables earlier detection remains untested.
[412]: Early Detection of Optic Nerve Changes on Optical Coherence Tomography Using Deep Learning for Risk-Stratification of Papilledema and Glaucoma	2024	A proof-of-concept analysis uses deep learning by retraining a VGG-19 model on OCT-derived retinal measurements to forecast progression to papilledema or glaucoma from prediagnostic scans, providing retrospective evidence of AI detecting early optic nerve changes when clinician-identified OCT findings are nonsignificant.	The results indicate that model performance depends on the OCT representation, with retinal nerve fibre layer thickness mapping for papilledema and extracted vertical tomograms for glaucoma, suggesting a data-driven route to earlier AI-based risk stratification in these optic nerve diseases.
[402]: Effect of patch size and network architecture on a convolutional neural network approach for automatic segmentation of OCT retinal layers	2018	A controlled architecture experiment investigates CNN-based retinal boundary segmentation in ophthalmic OCT B-scans by predicting per-pixel boundary probabilities from image patches and converting them to boundary paths with graph search, while systematically varying patch size and CNN architecture to assess segmentation quality under different labelling granularities.	The experiments show how architectural and input-window design choices, together with the number of boundary classes used for training, measurably influence segmentation agreement and robustness, providing practical design guidance for more reliable OCT layer boundary detection via CNN plus graph search.
[265]: Effective blood vessels reconstruction methodology for early detection and classification of diabetic retinopathy using OCTA images by artificial neural network	2020	For early DR categorisation, the authors use OCTA imaging with custom blood-vessel reconstruction and feature extraction, followed by a supervised artificial neural network (ANN) to classify normal, diabetic without DR, and mild-to-moderate NPDR using reported diagnostic results from 100 processed images.	The OCTA-to-ANN pipeline reconstructs and quantifies vascular features for automated early DR categorisation, reporting high overall accuracy (97%) and low misclassification error with potential relevance to screening-oriented decision support.
[462]: Efficacy and prognostic factors of anti-VEGF treatment for neovascular age-related macular degeneration: An OCTA imaging-based deep learning analysis	2025	Within a retrospective cohort of neovascular age-related macular degeneration, an improved LUNet deep learning model analyses OCTA images to extract FAZ and vessel/lesion vascular parameters and then uses statistical testing to evaluate how these imaging biomarkers relate to anti-VEGF treatment response at 6 months.	The analysis reports specific OCTA-derived predictors of inadequate response, such as Vdisp in macular neovascularisation, neovascular surface area, and pigment epithelial detachment, supporting risk stratification and more personalised anti-VEGF treatment decisions.

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Work	Year	Basis for inclusion	Principal contribution
[427]: End-to-End Deep Learning Model for Predicting Treatment Requirements in Neovascular AMD From Longitudinal Retinal OCT Imaging	2020	The model forecasts individualised anti-VEGF retreatment requirements from longitudinal, treatment-initiation OCT volumes using an end-to-end DenseNet plus recurrent neural network model that integrates temporal information across multiple time points, with evaluation on a held-out test set using regression and classification evidence derived from real-world pro-re-nata scheduling labels.	The study uses a dimension-reduction preprocessing pipeline based on star-shaped sampling planes to enable end-to-end learning from raw volumetric OCT and demonstrates that the approach can recover prognostic spatiotemporal patterns beyond feature-engineered machine learning baselines, supported by occlusion-sensitivity visualisations tied to treatment requirement strata.
[415]: Enhanced AMD detection in OCT images using GLCM texture features with Machine Learning and CNN methods	2025	For AMD detection, the authors compare supervised machine learning and convolutional neural networks on pre-processed retinal OCT image datasets, focusing on grey level co-occurrence matrix (GLCM) texture features and assessing how feature selection affects diagnostic outcomes.	Selected GLCM measures, particularly the sfGLCM group, outperformed the full GLCM feature set and greyscale pixels alone. The comparison informs the design of more accurate OCT-based AMD classifiers that are less prone to overfitting.
[135]: Enhanced Deep Learning Model for Classification of Retinal Optical Coherence Tomography Images	2023	For retinal OCT classification, the authors build an enhanced deep learning pipeline that combines a modified ResNet-50 with a random forest and optimises training using Adam (and dual stochastic gradient descent (SGD)/Adam), demonstrating multiclass performance across four OCT categories (CNV, DME, drusen, and normal).	The proposed ResNet-50 plus random forest pipeline, trained with Adam, is evaluated for automated OCT image categorisation. Reported multiclass metrics quantify performance on the study dataset, but clinical decision-support utility was not evaluated.
[154]: Enhancing pathological myopia diagnosis: a bimodal artificial intelligence approach integrating fundus and optical coherence tomography imaging for precise atrophy, traction and neovascularisation grading	2025	In this single-centre retrospective cross-sectional study, a bimodal deep-learning approach fuses paired fundus photographs and OCT images to perform ATN (atrophy, traction, neovascularisation) staging for pathological myopia. Multiclass and binary evaluations compare the method with single-modality approaches.	The authors introduce a ResNet-50-based multimodal multi-instance learning model with cross-modality interaction to grade ATN components from matched fundus-OCT inputs. Reported improvements over single-modality models require prospective evaluation before screening or referral benefits can be inferred.

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Work	Year	Basis for inclusion	Principal contribution
[103]: Ensemble Convolutional Neural Networks for the Classification and Visualization of Retinal Diseases in Optical Coherence Tomography Images	2023	For four-class OCT analysis, the authors apply a deep-learning CNN ensemble to OCT imaging to classify four retinal disease categories and generate positive/negative heatmaps for activity visualisation and region-of-interest estimation, reporting diagnostic performance metrics as evidence of feasibility.	The ensemble combines VGG19, ResNet152, and DenseNet121 with soft and hard voting plus a new positive/negative heatmap algorithm to support second-reader interpretation of retinal OCT by highlighting relevant areas.
[372]: Ensemble Deep Learning for Diabetic Retinopathy Detection Using Optical Coherence Tomography Angiography	2020	For referable DR classification, the authors evaluate an ensemble of fine-tuned VGG19, ResNet50, and DenseNet CNNs. Majority soft voting and stacking classify referable versus non-referable DR from en face OCTA and co-registered structural OCT. Grad-CAM provides localisation, and five-fold cross-validation compares performance with a handcrafted-feature classifier.	The cross-validation results indicate that combining multiple OCT/OCTA-trained CNNs yields higher referable DR discrimination than single-data-type networks and a conventional pipeline using handcrafted features, while Grad-CAM provides model interpretability by highlighting image regions linked to microvascular and structural disease signals.
[69]: Ensemble Learning Based on Convolutional Neural Networks for the Classification of Retinal Diseases from Optical Coherence Tomography Images	2020	The authors evaluate an ensemble of CNN-based architectures to classify spectral-domain OCT images into four retinal categories (CNV, DME, DRUSEN, and NORMAL), applying supervised training with image preprocessing and augmentation and evaluating performance against individual CNNs using metrics reported for the same dataset split.	The evaluation compares multiple ensemble configurations, then identifies the best-performing CNN plus DenseNet121 plus InceptionV3 averaged-prediction setup, demonstrating high multiclass diagnostic accuracy for OCT while emphasising feasibility on a limited dataset.
[327]: Epiretinal Membrane Detection in Optical Coherence Tomography Retinal Images Using Deep Learning	2021	The authors use deep learning with transfer learning to adapt four convolutional neural network architectures for automatic epiretinal membrane detection in optical coherence tomography retinal images, reporting quantitative evidence from an exhaustive evaluation with high reported classification performance.	The experiments refine CNN transfer learning for ERM detection by tuning network hyperparameters and comparing cross-entropy versus focal loss, demonstrating that the approach can work effectively with relatively small training datasets in OCT-based clinical imaging.
[453]: Evaluation of 2D and 3D Deep Learning Approaches for Automatic Segmentation of the Retinal External Limiting Membrane in Spectral Domain Optical Coherence Tomography Images	2022	The study compares 2D versus 3D deep learning networks for automated segmentation of the retinal external limiting membrane in spectral-domain optical coherence tomography, reporting quantitative evidence that 3D architectures better capture 3D structure, with performance differences across overlap and surface-distance measures.	The evaluation directly compares three contemporary model designs in both 2D and 3D forms on a public spectral-domain OCT dataset. The 3D variants typically improved Dice, mean surface distance, false positive rate, and surface smoothness while trading off Hausdorff distance.

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Work	Year	Basis for inclusion	Principal contribution
[65]: Evaluation of Convolutional Neural Networks (CNNs) in Identifying Retinal Conditions Through Classification of Optical Coherence Tomography (OCT) Images	2025	For diabetic macular disease classification, the authors train supervised convolutional neural networks (VGG16 and InceptionV3) to categorise greyscale OCT images into four diagnostic groups (normal, choroidal neovascularisation, diabetic macular oedema, and drusen) and report evidence from model evaluation using accuracy, sensitivity, specificity, and confusion-matrix-based analyses across varying dataset sizes.	The experiments provide a direct, experimentally tested comparison of VGG16 versus InceptionV3 on a four-class OCT dataset, showing how performance (including sensitivity and specificity) changes with training set size and clarifying practical limitations such as missing image augmentation and greyscale-only inputs.
[317]: Explainable Deep Learning Approach for Automated Classification of Retinal Diseases Using OCT Images and Grad-CAM	2025	A CNN is trained for automated four-class retinal OCT image classification (CNV, DME, drusen, normal), uses Grad-CAM to generate region-level visual explanations, and reports classification accuracy and related measures.	The explainable OCT classification pipeline combines extensive labelled training data, preprocessing and augmentation, and Grad-CAM highlighting to support interpretable decision-making for retinal disease triage.
[321]: Eye Disease Prediction Using Deep Learning and Attention on Oct Scans	2024	For eye-disease prediction, an end-to-end OCT pipeline combines a custom UNet for image segmentation followed by an attention-enhanced InceptionV3 classifier to categorise normal eyes versus conditions including DME, CNV, and drusen, with reported classification performance from extensive experimentation.	The authors implement an end-to-end web-based workflow that lets users upload raw OCT scans for automated segmentation and condition categorisation, supporting timely decision-making for screening and early detection based on the model's reported precision, recall, accuracy, and F1 score.
[62]: Feasibility study to improve deep learning in OCT diagnosis of rare retinal diseases with few-shot classification	2021	To address limited rare-disease data, the authors generate synthetic OCT images with CycleGAN from normal OCTs and train an Inception-v3 classifier, with validation on an independent test dataset to demonstrate improved classification accuracy versus several deep-learning and few-shot baselines.	The method uses disease-specific GAN augmentation for five rare retinal classes and shows that the resulting feature learning improves rare-class separability and diagnosis performance, addressing the data-scarcity challenge that limits conventional multiclass OCT deep learning.
[363]: Few-shot out-of-distribution detection for automated screening in retinal OCT images using deep learning	2023	In 2023, the authors evaluate uncertainty-based out-of-distribution (OOD) detection for automated age-related macular degeneration screening and staging from retinal OCT B-scans using a CNN and several OOD scoring/uncertainty method families, testing performance on near-OOD OCT outliers (DME, RVO, Stargardt) and far-OOD eye fundus images, with evidence from comparative AUC-based experiments including a few-shot outlier exposure strategy.	The comparative experiments indicate that few-shot outlier exposure (OE) combined with cosine-distance feature-space scoring yields improved near-OOD rejection (e.g., with eight exposed outliers per class) without reducing AMD classification performance, and argues for cosine-distance as a practical alternative to reject-bucket probability for OCT-based OE setups.

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Work	Year	Basis for inclusion	Principal contribution
[328]: Forme fruste keratoconus detection with OCT corneal topography using artificial intelligence algorithms	2024	In a retrospective monocentric case-control design, the authors evaluated machine learning classification (random forest and logistic regression) using swept-source OCT corneal topography from CASIA 2 to separate normal corneas from forme fruste keratoconus (FFKC) for refractive surgery risk screening, benchmarking algorithm outputs against the ectasia screening index (ESI).	The comparison reports a multiclass/automated approach that identifies FFKC with markedly higher sensitivity than the built-in ESI while maintaining high specificity, highlighting discriminative topographic parameters, particularly posterior surface features, that may improve clinical screening workflows.
[181]: From Machine to Machine: An OCT-Trained Deep Learning Algorithm for Objective Quantification of Glaucomatous Damage in Fundus Photographs	2019	Using a cross-sectional glaucoma dataset, the authors train a convolutional neural network on optic disc fundus photographs using spectral-domain OCT retinal nerve fibre layer thickness as the quantitative target, then evaluate prediction fidelity and discrimination of glaucomatous versus healthy eyes with correlation/agreement and ROC analyses in a held-out test set.	Trained against quantitative RNFL thickness rather than subjective optic-disc labels, the model demonstrates OCT-trained image-to-measurement transfer by mapping disc photographs to continuous RNFL thickness estimates, and validates that these predictions replicate SD-OCT-based structure-function diagnostic performance in the test cohort.
[314]: Fully Automated Detection and Quantification of Macular Fluid in OCT Using Deep Learning	2018	In 2018, the authors develop and validate a fully automated deep-learning semantic segmentation approach for conventional spectral-domain OCT to detect and quantify intraretinal cystoid fluid (IRC) and subretinal fluid across neovascular AMD, DME, and RVO on both Cirrus and Spectralis platforms, with evaluation against masked expert consensus using volume-level detection metrics and pixel-level localisation and extent comparisons.	The validation reports reproducible pixel-wise fluid typing and volume quantification in the evaluated Cirrus and Spectralis data, using an encoder-decoder pipeline with device-specific training and comparison with manual expert measurements. This supports consistent IRC versus SRF measurement for research, while clinical decision utility was not evaluated.
[63]: Fully automated detection and quantification of multiple retinal lesions in OCT volumes based on deep learning and improved DRLSE	2019	For automated OCT lesion analysis, the authors describe a fully automated deep learning approach that integrates improved distance regularised level set evolution (DRLSE) to detect and segment multiple retinal lesion types. They evaluate the approach with quantitative overlap metrics on annotated 3D OCT-derived B-scans.	The pipeline distinguishes five lesion classes (PED, SRF, drusen, CNV, and MH) and reports performance on 500 B-scans from 15 3D OCT volumes, demonstrating an end-to-end method for automated lesion quantification in retinal imaging.

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Work	Year	Basis for inclusion	Principal contribution
[88]: Fused-AETNet: A Variational Transformer-Based Framework for Diabetic Retinopathy Classification Using OCT Biomarkers	2025	For OCT-based DR classification, the authors segment 11 retinal layers with atlas-guided probabilistic methods, extracting layer-wise handcrafted morphological biomarkers, and using a variational autoencoder with transformer-inspired attention to learn and refine a latent representation for diagnostic decision-making with reported quantitative validation on 481 subjects.	Fused-AETNet combines clinically motivated handcrafted OCT biomarkers, including reflectivity distributions, Laplacian-derived thickness, and boundary tortuosity, in a variational latent space refined through attention. The study reports performance for this compact biomarker-driven DR classifier, but does not evaluate clinician interpretability.
[172]: Fusing Results of Several Deep Learning Architectures for Automatic Classification of Normal and Diabetic Macular Edema in Optical Coherence Tomography	2018	The authors evaluate a convolutional neural network ensemble approach on OCT imaging to classify diabetic macular oedema versus normal, using principal component analysis with voting and patient-independent Leave-Two-Patients-Out cross-validation as the evidence for an ophthalmic diagnostic task.	Two fusion strategies combine pretrained AlexNet, VggNet, and GoogleNet feature integration and majority/plurality voting, reporting volume-level performance to support automated screening potential on multi-institution OCT datasets.
[384]: GCC Aware Glaucoma Detection Using Macula OCT Image Analysis Based on Deep CNN	2024	A deep CNN pipeline is evaluated for glaucoma screening using macular SD-OCT B-scans. Segmentation first extracts the GCC, including the retinal nerve fibre layer and the ganglion cell–inner plexiform layer. A subsequent CNN distinguishes glaucomatous from non-glaucomatous eyes while modelling several macular GCC subregions. Performance is reported from training and validation on a single locally acquired OCT dataset.	The region-specific analysis indicates that GCC-aware, region-specific OCT analysis can improve glaucomatous discrimination beyond using OCT without targeted GCC focus, highlighting a previously unreported approach that accounts for how different macular GCC regions affect screening performance.
[387]: Generative adversarial network-based deep learning approach in classification of retinal conditions with optical coherence tomography images	2023	The authors evaluate a GAN-based data synthesis strategy for classifying retinal conditions from OCT images, using pretrained deep neural networks and testing on both a publicly available benchmark dataset and a hospital-collected clinical OCT set, providing comparative performance evidence in an ophthalmic workflow.	The experiments indicate that training with GAN-generated class-balanced OCT data can improve classification accuracy relative to training on the original imbalanced dataset across multiple evaluation settings, including a clinical test set, which supports the practical value of synthesis-based balancing for retinal OCT classification.

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Work	Year	Basis for inclusion	Principal contribution
[252]: GENERATIVE DEEP LEARNING APPROACH TO PREDICT POSTTREATMENT OPTICAL COHERENCE TOMOGRAPHY IMAGES OF AGE-RELATED MACULAR DEGENERATION AFTER 12 MONTHS	2025	To forecast posttreatment anatomy in neovascular age-related macular degeneration, the authors evaluate a conditional GAN-based generative deep learning model that uses pretreatment OCT plus fluorescein and indocyanine green angiography, along with treatment-related staging and clinical variables, to forecast 12-month posttreatment OCT anatomy, with evidence from validation on 533 eyes.	The results indicate that adding OCT acquired after loading doses and key regimen and treatment descriptors markedly improves compartment-level OCT prediction and wet versus dry macular status, providing evidence for a method to anticipate long-term anatomical response from baseline imaging and management context.
[438]: Glaucoma Detection From Raw Circumpapillary OCT Images Using Fully Convolutional Neural Networks	2020	The authors use fully convolutional neural networks in ophthalmic imaging to classify glaucoma versus normal from raw circumpapillary OCT B-scans around the optic nerve head, with evidence from an independent test set reported using ROC AUC and accuracy.	The comparison evaluates two CNN training strategies, training from scratch versus fine-tuning, and shows that fine-tuned VGG-family models perform best on the independent test set, which is particularly informative for settings with limited OCT data.
[431]: Glaucoma Diagnosis Through the Integration of Optical Coherence Tomography/Angiography and Machine Learning Diagnostic Models	2022	Using retrospective cross-sectional glaucoma-clinic data, the investigators derived OCTA-based structural and vascular parameters and built machine-learning classifiers to distinguish healthy from glaucomatous eyes (and among glaucoma severities), using a rigorous nested stratified group cross-validation framework with held-out test evaluation and interpretable decision-tree outputs.	The investigators also generated North Texas normative ranges across multiple OCT/OCTA metrics and showed which OCTA features most influenced the final model, providing clinician-facing, data-driven support for glaucoma assessment.
[209]: Glaucoma Diagnosis with Machine Learning Based on Optical Coherence Tomography and Color Fundus Images	2019	For glaucoma diagnosis, the model combines OCT-derived thickness and deviation maps with colour fundus images, using transfer-learned CNNs with an ensemble random forest to classify healthy versus glaucomatous eyes, evaluated by 10-fold cross-validated ROC AUC.	The multimodal pipeline builds separate VGG19-based classifiers for disc fundus and multiple OCT feature maps, then fuses their learned representations with a random forest to reach improved cross-validated discrimination of glaucoma from healthy controls.
[444]: Glaucoma Diagnostic Accuracy of Machine Learning Classifiers Using Retinal Nerve Fiber Layer and Optic Nerve Data from SD-OCT	2013	In a cross-sectional ophthalmic cohort, the authors assessed diagnostic accuracy of 10 machine learning classifiers for distinguishing early-to-moderate primary open-angle glaucoma from healthy eyes using SD-OCT-derived retinal nerve fibre layer and optic nerve parameters, with ROC/AUC and sensitivity/specificity computed under 10-fold cross-validation.	The study compared classifier performance against individual SD-OCT parameters and found that the best random forest model achieved good discrimination without producing statistically significant improvement over the top single structural metric, supporting the clinical value and limits of OCT-based AI for glaucoma screening in this setting.

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Work	Year	Basis for inclusion	Principal contribution
[299]: GlauNet: Glaucoma Diagnosis for OCTA Imaging Using a New CNN Architecture	2022	GlauNet performs glaucoma detection in an ophthalmic setting, designed to extract features from OCTA of the optic nerve head and retinal nerve fibre layer for a glaucoma versus non-glaucoma classification task, with evidence from training and separate testing and reported diagnostic metrics including robustness to artefact and poor-quality images.	GlauNet replaces handcrafted OCTA parameter pipelines with an end-to-end two-stage convolutional feature extractor and fully connected classifier and highlights vascularly relevant regions. Reported test results describe maintained sensitivity and specificity under artefact degradation, but clinical utility and deployment were not evaluated.
[302]: Harnessing deep learning for detection of diabetic retinopathy in geriatric group using optical coherence tomography angiography-OCTA: A promising approach	2024	In 2024, the authors develop an ophthalmic deep-learning system for diabetic retinopathy severity classification from OCTA imaging in a geriatric case-control cohort, using CNN-based feature extraction with multiple pretrained architectures and reporting model evaluation on a held-out test split with standard classification metrics.	The pipeline operationalises OCTA layer-focused preprocessing and compares several CNN backbone plus classical classifier combinations (including KNN and feed-forward neural networks) to grade DR into no DR signs, mild, or moderate categories, providing quantitative performance results that support OCTA-enabled automated screening for older adults.
[338]: HTC-retina: A hybrid retinal diseases classification model using transformer-Convolutional Neural Network from optical coherence tomography images	2024	The study evaluates a hybrid CNN and vision-transformer framework for automated multi-label retinal disease classification using optical coherence tomography images, incorporating a hierarchical convolutional patch and token embedding (CPTE) module and a parallel residual depthwise–pointwise convolutional network (RDP-ConvNet) to emphasise lesion-relevant local features and reporting comparative performance across three OCT datasets.	HTC-Retina replaces raw tokenisation with CPTE and adds an RDP-ConvNet branch, demonstrating high accuracy and sensitivity with computational efficiency for seven retinal diseases.
[417]: Human versus Artificial Intelligence: Validation of a Deep Learning Model for Retinal Layer and Fluid Segmentation in Optical Coherence Tomography Images from Patients with Age-Related Macular Degeneration	2024	An external validation study assesses a deep learning model for OCT retinal layer and total fluid segmentation in an ophthalmic setting of age-related macular degeneration by comparing automated outputs to a clinically derived gold standard based on manually adjusted Heidelberg Spectralis segmentation (and MATLAB fluid refinement), evaluating quantitative agreement across healthy, intermediate, and exudative AMD groups.	The validation found near-complete segmentation overlap for retinal boundaries overall and close concordance of key OCT-derived thickness and fluid area measurements, while also pinpointing reduced agreement for specific layers, notably the inner segment ellipsoid and increasing method discrepancies with later disease stage.

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Work	Year	Basis for inclusion	Principal contribution
[198]: Hybrid 3d-2d Deep Learning For Detection Of Neovascularage-Related Macular Degeneration Using Optical Coherence Tomography B-Scans And Angiography Volumes	2021	The authors use a CNN-based hybrid 3D-2D deep-learning approach on OCTA angiography volumes and OCT B-scans to classify retinal findings into non-AMD, non-neovascular AMD, and neovascular AMD, targeting automated detection of late-stage disease by capturing choroidal neovascularisation and sequelae evidence in an ophthalmic imaging setting.	The hybrid architecture provides automated multiclass staging for AMD based on hybrid 3D-2D CNNs using retinal angiography and structural OCT inputs, which matters for reducing clinician workload in interpreting OCTA-rich datasets for late-stage differentiation.
[176]: Hybrid deep learning and optimal graph search method for optical coherence tomography layer segmentation in diseases affecting the optic nerve	2024	For optic-nerve disease OCT, the authors test Deep LOGISMOS, a hybrid deep learning plus optimal 3D graph search method, for segmenting retinal layers and quantifying GCIPL thickness in optic nerve diseases with irregular layer topology, using reference-traced and longitudinal OCT cohorts across Cirrus and Spectralis and reporting segmentation accuracy and clinically oriented thickness robustness.	The evaluation indicates a two-stage true 3D workflow where learned probability maps drive in-region costs while built-in topology constraints avoid extensive dataset-specific post-processing, yielding more reliable GCIPL thickness measurement over time than baseline reference algorithms and deep-learning-only segmentation in the provided non-arteritic anterior ischaemic optic neuropathy (NAION), glaucoma, and multiple sclerosis test sets.
[207]: Hybrid Deep Learning/Machine Learning Model for Retinal Diseases Classifications Using OCT Images	2024	For eight-condition retinal diagnosis, the pipeline uses a hybrid deep learning and machine learning pipeline for OCT imaging, extracting features with a DenseNet and then performing multi-condition retinal disease classification using an SVM tuned by Bayesian optimisation, with testing on publicly accessible OCT data across eight retinal conditions.	The comparison quantifies the benefit of DenseNet feature extraction combined with Bayesian-optimised SVM hyperparameters for OCT-based classification across eight conditions, offering an end-to-end automated workflow that could support earlier, more consistent clinical screening and decision-making.
[146]: Hybrid deep learning models for the screening of Diabetic Macular Edema in optical coherence tomography volumes	2024	Within a real-world teleophthalmology DR screening program, the retrospective evaluation uses a hybrid CNN-RNN model to read complete OCT cubes (with bidirectional recurrence over cube-derived slice sequences) for binary diabetic macular oedema detection, with clinically oriented threshold tuning and diagnostic metrics reported against retina-specialist ground truth.	Analysis of unselected OCT volumes from 4 years of screening indicates how a tuned multi-CNN bidirectional RNN ensemble can support first-pass identification of diabetic macular oedema, particularly for extrafoveal cases, by optimising sensitivity and specificity tradeoffs relevant to population screening workflows.

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Work	Year	Basis for inclusion	Principal contribution
[437]: HyFormer: a hybrid transformer-CNN architecture for retinal OCT image segmentation	2024	HyFormer uses a hybrid Transformer-CNN architecture to perform multi-class delineation of retinal layers and lesion types (including traction maculopathy and age-related degeneration) on myopic traction maculopathy OCT data and the public AROI dataset, reporting evidence from comparative evaluations across these imaging tasks and datasets.	The HyFormer architecture combines parallel Transformer and convolutional encoders plus multi-scale gated attention, group positional embedding, and a three-path fusion decoder, and shows that this combination improves segmentation quality while maintaining computational efficiency through its network-level design and loss formulation with CAM-guided supervision.
[376]: Hyper-reflective foci segmentation in SD-OCT retinal images with diabetic retinopathy using deep convolutional neural networks	2019	Adapted deep convolutional classification networks perform pixel-wise segmentation of hyper-reflective foci in diabetic retinopathy spectral-domain OCT, evaluated with Dice-based evidence on a small-lesion task using retinal imaging rather than nonimaging signals.	The fused deep convolutional pipeline segments HRF from small patch-based pixel predictions and reports Dice similarity coefficients for foci area and count, providing quantitative support for lesion-focused delineation in SD-OCT B-scans.
[70]: Identification of diabetic retinopathy classification using machine learning algorithms on clinical data and optical coherence tomography angiography	2024	Across two centres, the cross-sectional study used machine-learning classifiers (random forest, gradient boosting machines, deep learning, and logistic regression) to assign multiclass diabetic retinopathy categories (DR, referable DR, and vision-threatening DR) from OCTA-derived quantitative vascular and retinal metrics combined with systemic clinical data, and it provides evidence from independent external validation using ROC/AUC.	Comparison of feature patterns across clinical-only, OCTA-only, and combined OCTA plus clinical/non-laboratory variables identifies OCTA-informed models, particularly those with non-laboratory factors and vascular density measures, as practical tools for screening and referral decision support for DR severity stratification.
[435]: Implementing Predictive Models in Artificial Intelligence through OCT Biomarkers for Age-Related Macular Degeneration	2023	The narrative review examines ophthalmic prediction of age-related macular degeneration progression by summarising AI methods that leverage multimodal retinal imaging, with particular emphasis on OCT-derived biomarkers used to forecast high-risk phenotypes and evolution towards late disease.	The review consolidates reported OCT feature predictors, such as retinal layer morphometrics, drusen metrics, reticular pseudodrusen, and hyperreflective foci, within the context of AI-based progression modelling to clarify what evidence currently supports and where future work may refine predictive efforts.
[269]: Improving OCT Image Segmentation of Retinal Layers by Utilizing a Machine Learning Based Multistage System of Stacked Multiscale Encoders and Decoders	2023	The authors evaluate a deep learning multistage stacked multiscale encoder-decoder segmentation pipeline for delineating retinal layers on OCT B-scans, using peripapillary OCT data plus additional publicly available datasets (Duke SD-OCT, UMN, and Heidelberg) to test robustness on noisier imagery and provide Dice-score-based evidence against prior single-stage approaches.	The architecture stacks a first local-context encoder-decoder with a second refinement and denoising stage plus paired attention-based decoders. It achieved the highest Dice score on the peripapillary dataset and was also examined qualitatively on datasets with limited or noisier annotations.

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[399]: Integrating Human Expertise With Artificial Intelligence (AI) Models for Optical Coherence Tomography (OCT) Retinal Fluid and Pathology Quantification: A Systematic Review	2025	A systematic review synthesises ophthalmology-focused evidence on clinician-in-the-loop hybrid workflows that pair human review with deep-learning AI architectures for OCT-based retinal fluid and biomarker quantification, assessing diagnostic and operational outcomes reported across included studies from 2021 to 2025.	The review summarises where the nine included studies reported expert-comparator biomarker performance and efficiency gains of more than 50%. Heterogeneity among the studies limits broad claims of expert-level reliability or practical adoption.
[195]: Integrating lightweight convolutional neural network with entropy-informed channel attention and adaptive spatial attention for OCT-based retinal disease classification	2025	LOCT-Net is a lightweight convolutional neural network for retinal disease classification using OCT B-scan imaging, combining nested residual multi-scale feature extraction with entropy-informed channel attention and adaptive spatial attention, and adds post-hoc explainable AI, with evidence from evaluation on six benchmark OCT datasets including reported classification metrics.	The architecture provides a parameter-efficient OCT classifier with attention mechanisms tailored to channel- and region-level feature refinement, reporting diagnostic performance at low computational cost while adding interpretability via explainable AI.
[233]: Integrating Retinal Segmentation Metrics with Machine Learning for Predictions from Mouse SD-OCT Scans	2025	In mouse SD-OCT data, retinal layer thickness segmentation metrics are used to train and optimise neural networks for categorical prediction of mouse age and species, with validation reported against multiple baseline machine learning methods using accuracy and AUC-like evidence.	The validation indicates that feature-level OCT segmentation metrics, rather than raw images, can yield high validation performance for mouse group differentiation, supporting automated retinal morphometry as a practical input for experimental predictive modelling.
[341]: Interactive Deep Learning-Based Retinal OCT Layer Segmentation Refinement by Regressing Translation Maps	2024	The authors evaluate an interactive deep learning method for retinal OCT layer boundary segmentation that refines an initial model output using clinician-guided point clicks or line scribbles, regressing pixel-wise translation maps to update boundaries and providing coordinate-wise confidence as evidence of failure-prone regions across multiple private and public datasets.	The method uses a translation-map-based interaction mechanism that turns subjective specialist corrections into localised boundary refinements and couples them with confidence estimates, enabling more reliable, user-supervised segmentation refinement for clinical OCT workflows.
[162]: Interpretable Deep Learning For Automated Retinal Damage Detection in Optical Coherence Tomography	2024	Transfer-learned CNNs (VGG16, EfficientNetB3, and ResNet50) are evaluated on optical coherence tomography images for automatic four-class retinal injury classification and generate Grad-CAM visual explanations.	The evaluation compares three pretrained architectures for Drusen, Normal, DME, and CNV detection and pairs the best-performing model with Grad-CAM region highlighting to visualise image regions associated with the model's predictions.

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Work	Year	Basis for inclusion	Principal contribution
[120]: Interpretable Retinal Disease Classification from OCT Images Using Deep Neural Network and Explainable AI	2021	For interpretable macular classification, the authors use a deep neural network with pretrained CNN backbones on macular OCT images to classify CNV, DME, and drusen, and apply LIME to examine the model's misclassifications.	The OCT classification pipeline couples CNN-based predictions with LIME visual explanations to identify why errors occur, supporting future refinement of model performance.
[420]: IoT based automatic detection of glaucoma disease in OCT and fundus images using deep learning techniques	2020	For glaucoma detection from retinal imaging, the authors apply a two-phase region-based convolutional neural network (R-CNN)-style deep learning workflow to first localise the optic nerve head/optic disc and then classify it as glaucoma or stable, with rule-based semi-automatic ground-truth support and evaluation on the ORIGA fundus dataset.	The method combines a semi-automatic, rule-assisted labelling strategy for optic disc localisation and couples this with glaucoma-versus-stable classification to show that AUC alone may be insufficient for assessing performance under dataset splits and class imbalance conditions.
[170]: JOINT MOTION CORRECTION AND 3D SEGMENTATION WITH GRAPH-ASSISTED NEURAL NETWORKS FOR RETINAL OCT	2022	Graph-assisted neural networks process 3D OCT volumes to jointly handle motion artefacts and produce consistent retinal layer segmentation across neighbouring B-scans, with evidence reported as experimental visual and quantitative improvements versus conventional and 2D deep-learning approaches.	The framework couples eye motion correction with 3D, cross-B-scan retinal layer segmentation, addressing how misalignment from involuntary eye movements undermines downstream OCTAngiography projection and disease analysis.
[64]: Joint Segmentation and Uncertainty Visualization of Retinal Layers in Optical Coherence Tomography Images Using Bayesian Deep Learning	2018	A Bayesian OCT segmentation model couples end-to-end pixel-wise labelling with a corresponding uncertainty map, with evidence from validation on 1487 OCT volumes from 15 subjects against contemporary uncertainty-agnostic segmentation methods.	The method provides an uncertainty visualisation mechanism that highlights likely mis-segmented retinal regions and reports comparable or improved accuracy with increased robustness to noise relative to methods that do not model uncertainty.
[114]: Linking Function and Structure with ReSensNet: Predicting Retinal Sensitivity from OCT using Deep Learning	2022	In a retrospective cross-sectional cohort, the authors use a deep learning model to predict microperimetry retinal sensitivity maps from spectral-domain OCT, evaluated with internal development and external validation across multiple retinal diagnoses by point-wise and mean error and correlation measures.	Internal and external evaluations quantify agreement and correlation between OCT-derived estimates and microperimetry for point-wise and mean retinal sensitivity. The results support further evaluation of an imaging-derived estimate, but do not validate it as a functional surrogate for monitoring or trials.

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Work	Year	Basis for inclusion	Principal contribution
[357]: Loss-Modified Transformer-Based U-Net for Accurate Segmentation of Fluids in Optical Coherence Tomography Images of Retinal Diseases	2023	For OCT fluid segmentation, the authors use a transformer-enhanced U-Net (Trans-U-Net) with B-scan imaging and a customised hybrid loss to delineate retinal fluid regions associated with macular oedema, providing evidence from quantitative evaluation including 5-fold cross-validation and robustness testing with added Gaussian noise.	The loss function combines Dice, focal Tversky, and weighted binary cross-entropy to address class imbalance and noisy OCT inputs, and reports improved segmentation quality and convergence relative to alternative loss functions on initial and noise-perturbed datasets.
[126]: Machine learning and optical coherence tomography-derived radiomics analysis to predict persistent diabetic macular edema in patients undergoing anti-VEGF intravitreal therapy	2024	From manually segmented spectral-domain OCT inner-retina layers, the authors derive OCT-omics radiomics and apply logistic regression, SVM, and backpropagation neural network classifiers to predict persistent versus non-persistent diabetic macular oedema after the first three anti-VEGF intravitreal injections, with evidence from separate training and test sets and decision-curve analysis to support clinical relevance.	The resulting non-invasive OCT-based prediction pipeline uses z-scored, reproducibility-filtered radiomics features and recursive feature elimination. OCT-omics scores differentiated persistent from non-persistent DME and tracked central subfield thickness change in the study data.
[359]: Machine learning-based prediction of persistent diabetic macular edema using OCT-derived structural biomarkers and clinical features	2026	Using retrospective single-centre data, the authors trained machine learning classifiers that combine baseline clinical variables with OCT-derived structural biomarkers to predict persistent diabetic macular oedema after an anti-VEGF loading phase, evaluating performance in an internal train/validation split and using SHAP for evidence-based interpretability of key predictors.	Comparison of eight model types indicates that multimodal OCT-and-clinical approaches outperform clinical-only or OCT-only baselines within the internal split. SHAP highlights diabetes duration, EZ/ELM disruption, and hyperreflective foci as influential predictors, while external validation and clinical decision-support utility remain untested.
[205]: Machine learning classifiers for glaucoma diagnosis based on classification of retinal nerve fibre layer thickness parameters measured by Stratus OCT	2010	For glaucoma diagnosis, the authors evaluate two machine-learning classifier families, ANNs and SVMs, using Stratus OCT RNFLT features, including transformed A-scan measurements, to distinguish glaucoma from healthy eyes. Evidence is reported as AUC comparisons across conventional and novel input parameterisations.	Classifier choice, comparing ANNs with SVMs, yielded no meaningful differences. Transformed A-scan-based RNFLT inputs produced the highest AUCs, indicating that imaging-feature and input engineering can be at least as important as the learning algorithm for OCT-based glaucoma diagnosis.
[128]: MACULA OCT Disease Images Classification using Deep Learning Techniques	2024	Convolutional neural networks are evaluated on macular optical coherence tomography images for automated retinal disease classification across multiple macular conditions, supported by reported classification accuracy with a top-performing ResNet model.	The comparison evaluates several CNN architectures, including ManualNet, AlexNet, and ResNet, and reports that ResNet reaches 90% accuracy, offering an actionable deep-learning approach for differentiating specified retinal diseases on macular OCT.

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Work	Year	Basis for inclusion	Principal contribution
[459]: Macular edema degeneration classification on OCT and fundus images with portable platform based on artificial intelligence methods	2022	The authors develop a portable AI computer-aided diagnosis workflow for ophthalmic staging of age-related macular degeneration, using machine learning neuronal networks with combined OCT and fundus images to classify normal retina, dry AMD, and wet AMD on a Jetson TX2 device, with evidence limited to evaluation on a provided three-class dataset.	The prototype shows how a Jetson TX2-based model can perform multi-class AMD categorisation from multimodal retinal imaging in a point-of-care setting, highlighting practical cost and portability advantages for clinician support.
[184]: Macular Holes Detection Using Deep Learning on Optical Coherence Tomography Images	2023	For automated macular-hole analysis, the authors apply deep learning to optical coherence tomography imaging to detect macular holes and automate their size measurement, addressing the clinically relevant task of staging MHs and reducing human judgment errors while reporting detection accuracy and measurement efficiency.	The OCT-based method couples macular hole detection with automated sizing, aiming to speed up and improve stage recognition by distinguishing stage 2 from stage 3 or 4 without the time-consuming caliper workflow.
[60]: Macular OCT Classification Using a Multi-Scale Convolutional Neural Network Ensemble	2018	For macular OCT computer-aided diagnosis, a convolutional neural network ensemble uses a multi-scale mixture-of-experts approach to classify OCT images into normal retina versus dry age-related macular degeneration and diabetic macular oedema, with evidence reported as classification results on two Heidelberg OCT datasets.	The multiscale mixture-of-experts model uses a new cost function to enable discriminative, fast feature learning from multiple OCT sub-image scales and reports high average precision and AUC for the three-way macular classification task.
[248]: MBT: Model-Based Transformer for retinal optical coherence tomography image and video multi-classification	2023	The authors apply a model-based transformer (MBT) framework to OCT data for multiclass retinal disease classification, extend transformer-based learning from OCT images to OCT videos, and report evidence from experiments on three real-world retinal datasets using standard classification metrics.	MBT uses approximate sparse representation to learn optimal features and improves OCT image and video classification relative to transformer baselines, highlighting a pathway to identify retinal pathologies from OCT videos as well as images.
[288]: MDLGO: Integrated Multimodal Deep Learning Framework for Enhanced Precision in Glaucoma Diagnosis Using OCT A-scans	2024	The authors develop a multimodal deep learning system for glaucoma diagnosis that fuses OCT A-scans with fundus images and visual field data, using transfer learning, attention-based segmentation, and anomaly detection to generate diagnostic evidence reported as improved metrics and reduced time to diagnosis.	The architecture combines pretrained feature extractors, V-Net and SegNet attention mechanisms for OCT A-scan delineation, and One-Class SVM and Isolation Forest anomaly detection. The reported experiments assess whether this combination captures glaucoma-related signals. They do not establish improved diagnostic precision in practice.

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Work	Year	Basis for inclusion	Principal contribution
[296]: Multi-Fundus Diseases Classification Using Retinal Optical Coherence Tomography Images with Swin Transformer V2	2023	The authors evaluate an ophthalmic deep learning approach that uses the Swin Transformer V2 architecture with windowed self-attention and PolyLoss to perform multi-disease classification from retinal OCT images, training and testing on two independent public datasets (OCT2017 and OCT-C8) with quantitative diagnostic performance and Grad-CAM visual evidence.	The modified Swin Transformer V2 pipeline combines localised attention computation with PolyLoss, then reports consistently high multi-class diagnostic results and interpretability heatmaps that support the model's decision basis across both datasets.
[219]: Multi-label classification of Retinal Disorders in Optical Coherence Tomography using Deep Learning	2021	A CNN is evaluated on OCT images to perform multi-label identification of diabetic macular oedema, choroidal neovascularisation, and drusen, with an additional normal-macula category, providing evidence of high reported classification performance for an ophthalmic screening task.	The OCT screening pipeline simultaneously flags the presence of each of three retinal disorders and labels the remaining cases as normal macula, emphasising early detection of potentially treatable conditions.
[255]: Multi-scale CNN-Swin transformer network with boundary supervision for multiclass biomarker segmentation in retinal OCT images	2025	A CNN-Swin transformer segmentation model is applied to retinal OCT images to localise and segment multiclass retinal biomarkers, using edge-focused boundary supervision and multiscale dual-encoder fusion, with evaluation reported on both local and public datasets as evidence for its effectiveness against noise, class imbalance, and small-marker errors.	MSCS-Net combines a parallel dual-encoder CNN-Swin architecture with an added edge detection path, a revised Swin window partition, and modules for dimensionality reduction of small biomarkers plus cross-fusion of global and local features, aiming to improve segmentation accuracy in challenging OCT biomarker settings.
[143]: Multi-scale convolutional neural network for automated AMD classification using retinal OCT images	2022	The authors develop a feature pyramid network (FPN)-style multi-scale CNN with feature fusion for automated classification of normal, dry AMD, and wet AMD (drusen and choroidal neovascularisation) from retinal OCT images, supported by evaluations on two OCT datasets and performance comparisons against other OCT classification frameworks, with heatmaps used as evidence for lesion localisation across scales.	The architecture fuses OCT features across scales and improves reported discrimination among normal, dry AMD, and wet AMD across the tested backbones. Heatmaps provide qualitative localisation evidence but do not validate clinical interpretation.
[333]: Multi-scale local-global transformer with contrastive learning for biomarkers segmentation in retinal OCT images	2024	The authors develop a multi-scale local-global transformer using contrastive learning for pixel-wise segmentation of retinal biomarkers in OCT. Experiments on one public dataset and one local dataset report performance improvements relative to the existing methods evaluated.	MsLGT-Net combines dedicated multi-scale fusion attention, a local-global transformer module, and a contrastive learning enhancement component to better capture small-structure detail and class-specific biomarker features for retinal OCT segmentation.

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Work	Year	Basis for inclusion	Principal contribution
[214]: Non-transfer Deep Learning of Optical Coherence Tomography for Post-hoc Explanation of Macular Disease Classification	2021	A non-transfer deep convolutional neural network classifies four macular disease conditions from monochromatic spectral-domain retinal OCT images, and it provides evidence through post-hoc kernel/feature analysis and expert review of misclassified cases to support explanation of model behaviour.	Feature analysis shows that only eight of 256 final-layer convolutional kernels remain active and that the resulting selective features maintain strong class separability in reduced 2D embeddings, while clinical interpretation of the specific misclassification patterns highlights where confounding imaging artefacts and textures drive errors.
[250]: Novel Deep Learning Model for Glaucoma Detection Using Fusion of Fundus and Optical Coherence Tomography Images	2025	A two-branch multimodal deep-learning model is evaluated that ingests paired fundus photographs and optical coherence tomography scans from the same eye to classify glaucoma, benchmarking against fundus-only and OCT-only baselines and reporting robustness via five-fold cross-validation on a private clinical dataset with fixed test-set results.	Comparison of a ResNet-based single-modality pipeline with a feature-level fusion network indicates that combining fundus and OCT representations yields higher glaucoma detection performance than either modality alone with statistical significance tests on validation predictions.
[206]: OCT(2)Former: A retinal OCT-angiography vessel segmentation transformer	2023	The authors introduce OCT2Former, an end-to-end transformer encoder-decoder model for semantic segmentation of retinal vessels in OCTA imaging, designed to better capture thin-vessel structure and preserve vessel connectivity. Jaccard-based comparisons on three public OCTA datasets evaluate the model against convolutional baselines.	The architecture combines a dynamic token aggregation transformer, an auxiliary convolution branch, and a lightweight decoder, and reports improved vessel-segmentation accuracy over the best convolutional approaches across multiple OCTA datasets, with the comparison confined to the reported benchmark datasets.
[271]: OCT-angiography based artificial intelligence-inferred fluorescein angiography for leakage detection in retina [Invited]	2023	For OCTA-based leakage assessment, the authors build an OCT-angiography-to-fluorescein angiography (FA) image-domain deep learning pipeline using a U-Net-type convolutional neural network to infer retinal vascular leakage, with training evidence derived from diabetic retinopathy FA and OCTA pairs and qualitative comparison to real FA in additional retinal vascular conditions.	The results demonstrate time-aware AI-inferred FA generation from same-day OCTA, including reproducible leakage depiction for diabetic retinopathy and evaluation by structural similarity against corresponding FA, providing evidence for OCTA-based leakage assessment without contrast-dye acquisition steps.

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Work	Year	Basis for inclusion	Principal contribution
[203]: OCT-based diagnosis of glaucoma and glaucoma stages using explainable machine learning	2025	For glaucoma diagnosis and staging, the authors develop explainable machine-learning classifiers using RNFL, ganglion cell–inner plexiform layer (GC-IPL), and macular thickness features together with frequency-domain temporal–superior–nasal–inferior–temporal (TSNIT) signals. The models distinguish normal, early, moderate, and advanced disease. Prospective clinical data, SHAP, and partial-dependence analyses are used to characterise model behaviour.	After model training with feature selection and an OCT-derived clinical feature set, the accompanying software translates SHAP global feature ranking and partial dependency cut-off estimates into feature-level decision support for glaucoma staging.
[56]: OCT DEEPNET 1—A Deep Learning Approach for Retinal OCT Image Classification	2023	The authors evaluate OCT DEEPNET 1, a deep learning model for retinal OCT imaging, to classify images into four clinically defined categories (normal, choroidal neovascularisation, diabetic macular oedema, and Drusen) using random-search hyperparameter tuning and reported classification metrics, supporting evidence for automated ophthalmic diagnosis from OCT data.	The hyperparameter-tuned OCT classifier reports an 88% accuracy at a batch size of 32, which helps quantify the feasibility of deep learning for four-way retinal OCT disease categorisation.
[446]: OCT Retinal and Choroidal Layer Instance Segmentation Using Mask R-CNN	2022	The authors evaluate a mask region-based convolutional neural network (Mask R-CNN) deep-learning approach for instance segmentation of retinal and choroidal layer boundaries in posterior-segment OCT images. The approach targets automatic delineation of seven retinal layers, with quantitative evidence reported through boundary error and Dice coefficients against a U-Net baseline.	The comparison indicates that a region-proposal-based instance-segmentation strategy can achieve performance comparable to U-Net while enabling faster boundary-position extraction, reducing inference time by 2.5-fold for the OCT layer-segmentation task.
[67]: OCT-Trans: A Novel Transformer Backbone with Multimodal Feature Extraction in OCT-Based Retinal Disease Classification	2025	For OCT-based retinal disease classification, OCT-Trans uses a transformer backbone that extracts multimodal features from OCT images acquired with two devices, with atlas-based eleven-layer segmentation feeding layerwise clinical indicators to classify normal retina versus DR and AMD, validated on 541 patients and benchmarked against established model families.	OCT-Trans integrates layer-specific morphologic and texture features from automated eleven-layer segmentation into transformer-derived contextual embeddings and a multilayer perceptron classifier, showing statistically significant gains over prior baselines and reporting high agreement and classification scores.

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Work	Year	Basis for inclusion	Principal contribution
[242]: OCT Vision: Retinal Disease Detection Using Deep Learning	2024	The authors use deep learning for retinal disease classification from OCT images, tested with VGG16 and ResNet50 in TensorFlow and reported as an image-based diagnostic decision-support workflow that outputs real-time, confidence-scored predictions via a Streamlit interface.	The evaluation compares OCTDL-trained classifiers for AMD, DME, and RVO and implements confidence-scored predictions in a Streamlit interface. Effects on screening timeliness or clinical interpretation were not tested.
[232]: OctNET: A Lightweight CNN for Retinal Disease Classification from Optical Coherence Tomography Images	2021	OctNET is a lightweight ophthalmic deep CNN that uses gradient-based class activation mapping to classify diabetic macular oedema, drusen, and choroidal neovascularisation from optical coherence tomography images, with evidence reported from training and validation on a public labelled dataset.	OctNET uses a six-block downsampling and weight-sharing architecture that markedly reduces learnable parameters while producing high classification performance and providing model explanations via class activation maps for OCT-based computer-aided diagnosis.
[409]: Optic Cup and Disc Segmentation of Fundus Images Using Artificial Intelligence Externally Validated With Optical Coherence Tomography Measurements	2025	For optic-nerve-head quantification, the study builds a deep learning fundus optic cup and disc segmentation pipeline to derive VCDR from retinal photographs, then externally validates VCDR measurements against OCT VCDR and benchmarks against electronic health record (EHR) and expert assessments on retrospective hospital data and public datasets.	An ONH detector, transformer-based segmentation model, and standardised crop-and-square-resize procedure are benchmarked against OCT, EHR, and expert VCDR measurements. The reported cross-dataset agreement and same-day repeatability do not establish near-expert clinical use.
[50]: Optical coherence tomography-based diabetic macula edema screening with artificial intelligence	2020	In retrospective DME screening data, the study uses convolutional neural networks to classify macular OCT cross-sectional images into DME versus non-DME, and it provides evidence from internal validation/verification with reported classification accuracy, sensitivity, and specificity.	The study trains two CNN architectures (VGG16 and InceptionV3) on labelled OCT images and packages the resulting models into an OCT-driven screening platform intended to reduce manual clinician grading of DME.
[153]: Optical coherence tomography image based eye disease detection using deep convolutional neural network	2022	For four-class OCT diagnosis, the authors build an attention- and transfer-learning-enhanced deep convolutional neural network for multi-class image classification of CNV, DME, drusen, and normal, with the clinical motivation of reducing diabetic retinopathy screening workload and evaluation reported with standard diagnostic metrics on a predefined training/testing split.	The OCT-based VGG16_Attention pipeline couples ImageNet transfer learning with attention and reports comparative results showing the model achieves substantially higher classification performance than several other CNN baselines on a 1000-image test set across the labelled disorders.

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Work	Year	Basis for inclusion	Principal contribution
[463]: Optical coherence tomography image classification for retinal disease detection using artificial intelligence	2024	The chapter reviews an ophthalmic deep learning approach for retinal disease classification from optical coherence tomography images, using convolutional neural networks with pretrained and transfer learning components to detect drusen, diabetic macular oedema, and choroidal neovascularisation, and discusses evidence from the reported diagnostic strategy and its potential clinical impact.	The chapter outlines a specific OCT-based pipeline that leverages convolutional neural networks plus pretrained image classification models and derived features to support automated screening and earlier identification of clinically important retinal conditions, aiming to streamline diagnostic workflows and reduce health care burden.
[326]: Optimizing Deep Learning Based Retinal Diseases Classification On Optical Coherence Tomography Scans	2023	The authors apply convolutional neural networks with transfer learning and fine-tuning to OCT B-scans for automated three-way retinal disease classification (CNV, drusen, DME) versus healthy eyes, using a publicly available OCT2017 dataset and reporting evaluation using standard multiclass metrics and confusion matrices.	The comparison evaluates EfficientNet variants B0, B3, and B5 under an imbalanced multiclass setting and identifies EfficientNet B0 as the best-performing model. The results show the effect of image enhancement and resolution scaling with ImageNet-pretrained weights on benchmark classification, but do not establish diagnostic or decision-support performance in clinical populations.
[142]: Optimizing glaucoma diagnosis using fundus and optical coherence tomography image fusion based on multi-modal convolutional neural network approach	2024	In 2024, the authors investigate glaucoma diagnosis using an ophthalmic multimodal deep learning setup that fuses segmented fundus images with segmented optical coherence tomography images, employing a multimodal convolutional neural network (MMCNN) and using GoogLeNet as a unimodal comparator, with evidence from reported classification experiments on paired, segmented fundus-image and OCT-image data.	The MMCNN pipeline combines features from segmented fundus and OCT streams and provides evaluation results showing improved glaucoma detection and classification performance versus unimodal processing, supported by segmentation-based preprocessing and ROC/accuracy-related reporting.
[51]: Optimizing the OCTA layer fusion option for deep learning classification of diabetic retinopathy	2023	From OCTA en-face images of the superficial capillary plexus, deep capillary plexus, and choriocapillaris, the study trains a CNN to classify diabetic retinopathy stages (control, NoDR, NPDR) and compares single-layer versus early-, intermediate-, and late-fusion architectures, with Grad-CAM used for interpretability, supported by quantitative evaluation in an ophthalmic DR setting.	The comparison identifies the intermediate-fusion strategy as the best OCTA layer-combination approach for DR staging with the CNN and provides activation-map evidence on which spatial features drive classification decisions.

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Work	Year	Basis for inclusion	Principal contribution
[419]: Patch-based CNN for corneal segmentation of AS-OCT images: Effect of the number of classes and image quality upon performance	2023	A patch-based CNN is evaluated on AS-OCT images to automatically delineate three corneal boundaries (epithelium, Bowman's layer, and endothelium), evaluating how label-class configuration and training image-quality distribution affect segmentation accuracy and boundary error using Dice/probability and boundary error metrics.	The experiments quantify how moving from 4 to 8 boundary classes and varying training data quality impacts corneal segmentation behaviour, providing evidence that patch-based learning can perform competitively while showing limited sensitivity to image-quality shifts in this AS-OCT setting.
[110]: Performance of Artificial Intelligence in Detecting Diabetic Macular Edema From Fundus Photography and Optical Coherence Tomography Images: A Systematic Review and Meta-analysis	2024	A systematic review and meta-analysis evaluates AI models for detecting diabetic macular oedema using fundus photography (FP) and optical coherence tomography, pooling diagnostic performance and assessing how internal versus external validation and training characteristics relate to evidence quality in ophthalmic imaging.	The synthesis of 53 studies reports pooled accuracy estimates for FP- and OCT-based approaches and identifies deep learning and training-data diversity as factors associated with higher reported performance. It also underscores the need for more external validation.
[89]: Performance of retinal fluid monitoring in OCT imaging by automated deep learning versus human expert grading in neovascular AMD	2023	In neovascular AMD, the authors evaluate a deep learning segmentation approach for intraretinal and subretinal fluid on SD-OCT volumes, comparing automated fluid presence/absence and 3D fluid volume quantification against expert reading-centre gradings and relating the results to central retinal thickness and volume-derived biomarkers during anti-VEGF monitoring in clinical-trial OCT data.	The head-to-head comparison indicates that automated fluid quantification from OCT can perform reliably relative to certified experts and shows that central subfield thickness is a weak surrogate for fluid activity, supporting objective, volume-based monitoring of anti-VEGF treatment response.
[308]: Performances of Machine Learning in Detecting Glaucoma Using Fundus and Retinal Optical Coherence Tomography Images: A Meta-Analysis	2022	In 2022, a meta-analysis evaluates machine learning diagnostic models for glaucoma detection in an ophthalmic setting by pooling evidence from fundus and retinal OCT imaging, estimating pooled sensitivity, specificity, and AUC using a bivariate random-effects approach across classifier categories and dataset types.	Synthesis of 105 studies with modality-specific pooled performance clarifies how ML accuracy varies between fundus and OCT and across clinical versus external testing, informing whether these imaging sources and model classes are suitable for glaucoma screening or diagnosis.

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Work	Year	Basis for inclusion	Principal contribution
[325]: Predicting 10-2 Visual Field From Optical Coherence Tomography in Glaucoma Using Deep Learning Corrected With 24-2/30-2 Visual Field	2021	Across multiple centres, the authors use macular spectral-domain OCT with a convolutional neural network to predict HFA 10-2 threshold sensitivities in glaucoma, including open-angle glaucoma (OAG), and then test whether a correction informed by the same eye's HFA 24-2/30-2 results improves the pointwise visual field prediction, with evidence from an independent testing cohort.	The analysis quantifies lower absolute and pointwise errors after applying an HFA 24-2/30-2-based adjustment to CNN-generated 10-2 thresholds in the independent test cohort. Whether the approach can replace or reduce 10-2 perimetry was not evaluated.
[436]: Predicting Clinician Fixations on Glaucoma OCT Reports via CNN-Based Saliency Prediction Methods	2024	In 2024, the glaucoma study uses a CNN-based saliency prediction model (TranSalNet) to learn how ophthalmologists fixate on specific regions of glaucoma OCT reports by training on eye-tracking-derived fixation heat maps and evaluating agreement between predicted and observed fixation saliency with quantitative image-similarity metrics via 5-fold cross-validation.	The resulting saliency maps provide an initial, OCT-specific mapping from gaze behaviour to model-predicted medically salient regions on glaucoma report sub-images, showing that fixation regions can be predicted with reasonable qualitative correspondence while highlighting that improved data quantity and quality are needed for stronger pixel-level accuracy.
[380]: Predicting HFA 30-2 Visual Fields with Deep Learning from Multimodal OCT-Fundus Feature Fusion and Structure-Function Discordance Analysis	2026	The authors use a ViT-B/32 feature-fusion approach on multimodal OCT and fundus-derived structural inputs, including disc and red-free photographs and RNFL thickness maps, to predict HFA 30-2 visual-field outcomes. These outcomes include MD, PSD, and pointwise threshold sensitivity (TS) in glaucoma and ocular hypertension (OHT). Uncertainty-aware regression and sensitivity analyses excluding cases with structure–function discordance (SFD) assess how this discordance affects prediction reliability.	Sensitivity analysis quantifies the practical impact of structure-function discordance on forecast accuracy by reporting improved VF metric agreement after excluding discordant eyes, thereby highlighting that low-performing predictions may reflect SFD rather than failure of the OCT-fundus fusion model.
[403]: Predicting macular hole surgery outcomes: Integrating preoperative OCT features with supervised machine learning statistical models	2025	In a retrospective cohort, the authors apply supervised machine learning, evaluating multiple classifiers with a random forest model, to preoperative spectral-domain OCT features from idiopathic full-thickness macular hole eyes to predict dichotomous postoperative anatomical closure status, using an 80:20 training/test split with internal validation and external testing.	With OCT-derived macular-hole area as its highest-importance preoperative feature, the random forest classifier delivered the best internal and external predictive performance for postoperative hole closure, offering a practical, evidence-based decision-support approach grounded in routine OCT measurements.

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Work	Year	Basis for inclusion	Principal contribution
[407]: Predicting optical coherence tomography-derived diabetic macular edema grades from fundus photographs using deep learning	2020	The authors use 2D colour fundus photographs to predict OCT-derived centre-involved diabetic macular oedema (ci-DME) grade and retinal fluid presence, with AUC and sensitivity/specificity reported against expert OCT grading as reference on retrospective clinical validation sets.	In comparison with retinal specialists, the model achieved similar sensitivity with substantially higher specificity for ci-DME detection from routine fundus imaging, and extends prediction to intraretinal and subretinal fluid to support screening-style referral decisions.
[282]: Predicting Optical Coherence Tomography-Derived High Myopia Grades From Fundus Photographs Using Deep Learning	2022	In a retrospective cohort, the authors use fundus photographs labelled from OCT to predict OCT-derived ATN classification and grading for high myopia, and report test-set discrimination and comparison against attending ophthalmologists and retinal specialists.	The model provides a fundus-only inference framework for OCT-referenced atrophy, traction, and neovascularisation grading in high myopia. Reported test-set performance is compared with retinal specialists, and inference uses only fundus photographs.
[410]: Predicting the central 10 degrees visual field in glaucoma by applying a deep learning algorithm to optical coherence tomography images	2021	In retrospective glaucoma data, convolutional neural networks use macular OCT retinal-layer thickness maps to predict Humphrey Field Analyzer 10-2 total deviation across the central 10 degrees. The maps comprise RNFL, GCC, and outer segment plus retinal pigment epithelium (OS+RPE) measures. Predictions are evaluated against measured HFA 10-2 fields and corrected using overlapping inner points from HFA 24-2 fields.	The results indicate that incorporating an HFA 24-2-based adjustment substantially reduces prediction error for HFA 10-2 total deviation compared with uncorrected CNN outputs, supporting an OCT-to-central visual-field modelling strategy informed by clinically available 24-2 data.
[370]: Predicting the Glaucomatous Central 10-Degree Visual Field From Optical Coherence Tomography Using Deep Learning and Tensor Regression	2020	For central visual-field prediction, the authors apply deep learning with CNNs plus tensor regression to predict Humphrey 10-2 pointwise visual-field sensitivity from OCT-derived retinal layer thicknesses, evaluated with 5-fold cross-validation using RMSE to benchmark against linear and support vector regression (SVR) baselines.	Cross-validation quantifies the central 10 VF prediction error achievable from macular OCT thickness inputs, showing lower RMSE and mean absolute error for CNN-tensor regression than CNN pointwise mapping, SVR, and multiple linear regression, supporting OCT-to-visual-field translation.

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Work	Year	Basis for inclusion	Principal contribution
[385]: Predicting treat-and-extend outcomes and treatment intervals in neovascular age-related macular degeneration from retinal optical coherence tomography using artificial intelligence	2022	In a post-hoc analysis of neovascular AMD, CNN-based segmentation derived quantitative OCT biomarkers from baseline and four-week follow-up scans. These biomarkers included intraretinal and subretinal fluid, PED, hyperreflective foci, and photoreceptor-layer thickness. Machine learning with ten-fold cross-validation then predicted visual response under a treat-and-extend regimen and the likelihood of extending treatment intervals. The retrospective analysis used data from a prospective randomised clinical trial.	The results indicate that features computed after the first injection from OCT fluid and retinal integrity maps can stratify patients into visual responders and an extendable-vs-non-extendable treatment-interval group, offering an imaging-driven decision-support approach for individualising treat-and-extend management from early follow-up data.
[144]: Predicting visual field global and local parameters from OCT measurements using explainable machine learning	2025	For glaucoma cases in which standard perimetry is impractical, the cross-sectional study uses interpretable supervised regression (XGBoost, SVM, and random forest) on CIRRUS spectral-domain OCT-derived thickness features (RNFL, GC-IPL, and macular thickness) to predict 24-2 visual field global indices and local pointwise sensitivities, with evidence from five-fold cross-validation and SHAP-based explanation of feature contributions.	The OCT-to-VF Predictor pairs model performance reporting with SHAP analyses to highlight clinically relevant OCT parameters and support transparent, practitioner-facing interpretation of predicted visual field outcomes.
[140]: Predicting Visual Field Worsening with Longitudinal OCT Data Using a Gated Transformer Network	2023	Using a retrospective longitudinal glaucoma cohort from the Johns Hopkins Wilmer Eye Institute, a gated transformer network (GTN) ingests serial OCT scans to predict the probability of future visual field worsening, using AUC-based comparisons across multiple trend- and event-based progression definitions (including a majority-of-6 consensus) and assessing effects of baseline severity.	GTNs trained with at least five longitudinal OCT images and benchmarked against linear mixed-effects models and naive Bayes classifiers under matched inputs and reference standards discriminate visual field decline across definitions, with performance declining as baseline glaucoma severity increases.
[61]: Prediction and Detection of Glaucomatous Visual Field Progression Using Deep Learning on Macular Optical Coherence Tomography	2024	In retrospective glaucoma data, the authors use a self-supervised vision transformer trained on unlabelled macular OCT B-scan images to both detect concurrent and predict future glaucomatous visual field progression, with evidence based on ROC-based diagnostic performance for predefined mean deviation progression outcomes.	Large-scale self-supervised pretraining supports concurrent progression detection and near-term progression-risk estimation within the retrospective cohort. Use for identifying progressing eyes in clinical practice was not evaluated.

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Work	Year	Basis for inclusion	Principal contribution
[396]: Prediction of age-related macular degeneration disease using a sequential deep learning approach on longitudinal SD-OCT imaging biomarkers	2020	The authors model sequential exudation risk in non-exudative AMD using longitudinal SD-OCT radiomic biomarkers, demographic variables, and visual measures. A recurrent neural network comprising stacked LSTM units generates predictions at three-month intervals up to 21 months. The model is evaluated by patient-level cross-validation in the retrospective HARBOR trial dataset and tested externally on a real-world SD-OCT dataset from Bascom Palmer Eye Institute.	The hybrid sequence model uses engineered OCT drusen features aggregated across visits to forecast future conversion timing, providing evidence of strong short-term discriminative ability internally and partial generalisation externally that could support risk-stratified follow-up intervals.
[423]: Prediction of Axial Length From Macular Optical Coherence Tomography Using Deep Learning Model	2024	In retrospective internal and external cohorts, the authors use a ResNet-152 model to predict axial length from horizontal and vertical macular OCT images, evaluated with internal and external test sets using mean absolute error, R-squared, and error-range analyses.	The comparison shows that a dual-input configuration combining horizontal and vertical OCT sections yields the most accurate axial-length estimates, supporting the clinical relevance of macular OCT structure for modelling axial elongation.
[451]: Prediction of Causative Genes in Inherited Retinal Disorders from Spectral-Domain Optical Coherence Tomography Utilizing Deep Learning Techniques	2019	For genotype-category prediction, the authors apply deep learning to central foveal spectral-domain optical coherence tomography for a four-class task that predicts the causative gene category (ABCA4, RP1L1, EYS, or Normal) in inherited retinal disorders, using repeated random train-test splits to report classification performance evidence.	Repeated train–test evaluations provide proof-of-concept evidence for OCT-based gene-category prediction with the Medic Mind platform in the molecularly confirmed cohort. The study does not establish gene-level screening performance in general clinics.
[450]: Prediction of Central Visual Field Measures From Macular OCT Volume Scans With Deep Learning	2023	For glaucoma structure–function modelling, the authors use a 3D DenseNet121 deep learning model to predict central 10-2 visual field mean deviation, threshold sensitivity, and total deviation from macular OCT volume scans, with evidence based on ground-truth comparison using correlations and mean absolute error across 10-fold stratified cross-validation.	The study refines structure-function inference by training and evaluating a segmentation-free network on raw macular OCT volumes. Within the reported cross-validation, its predictions for global and pointwise central field metrics outperform baseline log-log linear models based on GCIPL thickness. Whether the OCT-derived estimates can supplement central perimetry in practice was not evaluated.

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Work	Year	Basis for inclusion	Principal contribution
[199]: Prediction of coronary artery disease using retinal optical coherence tomography angiography imaging and electronic health records: a multimodal machine learning approach	2026	Using retrospective OCTA and EHR data, the authors evaluate a multimodal AI model for predicting coronary artery disease that combines OCTA-derived retinal microvascular biomarkers from 3x3 mm macular scans with EHR variables, using Transformer-enhanced multi-structure segmentation to quantify vascular metrics and an XGBoost classifier for coronary artery disease detection, with performance reported via internal hold-out testing using standard discrimination and calibration metrics.	The retina–coronary feature pipeline selects 4 OCTA biomarkers and 13 EHR predictors via LASSO and models nonlinearity with restricted cubic splines, then compares the integrated OCTA-EHR approach with EHR-only and multiple deep-learning baselines in the internal hold-out set. External validation and clinical risk-stratification utility remain untested.
[101]: Prediction of neovascular age-related macular degeneration recurrence using optical coherence tomography images with a deep neural network	2024	In a retrospective nAMD cohort, the authors use spectral-domain OCT images with a DenseNet201-based multi-instance architecture to predict neovascular age-related macular degeneration recurrence within 3 months (after 1 month of follow-up post-third loading anti-VEGF injections), and evaluate performance by comparing model prediction accuracy against six ophthalmologists on a held-out set of 100 cases.	The comparison indicates that incorporating the pre-injection OCT alone versus sequential OCTs immediately after the first, second, and third injections changes reported model performance. On the held-out set, model accuracy is reported as higher than clinician accuracy, and Grad-CAM highlights OCT regions associated with recurrence predictions.
[290]: Prediction of post-operative visual acuity in patients with age-related cataracts using macular optical coherence tomography-based deep learning method	2023	For cataract-surgery prognosis, the authors used macular OCT B-scans plus extracted macular morphology indices and preoperative best-corrected visual acuity to predict 1-month postoperative visual acuity (VA) and whether vision improved by at least two lines in patients with age-related cataracts, with evidence reported from regression and classification performance on held-out validation and test sets.	Comparison of multiple input configurations showed that fusing horizontal and vertical macular OCT with quantitative foveal and retinal morphological features together with preoperative BCVA produced the best post-operative VA prediction and improvement classification, supporting OCT morphology-based prognostication for cataract surgery planning.
[136]: Prediction of response to anti-vascular endothelial growth factor treatment in diabetic macular oedema using an optical coherence tomography-based machine learning method	2021	For DME treatment-response prediction, the authors use deep learning feature extraction and a random forest classifier to predict anti-VEGF response at baseline, comparing model outputs with ophthalmologists and using five-fold cross-validation to support evidence for classification of good versus poor responders.	Feature-importance analysis identifies the sum of hyper-reflective dots as the most informative OCT feature and evaluates whether the self-explainable RF model outperforms human assessment for anticipating treatment response, which could inform individualised anti-VEGF decision-making.

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Work	Year	Basis for inclusion	Principal contribution
[391]: Prediction of visual field from swept-source optical coherence tomography using deep learning algorithms	2020	For OCT-to-visual-field estimation, the authors use ophthalmic deep learning to predict 24-2 visual field metrics from wide-angle swept-source OCT structure maps in glaucoma, training and testing multiple Inception family networks with RMSE-based evidence comparing model performance across glaucoma severity and Garway-Heath sectors.	The comparison identifies Inception-ResNet-v2 as the top performer for OCT-to-VF prediction and shows that error increases with glaucoma progression. These results support technical visual-field estimation, but whether the approach can supplement standard perimetry was not evaluated.
[211]: Progressive Neural Networks for Continuous Classification of Retinal Optical Coherence Tomography Images	2023	For retinal OCT lesion classification, progressive neural networks tackle cross-device performance drop by casting multi-device learning as domain-incremental tasks and evaluating evidence from experiments on public medical image datasets for both stability and forgetting.	The method applies a multi-granularity feature space constraint implemented with progressive neural networks to balance stability and plasticity across incremental domains, improving average accuracy while reducing forgetting versus selected baselines.
[187]: Proof-of-Concept Analysis of a Deep Learning Model to Conduct Automated Segmentation of OCT Images for Macular Hole Volume	2022	A proof-of-concept analysis applies a fully three-dimensional convolutional neural network to swept-source OCT to segment macular holes and quantify volume, using correlation with surgeon-defined manual volumetry as the main evidence for task-relevant automated retinal imaging in idiopathic full-thickness macular holes.	Automated macular-hole volumes showed high agreement with surgeon-defined manual volumes. Within the 1-year follow-up, the study also compared volumetric change with minimum linear diameter.
[439]: Quantification of Nonperfusion Area in Montaged Widefield OCT Angiography Using Deep Learning in Diabetic Retinopathy	2021	Using cross-sectional DR data, a deep learning segmentation model processes montaged widefield OCT angiography (three 6x6 mm scans) to automatically detect and quantify nonperfusion area in the superficial vascular complex for staging clinically meaningful DR severity thresholds, with evidence from five-fold cross-validation against majority-voted manual ground truth and diagnostic comparisons using sensitivity/AUC with specificity fixed at 95%.	Training the network to distinguish true nonperfusion from widefield-specific low-signal shadow artefacts and aggregating results across nasal, macular, and temporal regions provides an automated, artefact-aware workflow for nonperfusion quantification that improves diagnostic sensitivity and correlates with DR severity (including performance superior to macular-only measurements for several clinically relevant comparisons).

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Work	Year	Basis for inclusion	Principal contribution
[320]: Quantification of Retinal Nerve Fibre Layer Thickness on Optical Coherence Tomography with a Deep Learning Segmentation-Free Approach	2020	The authors introduce a segmentation-free deep learning model that predicts global retinal nerve fibre layer thickness directly from raw Spectralis spectral-domain OCT B-scans, and evaluate it against conventional segmentation-based RNFL measurements across image-quality tiers including scans with segmentation errors and other artefacts, providing correlation-based agreement evidence.	The comparison shows that an RNFL thickness regression network trained on unsegmented, high-quality B-scans can match conventional outputs in good-quality imaging and yield closer estimates to an intra-eye, same-day best-available reference when conventional segmentation is unreliable.
[105]: Quantifying the Characteristics of Diabetic Retinopathy in Macular Optical Coherence Tomography Angiography Images: A Few-Shot Learning and Explainable Artificial Intelligence Approach	2025	In retrospective OCTA data, the authors use a few-shot learning framework with an intrinsic self-attention explainable AI component (MTUNet/SCOUTER) to stage diabetic retinopathy as PDR versus NPDR from macular OCTA images, supported by model-level evidence including standard classification metrics and class-wise matching/attention outputs.	The class-wise results illustrate an OCTA-based data-efficient DR staging approach by coupling prototypical few-shot classification with pattern-matching explanations, using ResNet-50 as the selected backbone and reporting class-specific matching scores that indicate how the model emphasises nonperfusion-related regions for NPDR and neovascularisation for PDR.
[342]: Real-time retinal layer segmentation of adaptive optics optical coherence tomography angiography with deep learning	2020	For en face retinal-layer analysis from adaptive optics optical coherence tomography angiography, the authors use a U-Net family deep-learning approach to segment 3D OCT volumes and generate real-time projections suitable for image-guided surgical workflows. The implementation prioritises low-latency operation rather than clinical comparative outcomes.	A compressed CNN deployed on GPU hardware enables very low-latency retinal-layer segmentation and en face OCT/OCTA projection generation, which can support near-instant visual feedback during acquisition.
[158]: Real-time retinal layer segmentation of OCT volumes with GPU accelerated inferencing using a compressed, low-latency neural network	2020	For real-time OCT, the pipeline segments eight retinal layer boundaries in volumetric OCT with a compressed U-Net family model accelerated on GPUs via TensorRT, enabling millisecond inference and low-latency en face OCT/OCTA visualisation during continuous scanning.	The end-to-end OCTViewer pipeline compresses and optimises the segmentation network and runs an integrated GPU workflow to update en face layer projections with only 41 ms total pipeline latency, supporting immediate operator feedback during in vivo imaging.

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Work	Year	Basis for inclusion	Principal contribution
[256]: Recent Advanced Deep Learning Architectures for Retinal Fluid Segmentation on Optical Coherence Tomography Images	2022	An OCT survey reviews deep learning approaches for automated retinal fluid segmentation, focusing on semantic/instance segmentation of OCT-derived retinal fluid classes (IRF, SRF, PED) using CNNs, fully convolutional networks, U-Nets, and related hybrid architectures and summarising evidence from published studies and public OCT benchmark datasets.	The survey organises the main deep learning model families used for retinal fluid delineation on OCT, outlines commonly employed datasets and labelling settings, and discusses future directions to support reproducible macular change assessment research.
[464]: Repeatability of Microperimetry in Areas of Retinal Pigment Epithelium and Photoreceptor Loss in Geographic Atrophy Supported by Artificial Intelligence-Based Optical Coherence Tomography Biomarker Quantification	2025	In a prospective reliability study of geographic atrophy, the authors co-registered fundus-controlled microperimetry test–retest data from CenterVue MAIA and NIDEK MP3 with AI-based OCT biomarker quantification of RPE and photoreceptor integrity loss. They then evaluated pointwise sensitivity repeatability using Bland-Altman coefficients alongside interdevice correlation, scotoma-detection repeatability, and fixation-stability metrics.	The analysis provides device-specific reference estimates of pointwise microperimetry repeatability across biomarker-defined retinal areas. Photoreceptor integrity-loss patterns, with or without RPE loss, and related OCT features are associated with greater test–retest variability, supporting more accurate longitudinal interpretation and trial-endpoint planning.
[169]: Research on deep learning-based intelligent diagnosis algorithms for OCT medical images of macular edema	2019	For OCT-based macular-oedema analysis, the authors apply Faster R-CNN for lesion area detection and a U-Net with multi-attention modules for lesion segmentation to support ophthalmic diagnosis with results reported on an OCT fundus dataset.	The pipeline refines Faster R-CNN label generation for OCT lesion localisation and introduces multi-attention fusion in U-Net decoding to segment oedema regions and enable severity quantification. The reported evaluation measures lesion localisation, segmentation, and severity quantification on the study dataset rather than effects on clinician speed or accuracy.
[192]: Residual Neural Network Based Classification of Macular Edema in OCT	2019	For multilabel macular-oedema biomarkers, the authors apply a deep residual neural network to classify macular oedema imaging biomarkers, SRF and PED, directly from OCT scans, using a multi-label formulation with data augmentation, pretrained fine-tuning, and label-correlation analysis, and report results on a large competition dataset (AI Challenger).	Benchmark results indicate that a residual-network approach, strengthened by augmentation and transfer from pretrained weights while accounting for co-occurring labels, recognises SRF and PED in macular oedema OCT images in the competition dataset. The evidence is confined to that benchmark evaluation.

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Work	Year	Basis for inclusion	Principal contribution
[346]: Retinal and choroidal microvasculature and structural analysis in OCTA for refractive amblyopia diagnosis using machine learning	2025	In a case-control study, a cross-validated random forest evaluates OCTA-derived retinal thickness, choroidal thickness, and capillary plexus perfusion measurements to classify amblyopic versus non-amblyopic eyes in adolescents with refractive amblyopia, reporting diagnostic classification evidence in a labelled dataset from macular 12 by 12 mm scans.	The diagnostic pipeline integrates region-wise structural OCTA metrics from the retina and choroid and shows that selected thickness features are most informative for amblyopia discrimination even when SCP and deep capillary plexus (DCP) perfusion-density differences are not significant. The result is confined to the evaluated case-control dataset.
[49]: Retinal Boundary Segmentation in Stargardt Disease Optical Coherence Tomography Images Using Automated Deep Learning	2020	The authors evaluate a fully automated deep learning approach for OCT retinal boundary segmentation in Stargardt disease, using a fully semantic network and graph-search (FS-GS) approach to retinal boundary extraction and measuring boundary accuracy against manual ground truth as well as agreement for retinal thickness and volume quantification.	The validation supports FS-GS as a high-throughput alternative to both manual correction and existing OCT segmentation tools by delivering accurate, ground-truth-validated inner and outer retinal boundary localisation and clinically relevant thickness/volume measurements across the central 6 mm macular zone.
[251]: Retinal disease classification based on optical coherence tomography images using convolutional neural networks	2023	The authors use supervised convolutional neural networks on optical coherence tomography images to classify retinal diseases, with evidence provided through test-set comparisons of multiple CNN architectures under hyperparameter tuning.	The comparison evaluates AlexNet, VGG, Inception, and residual networks for OCT-based retinal disease classification and reports that Inception1 with tuned optimisation settings achieved the highest accuracy in the evaluated test set. This establishes a benchmark rather than evidence of clinical diagnostic utility.
[81]: Retinal Disease Classification from Bimodal OCT and OCTA Using a CNN–ViT Hybrid Architecture	2025	For bimodal retinal classification, the authors apply a multimodal CNN-ViT hybrid deep learning approach to retinal disease classification using OCT B-scans and OCTA projection maps and report test-set results on an OCTA dataset spanning AMD, DR, and normal cases.	MultiModalNet combines vascular and structural representations for multiclass AMD/DR/normal discrimination and quantifies the performance of paired OCT and OCTA inputs on the evaluated dataset.
[424]: Retinal Disease Classification from OCT Images Using Deep Learning Algorithms	2021	The authors use CNN-based deep learning with transfer-learned feature extractors on OCT cross-sectional retinal images to classify four retinal conditions (CNV, DME, drusen, and normal) via multiple binary classifier schemes and report test-set performance using quantitative metrics and ROC/confusion analyses.	The study compares two extensible four-class OCT classification pipelines built from three or four binary CNNs, combined through probability multiplication after FCN-based denoising and retina-layer cropping, and reports performance exceeding that of a multi-class CNN in the study's comparisons.

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Work	Year	Basis for inclusion	Principal contribution
[355]: Retinal Disease Classification from Retinal-OCT Images Using Deep Learning Methods	2022	A convolutional neural network processes retinal OCT cross-sectional images to perform automated classification and prediction of retinal disorders into CNV, DME, drusen, and normal categories, providing performance evidence via reported training and validation accuracies.	The Retina CNN reports that, versus an existing comparison model, the proposed approach achieved the highest reported validation accuracy, supporting its use for OCT-based retinal disease categorisation.
[465]: Retinal Disease Classification in Optical Coherence Tomography Images Using a Convolutional Neural Network Algorithm	2024	For diagnostic categorisation, the authors use a convolutional neural network with quantitative feature extraction on retinal optical coherence tomography imaging to classify retinal OCT pathology for diagnostic categorisation, reporting model efficacy on an independent dataset with comparisons to additional classical ML baselines.	The OCT-to-label workflow combines OCT-derived features with logistic regression and random forest classifiers and contrasts them against the CNN, highlighting a quantitatively grounded approach to automated retinal disease identification and categorisation.
[109]: Retinal Disease Classification Using Custom CNN Model from OCT Images	2024	The authors use OCT retinal imaging to perform multiclass classification of macular disorders using a custom convolutional neural network, with results reported in comparison to conventional machine learning and transfer learning baselines, providing evidence of an ophthalmology-relevant AI method for OCT-based diagnostic categorisation.	The authors train a custom CNN on a public OCT dataset to distinguish normal eyes from CNV, DME, and Drusen and report improved test accuracy over the evaluated alternative models, supporting the practical value of their architecture for OCT-driven retinal disease screening.
[79]: Retinal Disease Detection by Optical Coherence Tomography (OCT)-Based Deep Learning	2024	The authors evaluate OCT-derived retinal image classification for four retinal disorder categories using CNN architectures and transfer learning (including VGG16 and ShuffleNet), with the evidence described as reported classification accuracy across model variants after OCT preprocessing for enhanced image quality and edges.	Preprocessing steps targeting noise reduction and edge and gradient-contour enhancement are associated with improved deep learning performance for four-class retinal disease detection from OCT images, highlighting how image enhancement can strengthen ophthalmic OCT classification.
[390]: Retinal Disease Detection: Using Deep Learning for Accurate Diagnosis with Optical Coherence Tomography Scans	2024	The authors evaluate a CNN-based deep learning approach for ophthalmic retinal disease classification from optical coherence tomography-based imaging, targeting CNV, DME, drusen, and normal labels with evidence reported as supervised training and validation and test-set metrics.	The four-class retinal classifier is trained on 968 evenly distributed images with high overall accuracy (96.9%) and class-specific precision/recall, highlighting where confusion between DME and drusen limits performance and motivating data or feature improvements.

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Work	Year	Basis for inclusion	Principal contribution
[99]: Retinal diseases classification based on hybrid ensemble deep learning and optical coherence tomography images	2023	Using hospital OCT data, the authors develop a five-disease hybrid deep-learning and ensemble machine-learning framework, where transfer learning extracts features and KNN and XGBoost outputs are fused via soft-voting to produce class probabilities, with performance assessed on a held-out test set and reported classification accuracy metrics.	A tuned OCT feature-extraction stage with a selected KNN/XGBoost soft-voting pair provides an OCT-specific multiclass diagnostic pipeline that improves upon standalone classifier results and is deployed as a web service for rapid external predictions.
[466]: Retinal Diseases Classification System Using OCT Images Combined with CNN Models	2023	The authors evaluate an OCT-based deep learning classification approach that uses convolutional neural networks to distinguish retinal disease categories, specifically diabetic macular oedema versus age-related macular degeneration, from retinal OCT images.	The CNN workflow provides automated retinal disease category classification, emphasising early detection of DME and AMD as the intended clinical use case.
[339]: Retinal fluid segmentation in OCT images using adversarial loss based convolutional neural networks	2018	On 3D OCT volumes from the ReTOUCH challenge, a voxel-level convolutional network targets segmentation of intra-retinal fluid, sub-retinal fluid, and PED, combining cross-entropy and Dice with an adversarial loss to support higher-order mask consistency, and evaluates results on a held-out validation set via overlap and volume-difference metrics.	The architecture extends a U-Net-inspired encoder-decoder with skip connections, multi-scale feature use, and adversarial training to deliver improved segmentation and separate detection of fluid presence across vendors, providing quantitative validation that the loss design and architecture modifications enhance retinal fluid delineation.
[125]: Retinal Fluids Segmentation Using Volumetric Deep Neural Networks on Optical Coherence Tomography Scans	2020	A volumetric encoder–decoder network performs semantic segmentation on OCT volumetric imaging to delineate retinal fluids, reporting Dice-based evidence of segmentation accuracy using the RETOUCH challenge dataset and external Cirrus scanner data.	The volumetric formulation extends retinal fluid quantification by treating the task in 3D rather than 2D and reports Dice results that include the strongest performance for intraretinal fluid, supporting more reliable measurement of fluid burden on clinically used OCT scans.
[75]: Retinal Nerve Fiber Layer Features Identified by Unsupervised Machine Learning on Optical Coherence Tomography Scans Predict Glaucoma Progression	2018	In retrospective primary open-angle glaucoma data, the authors apply unsupervised PCA to wide-angle swept-source OCT retinal nerve fibre layer thickness maps to extract latent structural RNFL features and evaluate those features for detecting glaucoma and predicting 2-year progression using leave-one-out, AUC-based comparison versus standard perimetry and circumpapillary RNFL (cpRNFL) metrics.	The analysis indicates that PCA-derived RNFL components from SS-OCT outperform mean cpRNFL thickness and perimetry-based measures for glaucoma discrimination and provide a higher baseline AUC for identifying eyes classified as progressing. These retrospective results support further evaluation of imaging-based risk assessment.
[467]: Retinal OCT analysis and prediction with deep learning	2020	Deep learning is used to examine imaging-biomarker quantification and disease-progression prediction for longitudinal ophthalmic monitoring.	The proposed workflow links OCT-based biomarker quantification with prognostic prediction for longitudinal retinal-disease analysis.

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Work	Year	Basis for inclusion	Principal contribution
[239]: Retinal OCT Image Classification Based on Domain Adaptation Convolutional Neural Networks	2021	A domain-adaptation CNN addresses computer-aided diagnosis from retinal OCT, using an end-to-end domain-adaptation convolutional neural network (Inception V3) to classify six OCT categories and provide evidence via quantitative diagnostic metrics (accuracy, precision, sensitivity, specificity) compared with clinical expert performance.	The framework combines two publicly available retinal OCT datasets and presents a domain-adaptation classification framework that aims to transfer prior knowledge from a related domain to improve reliable categorisation across six OCT classes, with reported results near or above expert levels on key measures.
[113]: Retinal OCT Layer Segmentation via Joint Motion Correction and Graph-Assisted 3D Neural Network	2023	The method segments retinal layers in 3D SD-OCT by combining an axial motion-correction network with a graph-assisted 3D CNN based on a graph pyramid. It was evaluated on datasets containing healthy eyes and retinal disease, with both simulated and real deformations. Pixel- and boundary-based errors were compared with those from commercial software and earlier conventional and deep-learning methods.	Comparative experiments indicate that jointly correcting motion and performing volumetric (rather than slice-wise) segmentation improves retinal layer delineation, especially under severe deformation, while also releasing/using a large dataset with manually corrected OCT annotations to support clinically relevant benchmarking.
[131]: Retinal Optical Coherence Tomography Classification Using Deep Learning	2023	A transfer-learned network classifies retinal findings from optical coherence tomography for screening treatable conditions such as choroidal neovascularisation, diabetic macular oedema, drusen, and normal retina, with evidence reported as results comparable to human retinal specialists.	The OCT classification pipeline leverages pretrained network weights to learn patterns across multiple age-related and diabetic retinal categories. Reported comparisons with retinal specialists do not establish effects on triage or earlier clinical detection.
[45]: Retinal pathologies detection in OCT images based on Bilinear convolutional neural network	2023	A bilinear CNN (B-CNN) detects and classifies CNV, drusen, and diabetic macular oedema in an image dataset of 6000 OCT scans and compares its classification results with alternative models.	The benchmark comparison reports that the proposed B-CNN attains the highest classification score among the evaluated alternatives on the OCT dataset. No prospective early-diagnosis endpoint was tested for CNV, drusen, or DME.
[373]: Retinal Specialist versus Artificial Intelligence Detection of Retinal Fluid from OCT: Age-Related Eye Disease Study 2: 10-Year Follow-On Study	2021	In a prospective AMD comparison, the investigators evaluated intraretinal and subretinal fluid detection on spectral-domain OCT macular cube scans by benchmarking a machine-learning AI tool (Notal OCT Analyzer) against retinal specialists, using reading-centre grading as the ground truth to provide diagnostic performance evidence across two common OCT devices.	The head-to-head comparison found that the AI achieved higher accuracy with substantially higher sensitivity than retinal specialists for overall retinal fluid detection, including improved intraretinal and subretinal fluid performance, supporting AI-assisted OCT interpretation for retreatment-relevant disease activity assessment.

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Work	Year	Basis for inclusion	Principal contribution
[280]: RetOCTNet: Deep Learning-Based Segmentation of OCT Images Following Retinal Ganglion Cell Injury	2025	In a rat OCT model, the authors develop RetOCTNet, a deep learning semantic segmentation model that extracts RNFL and total retinal thickness from radial and volumetric OCT scans after retinal ganglion cell (RGC) injury, validated by comparison to human manual ground truth on held-out scan sets and by longitudinal thickness changes after optic nerve crush.	The model automates RNFL and total retina segmentation for preclinical RGC injury monitoring and quantifies longitudinal thinning across scan types, demonstrating strong agreement with human annotations and enabling consistent structural measurements for experimental glaucoma research.
[266]: RETRACTION:MAC-Leonets-A novel optimized hybrid convolutional neural networks for the segmentation and diagnosis of Edema diseases using retinal OCT images	2023	The retracted report uses retinal optical coherence tomography imaging to describe a hybrid deep-learning framework that combines convolutional layers with a single feedforward training network (SLFTN) and Lion-based hyperparameter tuning to segment and classify three oedema disease types, evaluated on a Kaggle OCT dataset with standard diagnostic performance metrics reported from model testing.	The retracted article describes an SLFTN–convolutional framework optimised with Lion search and reports very high classification accuracy for oedema segmentation and diagnosis. Because the report has been retracted, these findings should not be treated as valid evidence.
[322]: Robust and Interpretable Convolutional Neural Networks to Detect Glaucoma in Optical Coherence Tomography Images	2021	For glaucoma OCT reports, the authors evaluate CNNs on report imagery using end-to-end and fine-tuned transfer learning and a CNN ensemble. They assess robustness on a separate “Field” dataset and interpretability through Testing with Concept Activation Vectors (TCAV), together with clinician eye-fixation tracking on the same OCT reports.	TCAV analysis quantifies which OCT report sub-images and visual concepts, including RNFL and retinal ganglion cell plus inner plexiform layer (RGCP) probability maps and RGCP thickness maps, drive CNN classifications. It then compares these concept scores with glaucoma-expert fixation patterns. This provides an interpretability analysis grounded in model concepts and clinician gaze. Effects on clinical decisions were not evaluated.
[66]: Robust Deep Learning-Based Approach for Retinal Layer Segmentation in Optical Coherence Tomography Images	2022	The authors apply a deep learning method for automated retinal layer segmentation on OCT images, aiming to support ophthalmic assessment by delineating retinal layers as potential computational biomarkers for neurodegenerative disorders.	The OCT segmentation pipeline outputs the main retinal layers to facilitate evaluation of clinically relevant ocular structures and derive computational biomarkers for disease analysis.

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Work	Year	Basis for inclusion	Principal contribution
[324]: Robust Fovea Detection in Retinal OCT Imaging Using Deep Learning	2022	The authors use a deep learning, pixel-wise regression approach (PRE U-net) applied to retinal OCT B-scans with spatial location encoding to automatically localise the fovea centralis, evaluated on OCT volumes from healthy subjects and patients with nAMD, DME, and RVO to provide evidence of robustness for clinically relevant macular landmark detection.	PRE U-net is trained and validated with sampled 2D B-scans concatenated with spatial location information for end-to-end fovea detection, reporting improved robustness and performance versus state-of-the-art methods in macular OCT settings where automated landmarking can guide treatment.
[306]: SAE-wAMD: A Self-Attention Enhanced convolution neural network for fine-grained classification of wet Age-Related Macular Degeneration using OCT Images	2022	The authors evaluate a self-attention enhanced CNN (SAE-VGG16) with contrastive self-supervised pretraining to fine-grainedly classify wet age-related macular degeneration subtypes (neovascular AMD vs polypoidal choroidal vasculopathy (PCV)) from OCT images, addressing small lesion-region localisation and high image similarity using an experimental comparison on its OCT dataset.	SAE-wAMD integrates self-attention layers into a VGG16 backbone and leverages contrastive pretraining to better separate OCT instances for subtype discrimination, which is clinically meaningful because the two wet AMD subtypes require different treatment approaches.
[244]: Segmentation of choroidal area in optical coherence tomography images using a transfer learning-based conventional neural network: a focus on diabetic retinopathy and a literature review	2024	For choroidal segmentation in DR, the authors use a transfer-learning DeepLabv3 model with squeeze-and-excitation (SE) for semantic segmentation and EfficientNetB0 to delineate choroidal area in EDI-OCT B-scans and support automated luminal area and choroidal vascularity index (CVI) measurement by reporting segmentation quality and automated-manual agreement via Bland-Altman analysis.	The comparison identifies DeepLabv3+SE paired with EfficientNetB0 as the top-performing variant among several CNN backbones and provides clinically oriented evidence that automated choroid-derived luminal area and CVI can reproduce manual measurements in the validation set.
[264]: Segmentation of retinal fluid based on deep learning: application of three-dimensional fully convolutional neural networks in optical coherence tomography images	2019	A fully convolutional model is evaluated on retinal OCT volumes to segment retinal fluid, contrasting modified 2D versus improved 3D U-Net architectures and evaluating segmentation outputs with standard metrics including accuracy, F1 score, and Kappa coefficient.	The improved 3D U-Net performs end-to-end volume segmentation aimed at leveraging inter-slice spatial correlations and addressing retinal OCT category imbalance, producing fluid delineations that closely match ophthalmologist annotations.
[229]: Segmentation of retinal fluids and hyperreflective foci using deep learning approach in optical coherence tomography scans	2020	A deep-learning ensemble preprocesses OCT B-scans and combines CNN-based segmenters to identify and delineate intraretinal fluid, subretinal fluid, and hyperreflective foci in expert-annotated images. Dice coefficients quantify segmentation agreement.	The ensemble combines SEUNet, a modified DRIU, and Pix2Pix to produce automated masks for fluids and hyperreflective foci. It improved reported segmentation of small lesions relative to the selected baseline architectures.

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Work	Year	Basis for inclusion	Principal contribution
[57]: Segmentation of Retinal Layers in OCT Images Using Deep Learning Algorithms for Medical Diagnostics	2024	The study compares classical machine-learning and deep-learning approaches for retinal-layer and abnormality segmentation in OCT images. It then develops and tests a neural-network model for this task.	The side-by-side comparison culminates in training and testing a selected U-Net 3+ variant on the OCTDL dataset, with the stated goal of improving automated delineation of retinal abnormalities for diagnostic support.
[406]: Segmentation of retinal low-cost optical coherence tomography images using deep learning	2020	For AMD home monitoring, a computer-aided diagnosis pipeline applies CNN-based segmentation to low-cost self-examination full-field OCT images for delineating the retina and pigment epithelial detachments, with refinement by a convolutional denoising autoencoder to address imaging artefacts.	The comparison indicates how pairing CNN segmentation with denoising autoencoder refinement can improve automated retinal and PED delineation on SELF-OCT, supporting more reliable OCT biomarker quantification for continuous patient monitoring.
[275]: Semantic uncertainty Guided Cross-Transformer for enhanced macular edema segmentation in OCT images	2024	The authors evaluate a transformer-based, uncertainty-guided semantic segmentation model for multi-class macular oedema on OCT, targeting boundary delineation in low-contrast, noisy regions and improving separation of IRF, SRF, and PED, with evidence reported via evaluations on public datasets and OCT device data.	SuGCTNet uses semantic uncertainty-guided attention and cross-transformer modules to focus learning on ambiguous pixels while reducing inter-class feature confusion, supporting more reliable multi-class lesion segmentation where current methods struggle.
[305]: Sensitivity and specificity of machine learning classifiers for glaucoma diagnosis using Spectral Domain OCT and standard automated perimetry	2013	An observational cross-sectional study evaluates several machine-learning classifiers for distinguishing glaucomatous from healthy eyes. Inputs combine spectral-domain OCT retinal nerve fibre layer parameters with global indices from standard automated perimetry. Diagnostic performance is assessed using receiver operating characteristic analysis and statistical comparisons of combined and single-modality inputs.	The analysis reports a head-to-head comparison across nine classifier families trained on OCT-only, perimetry-only, and OCT-plus-perimetry feature sets, showing that adding OCT measures to perimetry improves diagnostic discrimination over OCT parameters alone.
[93]: SF Net: A Pyramid-Based Feature Fusion Convolutional Neural Network With Embedded Squeeze-and-Excitation Mechanism for Retinal OCT Image Classification	2025	For retinal OCT classification, SF Net uses an end-to-end multi-scale convolutional neural network with pyramid-based intermediate feature fusion and embedded squeeze-and-excitation to classify normal eyes versus early/late AMD and DME, and it provides evidence from evaluations on two datasets, including a Noor Eye Hospital dataset and a public UCSD dataset, compared with established OCT classification frameworks.	SF Net fuses intermediate feature maps for more reliable OCT-based diagnosis, reporting that multi-scale feature fusion maintains strong accuracy even when training data are substantially reduced relative to comparable network approaches.

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Work	Year	Basis for inclusion	Principal contribution
[42]: Simplified Convolutional Neural Network Model for Automatic Classification of Retinal Diseases from Optical Coherence Tomography Images	2023	A compact convolutional neural network was trained to classify OCT B-scan images from a public dataset into normal retina versus diabetic macular oedema, drusen, and choroidal neovascularisation, providing multi-class image-based evidence from a train/test evaluation with standard classification metrics.	The four-layer architecture combines resizing and normalisation with end-to-end, OCT-to-disease categorisation among four macular classes, reporting strong test accuracy to support practical deployment for ophthalmic screening workflows.
[247]: Spatial-Aware Transformer-GRU Framework for Enhanced Glaucoma Diagnosis From 3D OCT Imaging	2025	For glaucoma detection, the authors use 3D OCT volumes, combining a pretrained Vision Transformer for slice-wise feature extraction with a bidirectional GRU to model inter-slice dependencies, and evaluate the resulting classification model with cross-validation using standard diagnostic performance metrics.	The ViT-large plus bidirectional GRU framework uses full 3D OCT information through sequential processing of 64 B-scans. Reported comparisons show improved glaucoma-versus-normal discrimination relative to baseline approaches that use 3D-CNN features or slice-wise RETFound adaptation.
[226]: State-of-the-Art Deep Learning Strategies for Multi-Class Classification of Retinal OCT Images	2024	For four-class OCT diagnosis, the authors compare CNN, VGG-19, and DenseNet-121 plus ensemble learning to perform multi-class classification of CNV, DME, drusen, and normal, reporting accuracy evidence via comparisons on balanced versus imbalanced training datasets.	The experiments examine undersampling and model ensembling for multiclass OCT categorisation across four retinal-image categories. Reported balanced-accuracy results permit comparison among the backbone models on balanced and imbalanced training data.
[107]: Statistical Analysis of Deep Learning Models for Diabetic Macular Edema Classification using OCT Images	2022	The authors evaluate deep learning architectures for classifying diabetic macular oedema from OCT images, using statistical analysis of accuracy measures to support automated DME detection in an ophthalmic diagnostic setting.	The statistical comparison evaluates OpticNet and DenseNet on a standard OCT dataset and reports statistically evaluated differences that can guide selection of a model for DME classification.
[160]: Strategies to Improve Convolutional Neural Network Generalizability and Reference Standards for Glaucoma Detection From OCT Scans	2021	The authors evaluate CNNs for glaucoma detection on OCT imaging using RNFL probability maps and circumpapillary disc B-scans, testing generalisability on an unseen dataset from a different site and quantifying how different reference-standard schemes affect accuracy and robustness across models.	External-site testing reports practical gains in out-of-site performance by combining data augmentation, multimodal OCT inputs, and training on confidently labelled images, while finding that using a consistent, well-specified reference standard for both label creation and evaluation yields the best accuracy.

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Work	Year	Basis for inclusion	Principal contribution
[141]: Structural-prior guided and feature-enhanced transformer with masked image modeling pretraining for retinal layers and fluid segmentation in macular edema OCT images	2025	The authors develop a transformer-based OCT segmentation pipeline for macular oedema that uses masked image modelling self-supervised pretraining (SimMIM) alongside a structural prior-guided, feature-enhanced architecture to delineate retinal layers and intraretinal/subretinal fluid in B-scans, with evidence from reported overlap-based and quantitative thickness/volume assessments on both a public and a private macular oedema dataset.	A structural-prior constraint implemented through customised multi-class synergistic segmentation (MCSS) loss, together with dual self-attention and axial feature enhancement, targets layer order and topological continuity to improve segmentation robustness to low-contrast lesions and artefacts, which is reflected in substantially improved macular layer and fluid delineation outcomes across the evaluated datasets.
[157]: Studies of Plausibility Using Deep Learning Techniques to Automatically Detect Retinal Neovascularization in Enface Optical Coherence Tomography Angiography Images	2025	The authors evaluate a deep learning binary classification approach using en face OCT angiography to automatically distinguish retinal neovascularisation (RNV) from non-neovascularisation in diabetic retinopathy, addressing a clinically relevant prescreening task with evidence from model performance on a small, imbalanced OCTA dataset.	The feasibility assessment compares five deep learning architectures and identifies Inception V3 as the top performer under limited RNV examples. The study provides preliminary benchmark evidence under class imbalance. External validation is required before screening claims.
[300]: The Classification of Common Macular Diseases Using Deep Learning on Optical Coherence Tomography Images with and without Prior Automated Segmentation	2023	The authors compare deep learning for macular disease classification from spectral-domain OCT in an ophthalmic imaging setting, contrasting end-to-end classification against segmentation-assisted classification using automated OCT segmentation (RelayNet and a graph-cut Caserel approach) across seven disease categories plus normal/other, with performance reported using internal 5-fold cross-validation.	The cross-validation results indicate that introducing prior automated segmentation of OCT, especially with the graph-cut method, improves validation sensitivity while maintaining a high overall AUC, offering an evidence-based preprocessing strategy for OCT-based multiclass macular screening.
[106]: The classification of optical coherence tomography images using machine learning for macular hole and diabetic macular edema diagnoses	2023	The authors use a Support Vector Machine to classify retinal optical coherence tomography images for macular hole versus diabetic macular oedema, evaluated on both varied training data and real images with reported classification accuracy.	The SVM distinguishes the two retinal conditions in the evaluated OCT images and reports classification accuracy. Effects on detection timing or treatment planning were not assessed.

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Work	Year	Basis for inclusion	Principal contribution
[400]: The effect of optical degradation from cataract using a new Deep Learning optical coherence tomography segmentation algorithm	2024	The authors evaluate a freely available online deep learning segmentation method for Spectralis SD-OCT retinal images, testing how cataract-related optical degradation affects segmentation validity by comparing agreement with ETDRS-derived measurements and reviewing sensitivity via post-cataract-surgery imaging.	Agreement analyses report excellent agreement with ETDRS measures and suggest that, within the evaluated preoperative and postoperative sample, cataract-related degradation did not preclude use of the online segmentation tool for retinal layer quantification.
[217]: The possibility of the combination of OCT and fundus images for improving the diagnostic accuracy of deep learning for age-related macular degeneration: a preliminary experiment	2019	Using Project Macula imaging, the authors evaluate multimodal deep learning for age-related macular degeneration by fusing OCT and fundus images and extracting features with transfer learning from a pretrained VGG-19 before random-forest classification, providing experimental evidence that combined imaging improves diagnostic discrimination versus single-modality models.	Postmortem Project Macula data indicate that OCT-fundus combination produces higher AUC and accuracy and is statistically superior to OCT-only and fundus-only deep learning pipelines, indicating that multimodal fusion can materially boost AMD diagnostic performance in this setting.
[397]: The Predictive Capabilities of Artificial Intelligence-Based OCT Analysis for Age-Related Macular Degeneration Progression-A Systematic Review	2023	A PRISMA-guided systematic review evaluates AI-based analyses of spectral-domain OCT to predict future AMD outcomes in ophthalmic care, covering both natural-history progression and post-anti-VEGF follow-up (structure and visual function) using machine learning and deep learning evidence from 19 included studies.	The review consolidates 19 OCT prediction studies across clinically relevant tasks (anti-VEGF treatment needs and response, conversion to nAMD or GA, GA growth, and subsequent visual acuity) and critically summarises their strengths and limitations to inform where OCT-based AI prognostication is most mature and what gaps remain.
[124]: The role of artificial intelligence in monitoring glaucoma progression using optical coherence tomography	2025	The review examines AI-assisted glaucoma progression monitoring in an ophthalmic setting by synthesising OCT-based workflows that use deep learning for tasks such as segmentation, detection of subtle structural change, and progression prediction from RNFL, GCC, and ONH imaging biomarkers, drawing on reported diagnostic evidence and emphasising the need for robust clinical validation.	The synthesis distills practical considerations for translating AI-enhanced OCT into care, focusing on sources of performance risk such as device/protocol variability and limited data diversity, plus priorities like interpretability, privacy, and workflow integration, so readers can gauge how and when AI outputs may be trusted for longitudinal monitoring.
[411]: The utility of artificial intelligence in characterization and detecting causes of macular edema: A spectral-domain OCT-based algorithm study	2025	The authors use spectral-domain OCT imaging with pretrained CNNs and Grad-CAM to classify DME, AMD, and normal macular status, reporting retrospective evidence from an institutional dataset from King Abdullah University Hospital and a public benchmark with interpretability checked against annotated data.	The CNN-based framework reports classification accuracy for DME, AMD, and normal OCT images. Grad-CAM visualisations permit comparison of highlighted regions with annotations. Clinician interpretation and decision impact were not evaluated.

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Work	Year	Basis for inclusion	Principal contribution
[84]: Three-dimensional transformer-enhanced multi-scale global co-attention network for precise diabetic macular edema segmentation in OCT volumes	2025	For volumetric DME segmentation, the authors use a 3D transformer-based encoder-decoder segmentation approach augmented with non-local, multi-scale feature aggregation, and global channel-spatial co-attention to delineate DME lesions while leveraging 3D context, and they report evidence from quantitative test-set comparisons plus consistency between predicted and manual 3D lesion measurements.	The transformer-enhanced 3D framework models long-range dependencies and 3D morphology in retinal OCT and reports lesion-measurement consistency relative to manual annotations, together with performance improvements in challenging cases from comparisons with baseline 3D architectures and ablation of key attention modules.
[309]: Towards label-free 3D segmentation of optical coherence tomography images of the optic nerve head using deep learning	2020	In 2020, the authors address glaucoma-relevant 3D segmentation of optic nerve head tissues in spectral-domain OCT by pairing an enhancer for cross-device image harmonisation with ONH-Net, an ensemble-inspired 3D deep learning model. ONH-Net segments six tissue layers, and the study evaluates cross-device performance through device-wise training and testing against manual slice-wise reference labels.	Cross-device testing showed that enhancer-based harmonisation allowed ONH-Net trained on one OCT device to produce high-overlap 3D segmentations of six optic nerve head tissues from two unseen devices. The results support technical device transfer without new manual segmentations, but do not by themselves establish clinical deployment readiness.
[150]: TransDualSegNet: transformer dual segment network for retinal vasculature segmentation in OCT	2025	TransDualSegNet addresses retinal vasculature segmentation in OCT, using a Transformer-based encoder-decoder architecture with Pyramid Vision Transformer encoding plus a dual-branch, edge-aware decoder and feature fusion to predict vessel inner and outer boundaries, with evidence from comparative experiments on an ophthalmic OCT vessel dataset including cross-validation and ablation results.	TransDualSegNet uses a dual-branch framework that fuses global OCT features with edge-enhancement cues. Reported cross-validation and ablation results compare its segmentation accuracy and efficiency with U-Net, Attention U-Net, and Trans U-Net.
[238]: Transfer Learning and Multi-Feature Fusion-Based Deep Learning Model for Idiopathic Macular Hole Diagnosis and Grading from Optical Coherence Tomography Images	2025	In a retrospective single-centre cohort, the authors use transfer learning with multi-feature fusion on OCT images to perform idiopathic macular hole (IMH) diagnosis and grading, supplemented by clinical-factor modelling to estimate surgical risk, with evidence reported from ROC/AUC-based diagnostic evaluation and a surgical-risk nomogram.	The authors develop an OCT radiomics and deep-feature fusion framework using a ResNet101 transfer-learning backbone to classify IMH grades and identify key drivers for surgical decision-making through a logistic-regression nomogram.

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Work	Year	Basis for inclusion	Principal contribution
[91]: Transformer-Based DME Classification Using Retinal OCT Images Without Data Augmentation: An Evaluation of ViT-B16 and ViT-B32 With Optimizer Impact	2025	The authors assess Vision Transformer models (ViT-B16, ViT-B32) for classifying diabetic macular oedema from retinal OCT images, using transfer learning without data augmentation and evaluating fine-tuning behaviour across Adam, SGD, and RMSProp with evidence based on 5-fold cross-validation and Grad-CAM visual explanations in an ophthalmic diagnostic task.	Cross-validation shows that optimiser choice and transformer resolution affect DME OCT classification outcomes, with ViT-B16 outperforming ViT-B32 and transformer models outperforming CNN baselines while providing interpretable Grad-CAM heatmaps that support decision understanding.
[245]: Two-step hierarchical neural network for classification of dry age-related macular degeneration using optical coherence tomography images	2023	For dry AMD staging, the authors use a two-step hierarchical CNN (with OCT image enhancement) to perform multi-class classification of dry age-related macular degeneration phases (normal, drusen, nascent geographic atrophy (nGA), and geographic atrophy) on Heidelberg Spectralis OCT macular B-scans, assessed with five-fold cross-validation and supported by ablation experiments.	The OCT-based hierarchical pipeline isolates nascent geographic atrophy from drusen using a second-stage discriminator, and the ablation results indicate that combining image enhancement with the hierarchical design substantially improves nGA performance over a flat baseline.
[188]: Unsupervised machine learning analysis of optical coherence tomography radiomics features for predicting treatment outcomes in diabetic macular edema	2025	Unsupervised machine learning clusters 234 treatment-naive diabetic macular oedema eyes from pretreatment OCT radiomics into imaging-defined subtypes and compares residual/recurrent disease and 6-month visual outcomes across clusters, with evidence supported by clustering robustness checks (bootstrap stability) and downstream multivariate feature/predictor analyses.	The clustering identifies four distinct DME radiomic clusters, including a lower-residual/recurrent and better-visual subgroup, and reports outcome-associated OCT radiomic biomarkers, most notably logarithm_gldm_DependenceVariance for residual/recurrent DME and wavelet-LH_firstorder_Mean for visual prognosis, offering noninvasive, OCT-omics-based stratification for treatment response heterogeneity.
[46]: Using Deep Learning and Transfer Learning to Accurately Diagnose Early-Onset Glaucoma From Macular Optical Coherence Tomography Images	2019	Across multiple institutions, the authors apply deep learning with transfer learning to spectral-domain macular OCT retinal-layer thickness features. They aim to distinguish early open-angle glaucoma from normal eyes, evaluating accuracy in a testing cohort using the AUC and comparing it with random forests and support vector machines.	A convolutional neural network trained with two-stage pretraining improved discrimination of early glaucoma on OCT and achieved a higher AUC than the conventional classifiers evaluated. These results support technical feasibility in the study cohorts, while prospective diagnostic utility was not assessed.

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Work	Year	Basis for inclusion	Principal contribution
[190]: Using Machine Learning to Detect Different Eye Diseases from OCT Images	2023	The authors use hyperparameter-tuned deep convolutional neural networks to classify eight retinal disease categories plus normal from OCT tomography images (OCT-C8 dataset), evaluating evidence through reported test-set classification metrics and confusion-matrix analysis.	A comparative experiment tunes six CNN architectures and highlights a fine-tuned DenseNet121 pipeline as the top-performing approach for multi-disease OCT classification, offering an OCT-based machine-learning alternative for early retinal disease detection.
[82]: Utilization of deep learning to quantify fluid volume of neovascular age-related macular degeneration patients based on swept-source OCT imaging: The ONTARIO study	2022	In a phase IV retrospective proof-of-concept analysis of neovascular AMD, the authors used a U-Net to segment intraretinal fluid, subretinal fluid, and serous pigment epithelial detachments from baseline swept-source OCT. Optional OCT angiography provided choroidal neovascular membrane (CNVM) lesion size. The imaging-derived measurements were correlated with change in BCVA at week 52 to assess their predictive value.	Baseline total retinal fluid volume had the strongest reported association with 12-month visual improvement, and adding mean CNVM size from OCTA increased the correlation. These retrospective findings support further evaluation of automated fluid volume as a candidate prognostic biomarker rather than an established clinical metric.
[276]: Utilizing Swin Transformer for the Classification of Ophthalmic Diseases in Optical Coherence Tomography (OCT) Images: A Novel Approach	2024	A Swin Transformer is evaluated for AMD and diabetic retinopathy detection using a Kaggle dataset and compared with CNN and autoencoder approaches using standard diagnostic metrics (accuracy, precision, recall, and F1 score) in an experimental benchmarking study.	The benchmark demonstrates how transformer-based representations can classify ophthalmic diseases in OCT images, and the reported metric-based comparison supports technical feasibility relative to CNN and autoencoder baselines without establishing improvement in clinical diagnostic workflows.
[197]: Utilizing Transformers on OCT Imagery and Metadata for Treatment Response Prediction in Macular Edema Patients	2023	A transformer combines OCT imaging with patient clinical metadata to predict treatment response for diabetic macular oedema and to estimate comparative effectiveness among anti-VEGF injections, with reported evidence of model accuracy relative to clinician performance.	The authors report that combining clinical-record metadata with OCT improves response-prediction accuracy by 12% over OCT-only training and produces drug-specific response probabilities for ranibizumab, aflibercept, and bevacizumab. These retrospective predictions do not establish comparative treatment effectiveness or clinical decision utility.
[428]: Validation and Clinical Applicability of Whole-Volume Automated Segmentation of Optical Coherence Tomography in Retinal Disease Using Deep Learning	2021	For whole-volume OCT validation, the authors use deep learning whole-volume OCT segmentation to quantify four macular pathological features in patients with wet age-related macular degeneration and diabetic macular oedema across real-world device types, comparing automated outputs with adjudicated expert segmentations via voxelwise agreement, volumetric metrics, and expert qualitative clinical applicability ratings.	Specialist ratings and grader agreement indicate that the automated model can deliver clinically acceptable, rapid retinal disease measurements in most scans by showing favourable specialist stack-ranking and Likert satisfaction alongside grader agreement, while also highlighting that qualitative assessment is necessary because quantitative concordance alone does not fully predict clinical usability.

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Work	Year	Basis for inclusion	Principal contribution
[234]: VALIDATION OF A DEEP LEARNING-BASED ALGORITHM FOR SEGMENTATION OF THE ELLIPSOID ZONE ON OPTICAL COHERENCE TOMOGRAPHY IMAGES OF AN USH2A-RELATED RETINAL DEGENERATION CLINICAL TRIAL	2022	In a clinical-trial imaging cohort, the authors evaluated DOCTAD, a fully automatic deep learning en face segmentation algorithm on spectral-domain OCT volumes, to delineate ellipsoid zone on USH2A-related retinal degeneration (RUSH2A clinical trial) with performance tested against trained reader manual segmentations using overlap and measurement-agreement evidence.	The external evaluation tests the EZ segmentation pipeline, developed for macular telangiectasia type 2, on 127 baseline USH2A-related OCT volumes and compares Dice overlap plus EZ area and length differences with reader measurements. These results assess cross-disease, multicentre performance of quantitative EZ biomarkers but do not establish clinical trial utility.
[123]: Validation of a deep learning model for automatic detection and quantification of five OCT critical retinal features associated with neovascular age-related macular degeneration	2024	The authors validate a U-Net deep learning model to automatically segment five OCT-derived retinal biomarkers (intraretinal fluid, subretinal fluid, subretinal hyperreflective material, drusen/PED, and neovascular PED) linked to neovascular age-related macular degeneration, with evidence based on performance measured against manual annotations using detection (ROC/AUC) and segmentation agreement (correlation and Dice) on sequestered data.	Detection and segmentation metrics quantify agreement between model outputs and manual annotations for these biomarkers in sequestered data. The study evaluates standardised OCT feature measurement, not effects on treatment-response assessment or clinical workflow outcomes.
[421]: Validation of a deep learning model for the automated detection and quantification of cystoid macular oedema on optical coherence tomography in patients with retinitis pigmentosa	2025	External test data validate an nnU-Net deep-learning segmentation model for automated CME assessment on spectral-domain OCT in RP. The study uses expert-annotated external test scans and reports overlap and area-agreement evidence, based on Dice coefficients and intraclass correlation, against human graders.	External testing on RP OCT volumes against three graders under a rotated-reference framework indicates that the model can provide near-human CME detection and quantification, potentially reducing manual variability in ophthalmic monitoring and clinical-trial workflows.
[261]: Validation of an Automated Artificial Intelligence Algorithm for the Quantification of Major OCT Parameters in Diabetic Macular Edema	2023	Across multiple centres, the study tests a semi-supervised adversarial deep learning algorithm on spectral-domain OCT (Spectralis map and linear scans) to automatically detect and quantify intraretinal and subretinal fluid, external limiting membrane and ellipsoid zone interruption, and hyperreflective retinal foci, using comparisons against blinded expert manual examination and reporting agreement and accuracy-based evidence.	The algorithm provides reproducible, automated quantification of major OCT biomarkers with expert-benchmarked concordance for fluid and retinal integrity measures and excellent reliability for hyperreflective foci counts, supporting more objective grading and monitoring in clinical DME workflows.

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Work	Year	Basis for inclusion	Principal contribution
[311]: Validation of Deep Learning-Based Automatic Retinal Layer Segmentation Algorithms for Age-Related Macular Degeneration with 2 Spectral-Domain OCT Devices	2025	A prospective study validates UNet and DeepLabv3 retinal layer segmentation on spectral-domain OCT in an ophthalmic AMD cohort, using expert-labelled ground truth to quantify ILM, RPE, and related thickness and volume outputs, with testing across two OCT devices to provide evidence of device-robust performance rather than single-device accuracy.	The authors train on Heidelberg Spectralis B-scans and evaluate on both Spectralis and Zeiss Cirrus, reporting segmentation metrics on the held-out device. The cross-device comparison assesses transfer for retinal-layer biomarker generation but does not establish clinical utility or device independence.
[398]: VCT-NET: An Octa Retinal Vessel Segmentation Network Based on Convolution and Transformer	2022	VCT-Net addresses ophthalmic OCTA retinal vessel segmentation using a U-shaped deep learning architecture that combines convolution with a Swin-transformer branch to leverage global and local context, with evidence drawn from comparative experiments on an OCTA-6M dataset.	VCT-Net combines transformer and convolutional features to handle weak vessel boundaries and complex vascular patterns in OCTA, reporting improved segmentation performance relative to other deep learning approaches.
[55]: Vessel Density Features of Optical Coherence Tomography Angiography for Classification of Glaucoma Using Machine Learning	2023	For glaucoma detection, the authors apply machine learning classifiers to peripapillary OCT angiography vessel density features generated via threshold-based segmentation, using a supervised classification task (glaucoma versus healthy) and reporting evidence from a train/test split with AUC and accuracy (including mild-to-moderate subgroup results for SVM).	The authors report test-set AUC and accuracy of 1 for an SVM using threshold-derived vessel-density features. Because perfect discrimination can be sensitive to sample size, split design, and overfitting, the result requires independent external validation before clinical interpretation.
[257]: Vision Transformers for Retinal Disease Classification using Optical Coherence Tomography Images	2025	The authors evaluate vision transformer architectures for ophthalmic retinal OCT image categorisation, treating retinal disease labelling as a seven-class classification task and reporting comparative evidence between ViT and BEiT versus CNN baselines.	The evaluation defines an OCT-based seven-label retinal disease classification pipeline using ViT and BEiT and provides reported accuracies that quantify the potential benefit of transformer models relative to CNN approaches for this diagnostic task.
[379]: Visual acuity prediction on real-life patient data using a machine learning based multistage system	2024	Using real-world patient data from heterogeneous clinical information systems, the authors combine OCT biomarkers with machine-learning and deep-learning components to predict visual-acuity progression in age-related macular degeneration, diabetic macular oedema, and retinal vein occlusion during intravitreal operative medication therapy, with evidence from a multistage system evaluated against ophthalmologist annotations.	The research workflow aggregates more than 49,000 patients, completes incomplete OCT biomarker documentation with a deep neural network ensemble, and predicts winners, stabilisers, and losers of visual-acuity progression from routine-care data. This supports large-scale longitudinal modelling, while effects on clinical decisions and patient outcomes remain untested.

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